Complete AI Study Notes

Source rule used in these notes: these notes are based only on the supplied PDFs listed by the user. No outside theory has been added. Links and names that appear here are included only when they appear in the source slides.

Companion coverage file: see `AI_page_by_page_coverage_index.md` for a page-by-page source map.

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01 Introduction-rrm2024.pdf

INTRODUCTION

This chapter introduces Artificial Intelligence, its history, major types, machine learning, deep learning, generative AI, applications, and risks.

Content

Source page(s)	Topic covered
1-2	Introduction and chapter content
3-4	What AI is and what intelligence means
5-11	Historical context: Turing Test, Dartmouth Workshop, AI winters, expert systems, neural networks
13, 18-19	Brain inspiration and neural networks
20-24	Types of AI and machine learning basics
27-35	AI applications and AI literacy
36-42	Discriminative AI, generative AI, and industry/academia examples
45-51	AI risks, job replacement, hallucination

What is Artificial Intelligence?

Artificial Intelligence is the part of computer science that creates systems or machines that can do tasks that normally need human intelligence. In the slides, these tasks include recognizing patterns, perception, understanding natural language, learning from experience, solving problems, reasoning, and making decisions.

The source example is a self-driving car. It uses cameras to perceive the road, signs, obstacles, and pedestrians, then uses that information to make driving decisions such as stopping or following traffic signals.

Intelligence

Level of intelligence from the slides	Simple meaning
Performing complex tasks	Solving difficult problems and handling many things at the same time
Learning new skills	Picking up new information and improving over time
Innovation	Creating new ideas, technologies, or solutions
Taking over	Handling tasks that used to be done manually or at a lower level

Historical Context

Period / item	What the slides say	Simple study meaning
Alan Turing Test, 1950	The test was described in Alan Turing's 1950 article.	A machine is judged by whether a questioner can tell it apart from a human through answers.
Dartmouth Workshop, 1956	Described as the birth of AI.	AI became a formal research field.

Period / item	What the slides say	Simple study meaning
1960s-1980s	Symbolic AI, reasoning and logic, inference, knowledge representation, search, first AI winter, expert systems, brute-force AI, IBM Watson, Deep Blue.	AI focused strongly on symbols, rules, search, and expert-like decision making.
First AI winter	Progress slowed and interest/funding decreased because expectations were not met.	AI became less popular for a time.
Expert systems	Computer systems designed to imitate human expert decision-making.	Systems that use expert rules to make decisions.
Brute-force approach	Trying every possible solution until the correct one is found.	A try-everything method.
1980s-1990s	Connectionist AI, neural networks, second AI winter, backpropagation paper in 1986.	AI interest returned to brain-inspired networks, then slowed again.
Backpropagation paper, 1986	Introduced a method to improve neural network training.	Helped make modern deep learning possible.

Turing Test

The Turing Test asks whether an interrogator can tell which respondent is a human and which is a machine by asking questions through a terminal. If the computer can fool the interrogator so that the interrogator cannot distinguish it from the human, the machine may be assumed to be intelligent according to the slide.

The Solution is in the Brain

The slides connect learning and intelligence to the human brain. The brain is made of neurons, and neurons send signals and connect with each other. This idea inspires artificial neural networks.

The Neural Network

Biological idea	AI version in the slides
Neurons and synapses help the brain learn.	Artificial neural networks are inspired by this learning structure.
Synapses connect neurons.	Parameters and weights act like synapses in a neural network.
The brain has about 100 trillion synapses or more.	ChatGPT is stated in the slide as having 150 billion parameters.

Important caution from the slide: an artificial neural network is not the same as a brain.

Types of AI

Type named in the slide	Short source-based meaning
Artificial Intelligence (AI)	The broad field.
Artificial Narrow Intelligence (ANI)	Listed as a type of AI in the slide.
Artificial General Intelligence (AGI)	Listed as a type of AI in the slide.

Machine Learning

Machine learning is described in the slides as programming computers so they can learn from data, and as giving computers the ability to learn without being explicitly programmed.

The Basic Idea

Approach	What is given	What is produced
Traditional programming	Data + rules	Results
Machine learning	Data + results	Rules

Simple meaning: in normal programming, humans write the rules. In machine learning, the system learns the rules from examples.

When Should We Use ML?

Condition from the slide	Simple meaning
A pattern exists	There is something repeatable in the data.
We cannot pin it down mathematically	It is hard to write exact rules by hand.
We have a representative data set	We have examples that fairly show the problem.

Example Applications of AI

Area	Examples from the slides
Images and vision	Product image analysis, tumor detection, tumor type detection
Text and language	Classifying news, flagging offensive comments, summarizing documents, chatbots, translation
Business and finance	Forecasting revenue, detecting credit card fraud, market forecasting, recommender systems
Voice	Reacting to voice commands
Customer/data grouping	Segmenting clients based on purchases, profiling
Medicine	Medical diagnosis

Deep Learning

Deep learning is presented as a branch of AI that uses neural networks with multiple hidden layers, needs lots of data, uses less feature engineering, and is described as the most successful branch of AI.

Shallow neural network	Deep neural network
Fewer hidden layers.	Multiple hidden layers.
Less depth for learning patterns.	More layers for learning richer patterns.

Why Now?

The slides give three reasons deep learning became powerful now: data, models, and computation.

Killer Applications

Area	Examples from the slides
Computer vision	Image classification, object detection, image segmentation, tracking
Speech	Speech-to-text, text-to-speech, trigger/wake-word detection, speaker ID
Natural language processing	Text classification, sentiment recognition, machine translation
Information retrieval	Web search

Other Applications of AI

AI can be used for demand forecasting, product placement, supply chain, and quality control. The slides give examples of checking fabric defects, cake quality, vegetable quality, and wood quality.

Applications in the Medical Field

The slides mention personalized medicine, remote medicine, and detecting a signal that would otherwise be buried in noise.

AI Literacy

Low-code and no-code tools, plus platforms such as LandingLens by Landing AI, may allow more people to build AI systems, including doctors, store managers, accountants, quality inspectors, retailers, restaurants, and small business owners.

Sample FYP/Master Projects

Project area	Examples from the slides
Human Activity Recognition (HAR)	Sports, fitness, lifestyle, healthcare
Gesture recognition	Hand/head gesture recognition
Smart environments	Smart home applications
Interfaces	Machine user interface
Roads	Road surface monitoring, road maintenance, road safety, comfort driving

Discriminative versus Generative

Discriminative	Generative
Given a picture, tells who or what it is.	Creates something new, such as a non-existing person's face.
Used for recognition or classification.	Used for creating text, images, or other media.

Generative AI

Generative AI creates text, images, or other media using generative models. It learns patterns and structure from training data, then creates new data with similar characteristics.

The slides mention transformer-based deep neural networks and examples such as ChatGPT, Bing Chat, Bard, LLaMA, Stable Diffusion, Midjourney, and DALL-E.

Academia versus Industry

Example	What the slide says
AlexNet, 2012	Started the current AI revolution; done by a graduate student in a university.
GAN model, 2014	Started the generative AI revolution; done by a graduate student in a university.
AlphaFold, 2021	DeepMind.
ChatGPT, 2022	OpenAI.
Parameter scale	From less than 100 million parameters to more than 100 billion parameters.

The Risks of AI

Risk area	Examples from the slides
Malicious use	Disinformation, scams, hacking, cybercrime
Fake media	Fake news and deepfakes
Attacks	Adversarial attacks
Ethical and societal concerns	Responsible AI, bias, fairness, privacy
Human impact	Mental health, depression, anxiety, job replacement
Severe future concern	Existential threat question
Reliability	Hallucination

On Job Replacement

The slides say the AI revolution may happen much faster than the industrial revolution. They also state that AI can automate tasks requiring routine intelligence, with examples in transportation, customer service, healthcare, and marketing.

02 Logic and inference- sem 2 202526 (2).pdf

LOGIC AND INFERENCE

This chapter explains how AI uses knowledge, logic, structured representation, inference, propositional logic, predicate logic, and reasoning rules.

Content

Source page(s)	Topic covered
1-2	Chapter title and content
3-6	Introduction to logic in AI and knowledge representation methods
7-13	Domain knowledge, inference mechanism, syntax and semantics
14-16	Logic as a formal AI language and modus ponens
17-23	Propositional logic and sentences
24-31	Arguments, syllogism, deductive reasoning, rules of inference
32-45	Predicate calculus / FOL and predicate logic exercise

Introduction to Logic in AI

Knowledge is necessary for intelligent behavior. The slides give simple examples: if you do not know the features of two products, you cannot choose smartly between them; if you do not know fire burns, you could hurt yourself.

AI systems use knowledge to decide and solve problems. A smart assistant uses knowledge about user preferences, while a self-driving car needs knowledge about road objects to make safe decisions.

Need for Knowledge Representation

AI needs a structured way to store and process information. The slides mention knowledge graphs, where entities are nodes and relationships are edges.

Logic as a Technique

Logic provides a systematic way to represent facts, rules, and relationships in a formal language that AI can interpret. It also helps AI reason about what it knows.

Knowledge Representation Methods

Method	Strengths	Weaknesses
Logic	Precise, clear, and allows automated reasoning and new knowledge inference.	Can be complex for real-world knowledge and may not capture common sense well.
Semantic Networks	Clear visual representation of relationships and efficient for specific tasks.	Limited expression power and limited reasoning ability.
Production Rules	Good for cause-and-effect relationships.	Can become difficult to manage with complex knowledge and exceptions.

Knowledge Representation - How to represent Knowledge in Machine?

Component	Simple meaning	Source example
Domain-specific knowledge	Knowledge built for one field or problem area.	Medical diagnosis AI uses disease, symptom, and treatment knowledge.
Inference mechanism	A method for interpreting knowledge after sensing facts from the environment.	Smart home turns on heat if temperature drops; turns on lights if motion is detected at night.
Syntax and semantics	Syntax controls valid symbol structure; semantics controls meaning.	NLP grammar vs actual meaning; image pixels linked to "stop sign."

!!! EXAM TOPIC: Predicate and Propositional Logic !!!

This is one of the most important exam areas. Keep the difference clear:

Logic type	What it can talk about	Simple meaning
Propositional logic	Whole statements that are true or false.	It treats statements like blocks: "It is raining" is either true or false.
Predicate logic / First-Order Logic (FOL)	Objects, properties, relationships, and quantities such as "all" or "some."	It can say who or what a statement is about, such as "Aminah likes pizza."

Logic as a Foundation for AI

Logic is a formal language for representing knowledge at the symbol level. It includes:

Part of logic	Simple meaning
Syntax	Rules for forming valid logical sentences.
Semantics	Meaning of the sentences.
Deduction	Deriving conclusions from premises.

In logic, letters such as P, Q, and R can stand for propositions or statements. A statement must have a truth value: True (T) or False (F).

Modus Ponens

Piece	Source scheduling example
Premise 1	If there is a conflicting meeting, then I cannot schedule a new meeting at that time. P -> Q
Premise 2	There is a conflicting meeting at 2 PM. P
Conclusion	Therefore, I cannot schedule a new meeting at 2 PM. Q

Simple meaning: if "P causes Q" is true, and P happens, then Q must happen.

Propositional Logic and Sentences

Propositional logic deals with statements that can be either true or false. The simplest knowledge pieces are propositions.

Example from the slide: "It is raining" can be True or False depending on the weather.

The question "Please work AI before AI works us" is not treated like a normal proposition in the same way, because it is a request/command rather than a statement with a clear true/false value.

How to Form Propositional Sentences

Rule from the slide	Simple meaning
Each symbol is a sentence.	P by itself can be a valid logical sentence.
If P and Q are sentences, then (P) is a sentence.	Parentheses can group a sentence.
P □ Q is a sentence.	P AND Q.
P □ Q is a sentence.	P OR Q.
¬P is a sentence.	NOT P.
P -> Q is a sentence.	IF P, THEN Q.
P <-> Q is a sentence.	P IF AND ONLY IF Q.
Nothing else is a sentence.	Only expressions built by the rules count as well-formed formulas.

Sentences are also called well-formed formulas (WFF).

Logical Connectives

Symbol	Word meaning	Everyday source-style meaning
□	AND	Both parts must be true.
□	OR	At least one part is true.
¬	NOT	Reverses the truth value.
->	IMPLIES	If the first part is true, the second follows.
<->	IF AND ONLY IF	Both sides match each other.

Truth Values and Semantics

When we interpret a logical sentence, we give it meaning, then it becomes either True or False. For compound sentences, first assign truth values to the smaller statements, then calculate the truth value of the whole expression.

English Sentences to Propositional Logic Sentences

English form	Logic form	Example from slides
and, but	P □ Q	It is hot and cloudy.
not	¬P	It is not hot.
inclusive or	P □ Q	It is hot or cloudy, or both.
exclusive or	(P □ Q) □ ¬(P □ Q)	It is either hot or cloudy, but not both.
neither... nor	¬P □ ¬Q	It is neither hot nor cloudy.

Propositional Exercise

Let P = "King is healthy", Q = "King is wealthy", and R = "King is wise".

Statement	Propositional expression
King is healthy and wealthy but not wise.	P □ Q □ ¬R
King is not wealthy, but he is healthy and wise.	¬Q □ P □ R
King is neither healthy nor wealthy nor wise.	¬P □ ¬Q □ ¬R
King is healthy, wealthy and wise.	P □ Q □ R
King is wealthy, but he is not both healthy and wise.	Q □ ¬(P □ R)

Arguments, Premises, and Conclusions

Part	Meaning	Source example
Premise	A statement used as support.	All humans are mortal. Socrates is a human.
Conclusion	The statement supported by the premises.	Therefore, Socrates is mortal.

AI uses logical reasoning to draw new conclusions from known premises.

Syllogism and Deductive Reasoning

A syllogism has two premises and one conclusion. It helps us reason based on clear evidence.

Type	Meaning
Valid syllogism	The conclusion logically follows from true premises.
Invalid syllogism	The conclusion does not logically follow from the premises.

AI can detect invalid syllogisms by checking logical structure, truth conditions, and reasoning patterns against known logical fallacies.

Rules of Inference

Rules of inference are logical rules for deriving valid conclusions from premises. They make sure the reasoning process has the correct structure.

Rule named in slides	Simple meaning
Modus Ponens	If P -> Q and P is true, conclude Q.
Modus Tollens	Named as a rule used by AI systems in the slide.
Hypothetical rule	Named as another inference rule.
Disjunctive rule	Named as another inference rule.

Predicate Calculus / FOL

Predicate logic is usually used as another name for first-order logic. It is needed because propositional logic has limited expressive power.

Need	Source example
Talk about "some" humans.	"Some humans are rich."
Talk about a relationship involving a named object/person.	"Aminah likes pizza."

FOL - Introduction

First-order logic is another way to represent knowledge in AI. It extends propositional logic and can express natural language statements more meaningfully and concisely.

FOL Contains

Element	Simple meaning	Source examples
Objects	Things being talked about.	A, B, people, numbers, colors, wars, theories, squares, pits
Relations	Ways objects connect or properties they have.	red, round, adjacent, sister of, brother of, has color, comes between
Functions	A relation-like expression that gives an object.	father of, best friend, sons of
Syntax	Rules for writing valid FOL expressions.	Listed as one of the two main parts.
Semantics	Meaning of FOL expressions.	Listed as one of the two main parts.

Atomic Sentences and Complex Sentences

Sentence type	Form	Example
Atomic sentence	Predicate(term1, term2, ..., term n)	Brothers(Ravi, Ajay), cat(Chinky)
Complex sentence	Atomic sentences combined using connectives.	Built by joining smaller logical parts.

Subject and Predicate

In the source example "x is an integer", x is the subject and "is an integer" is the predicate.

Quantifiers in First-Order Logic

Quantifier	Symbol	Simple reading	Main connective from slides	Source example
Universal quantifier	□	For all / for each / for every	Implication ->	□x students(x) -> learn(x, AI)
Existential quantifier	□	There exists / for some / at least one	AND □	□x staff(x) □ busy(x)

Properties of Quantifiers

Property	Meaning
□x□y is similar to □y□x	Two universal quantifiers can be switched.
□x□y is similar to □y□x	Two existential quantifiers can be switched.
□x□y is not similar to □y□x	Mixing existential and universal order changes meaning.

03 CSNB4133_Chap3- sem 1 2024 2025.pdf

State Search Representation

This chapter explains search as the process of finding a sequence of actions that reaches a goal.

Content

Source page(s)	Topic covered
1-3	Outline and search introduction
4-9	State space, paths, goal tests, travelling salesperson example
10-14	Search trees, evaluation criteria, search method overview
15-36	Blind search and depth-first search
37-42	Breadth-first search
43-44	Heuristic search and benefits
45-56	Hill-climbing search, problems, and solutions
57-62	Best-first search

Introduction

Predicate calculus can describe objects and relationships. Inference rules such as modus ponens let us infer new knowledge. These inferences create a space that can be searched for a problem solution.

Search means looking for a sequence of actions that reaches a goal. A search algorithm takes a problem as input and returns a solution as an action sequence.

State Search Representation

Part of state space	Simple meaning
Initial state space	Where the problem starts.
Possible actions / operators	Moves the system can take.
Goal state	The state we want to reach.
Nodes	States in the tree/graph.
Links	Actions between states.

State Space Terms

Term	Meaning
Path	A sequence of actions from one state to another, hopefully ending at the goal.
Path cost	A numeric cost assigned to a path. The desired path is usually the shortest.
Goal test	A check to see whether a state is a goal state.

Why "Search"?

Search helps explore alternatives in a tree and find a sequence of steps in planning. Goal-driven activities happen in a state space, so they are state space problems.

Classical Search Domains

Domain listed in slides	Simple study meaning
8-Puzzle	A puzzle search problem.

Domain listed in slides	Simple study meaning
Water Jug	A problem involving possible water states.
Blocks World	A block-arrangement search problem.
Travelling Salesman	Find shortest route visiting cities and returning home.
Maze	Find a route through a maze.
Chess	Search possible moves.
Tower of Hanoi	Search move sequences between pegs.

Travelling Salesman Example

The goal is for a salesperson to visit five cities, return home, and find the shortest path.

!!! EXAM TOPIC: Search Trees !!!

A tree is a graph that is connected, becomes disconnected if any branch is removed, and has exactly one path between any two nodes.

When judging a search method, ask these exam-style questions:

Question	What it checks
Does the method actually find a solution?	Completeness.
Is it a good solution?	Path cost and search cost.
How much time and memory are needed?	Search cost.
Does it find the best possible solution?	Optimality.

Evaluating a Search

Criterion	Simple meaning
Completeness	Guaranteed to find a solution if one exists.
Time complexity	How long it takes to find a solution.
Space complexity	How much memory is needed.
Optimality	Whether it finds the best solution when several solutions exist.

Search Methods

Category	Methods listed
Blind search / uninformed algorithms	Depth-first, breadth-first, bi-directional, iterative deepening, depth-limited, uniform cost
Heuristic search / informed algorithms	Hill-climbing, best-first, generate and test, induction, greedy search
Optimal search	A* search

Search Methods at the First Glance

Method	Simple source-based explanation
Depth-first search	Explores the search tree branch by branch.
Breadth-first search	Examines the search tree row by row.
Hill-climbing	Like depth-first, but examines the most promising child first.
Best-first search	Expands the most promising partial path found so far.
A* search	Described as best-first plus branch-and-bound.

Blind Search Methods

Blind search examines the search tree in an orderly way. It is also called uninformed search because it has no domain knowledge. It can only tell whether a state is a goal state or not a goal state.

Blind search idea	Meaning
Exhaustive / complete	Checks all possible solutions.
Partial	Checks only some alternatives.

Depth-First Search (DFS)

DFS begins at the root and works downward into deeper levels. It continues until it finds a solution or reaches a dead end and must backtrack.

DFS algorithm step	Simple meaning
Delete FIRSTNODE from start of QUEUE	Take the next node to process.
Take children of FIRSTNODE	Look at the node's next possible states.
Add to the front of QUEUE	Put children first so the search goes deep.
Put result in NEWQUEUE	Continue with the updated queue.

For the source tree, DFS traversal order is:

Source tree	DFS order
S with children A and B; A has C and D; B has E and F; E has G and H	S, A, C, D, B, E, G, H, F

DFS exercise answer from the slide:

Initial state	DFS output
A	A, B, L, C, D, E, H, G, F

!!! EXAM TOPIC: Breadth-First Search (BFS) !!!

BFS examines all nodes in the search tree starting from the root. It finishes one level before moving to the next level.

BFS algorithm step	Simple meaning
Take children of FIRSTNODE	Find all next states from the current node.
Delete FIRSTNODE from start of QUEUE	Remove the node that has just been handled.
Append children to the end of QUEUE	Add new nodes at the back so older same-level nodes go first.
Put result in NEWQUEUE	Continue with the updated queue.

For the source tree, BFS traversal order is:

Source tree	BFS order
S with children A and B; A has C and D; B has E and F; E has G and H	S, A, B, C, D, E, F, G, H

BFS exercise answer from the slide:

Initial state	BFS output by levels
A	A, BL, CG, DFEH

Heuristics Search

Heuristic search reduces the amount of searching. It uses a rule-of-thumb approach, prunes non-promising or weaker nodes, and usually speeds up the process of getting a good-enough solution.

Benefits of Heuristics

Benefit / use case	Simple meaning
Used when there is no exact solution, such as medical diagnosis.	It helps when perfect certainty is not available.
Used when exact solution is too expensive, such as chess.	It helps when checking everything would take too much time.
Flexible.	It can adapt to problem situations.
Better when the optimal solution is too costly to generate.	A good-enough answer may be practical.
Simpler for decision makers to understand.	Managers may think in a similar rule-of-thumb way.

!!! EXAM TOPIC: Hill-Climbing Search !!!

Hill-climbing combines depth-first search with a way to order alternatives by measuring likely success at each decision point. It tries to reach the goal by choosing nodes predicted to be nearest to the goal.

Hill-climbing algorithm step	Simple meaning
Delete FIRSTNODE from start of QUEUE	Pick the current node.
Take children of FIRSTNODE	Look at possible next nodes.
Order children with most promising first	Rank children by how close they seem to the goal.
Place them at the front of QUEUE	Search the best-looking child first.
Put result in NEWQUEUE	Continue.

Hill-climbing is similar to DFS, but paths are not selected randomly. They are selected based on closeness to the goal.

In the source example for finding target node "1", hill-climbing compares B, C, and D, starts with branch I because it has the lowest value among B, C, and D, then backtracks and eventually reaches path A, B, E, I, D, H, G, J. The slide says the path A-C-F was not visited, saving time and cost.

Problems with Hill-Climbing

Problem	Simple meaning from the slides
Foothill problem	A local peak is better than nearby states, but another higher state exists elsewhere.
Plateau problem	A flat area where neighboring states have the same value, so the algorithm cannot find a best direction.
Ridges problem	A high area with a slope that cannot be reached in a single move.

Solution to Hill-Climbing

Solution	Simple meaning
Random restart hill-climbing	Generate random initial states and try again.
Simulated annealing	Like hill-climbing, but it can accept random moves. If the move improves the solution, accept it; bad moves can also be allowed with a probability that decreases as the move gets worse.

Best-First Search

Best-first search combines depth-first and breadth-first search. It uses a heuristic evaluation function to score candidate nodes, then chooses the best value node.

Best-first feature	Simple meaning
More flexible than hill-climbing	It can switch between paths.
Uses an evaluation function	Each candidate node gets a score.

Best-first feature	Simple meaning
Chooses the most promising node	It follows the best-looking option at each step.
Can become shortsighted	It may focus on a promising partial path too strongly.

Source examples include games and web crawlers. In a web crawler, pages are nodes and hyperlinks are unvisited successor nodes. Priority can be based on how closely a page matches a search query.

Best-first algorithm step	Simple meaning
Delete FIRSTNODE from start of QUEUE	Take the current best node.
Take children of FIRSTNODE	Generate next possible nodes.
Append children to QUEUE	Add them to possible choices.
Order result with most promising first	Sort by best score.
Put ordered result in NEWQUEUE	Continue with the sorted queue.

In the source example for finding target "3", best-first search path is A, B, D, H, G, J. The slide says hill-climbing would use A, B, E, I, D, H, G, J, which is longer.

05_CSNB234_Chap5_Notes_Knowledge_representation_sem_1_2024_2025.pdf

Knowledge Representation

This chapter covers formal logic, semantic networks, conceptual graph, frames, scripts, production rules, and reasoning methods in production rule systems.

Content

Source page(s)	Topic covered
1-4	Introduction to knowledge representation and reasoning
5-8	Types of knowledge and representation techniques
9-12	Semantic networks
13-15	Frames and frame advantages
16-22	Production rules and rule firing
23-32	Forward chaining
33-37, 40-43	Backward chaining
38-39	Summary and supplementary

Conceptual Graph and Scripts are named in the source outline and summary. The extracted source pages do not provide separate explanatory body text for them, so these notes do not add outside explanation for those two headings.

Introduction

Humans understand, reason, and interpret knowledge. Knowledge representation asks how machines can represent information about the real world so they can use it to solve complex problems such as medical diagnosis or natural language communication.

Knowledge representation is not just storing data. It lets an intelligent machine learn from knowledge and experience so it can behave intelligently.

Knowledge and KRR

Term	Simple meaning
Knowledge	Awareness or familiarity gained from facts, data, situations, and experience.
Knowledge representation and reasoning (KR, KRR)	The AI area about how agents think and how thinking supports intelligent behavior.

Types of Knowledge to Represent

Type	Simple meaning	Source example
Object	Facts about objects in the world.	Guitars have strings; trumpets are brass instruments.
Events	Actions that happen in the world.	Events are actions.
Performance	Knowledge about how to do things.	Behavior/action knowledge.
Meta-knowledge	Knowledge about what we know.	Knowing about knowledge.
Facts	Truths about the real world.	What is represented.
Knowledge-base (KB)	A group of sentences used by a knowledge-based agent.	Central component of knowledge-based agents.

Knowledge Representation Techniques

Technique	Simple source-based meaning
Logical representation	Uses rules, propositions, syntax, and semantics with no ambiguity.
Semantic networks	Concepts as nodes and relationships as links.
Frames	Information stored in meaningful chunks using slots and slot values.
Production rules	Knowledge captured as IF condition THEN action rules.

Formal Logic

Logical representation is a language with concrete rules. It draws conclusions based on conditions, uses precise syntax and semantics, and supports sound inference. The slides categorize it into propositional logic and predicate logic.

Syntax and Semantics

Concept	Simple meaning
Syntax	Rules that decide how legal sentences are built and which symbols can be used.
Semantics	Rules for interpreting the sentence and assigning meaning.

!!! EXAM TOPIC: Semantics / Semantic Networks !!!

Semantic networks show meaning by drawing concepts and relationships. Think of them as a meaning map.

Semantic network part	What it means
Node	A concept, such as bird, person, book, famous, intelligent.
Link / arc	A relationship between concepts.
Label on arc	The type of relationship, such as is_a, has_a, has_part.
Binary relation	A relationship connecting two concepts.

How meaning and relationships are mapped:

Source statement	Semantic network mapping
Jerry is a cat.	Arrow from Jerry to cat, labeled is_a.
Jerry is a mammal.	Arrow from Jerry to mammal, labeled is_a.
Jerry is owned by Priya.	Relationship link between Jerry and Priya.
Jerry is brown colored.	Relationship link from Jerry to brown colored.
All mammals are animal.	Arrow from mammal to animal, labeled is_a.

Example 2 from the source:

Description	Relationship
Lab is a room.	Lab is_a room.
Lab has a door.	Lab has_a door.
Lab has many computers.	Lab has computers.
Printer is in lab.	Printer in lab.
Laser printer is a printer.	Laser printer is_a printer.

Drawbacks of Semantic Networks

Drawback	Source example
Disjunctive information is hard to include.	Apple can be either green or red.

Drawback	Source example
Conjunctive information is hard to include.	Panda has color black and white.

Frames

A frame stores information in meaningful chunks. It has slots and slot values.

Frame idea	Simple meaning
Slot	A field/category, such as title or author.
Slot value	The value stored in that field.

Book frame example from the slide:

Slot	Value
Title	Qualitative Reasoning
Author	Ken D. Forbus
Publisher	Prentice-Hall
Year	2000

Advantages of Frame Representation

Advantage	Simple meaning
Groups related data.	Makes programming easier.
Flexible and used in many AI applications.	Can fit different situations.
Easy to add slots.	New attributes and relationships can be added.
Easy to include default data.	Missing values can be searched for or filled.
Easy to understand and visualize.	Humans can read the structure more easily.

Production Rules

Production rules use simple IF-THEN form. Expert systems often use large sets of production rules.

Production rule part	Other names in slide	Simple meaning
IF part	Antecedent, premise, condition	What must be true.
THEN part	Consequent, conclusion, action	What happens after the condition is true.

Traffic-light examples:

Rule	Meaning
IF traffic-light is green THEN action is go	Green means go.
IF traffic-light is red THEN action is stop	Red means stop.

Multiple Conditions

Connector	Meaning
AND	All conditions must hold.
OR	At least one condition must hold.

Example from the slide:

IF conditions	THEN action
Age of student < 21 AND SPM number of A's >= 8	Admit student to BIT

Types of Things Rules Can Represent

Type	Source example
Relations	IF fuel tank is empty THEN car will not start.
Recommendation	IF you study hard AND smart AND never absent THEN you will get an "A".
Strategy	Step-by-step car troubleshooting rules.
Heuristics	IF spill is liquid AND pH < 6 AND smell is vinegar THEN spill material is acetic acid.
Directive	IF fuel tank is empty THEN refuel the car.

"Firing" of Rules

A rule fires when its condition part is satisfied. Then its action part is executed. The inference engine links rules in the KB with facts in the DB to reach a solution. The explanation facility lets users ask "why" and "how".

Reasoning Methods in Production Rule Systems

Method	Simple name	Direction
Forward chaining	Data-driven reasoning	Starts from known facts and moves forward to conclusions.
Backward chaining	Goal-driven reasoning	Starts from a goal and works backward to prove it.

!!! EXAM TOPIC: Forward Chaining !!!

Forward chaining is data-driven reasoning. It begins with known facts/data, then uses rules to add new facts until it solves the problem or no rule can fire.

Forward Chaining Rules

Forward chaining rule from slides	Simple meaning
Starts from known fact/data.	Begin with what is already true.
Proceeds forward with the data.	Use facts to trigger rules.
Only the topmost rule is executed each time.	Rules are checked in order.
Fired rule adds a new fact to the database.	New knowledge is created.
Any rule can be executed only once.	Do not fire the same rule repeatedly.
Stops when no further rules can fire.	End when nothing new can be concluded.

Forward Chaining - Simplest Process

Step	What to do
1	Collect rules in the KB whose conditions match facts in the DB.
2	Do the action stated by the rule.
3	Add facts to the DB or delete facts from the DB.
4	Repeat until the problem is solved or no condition matches.

Forward Chaining Example: Prove If A and B true, then D is true

Given rule	Meaning
Rule 1: If A and C then F	A plus C gives F.
Rule 2: If A and E then G	A plus E gives G.
Rule 3: If B then E	B gives E.
Rule 4: If G then D	G gives D.

Step	Known facts / fired rule	Result
Start	A and B are true.	DB has A, B.
1	Rule 3 fires because B is true.	Add E.
2	Rule 2 fires because A and E are true.	Add G.
3	Rule 4 fires because G is true.	Add D.
End	D is true.	Goal reached.

Forward Chaining Health Exercise

Starting facts	Fired rules	Final recommendation
Catholic, eats poultry, works 4 hours today	R9, R4, R2, R6	Healthy
Eats veal	R8, R1, R7	Unhealthy

Why rules did or did not fire in the Catholic/poultry example:

Rule	Used?	Reason from slide
R9	Yes	Working 6 hours or less today means it is Friday.
R4	Yes	Catholic and Friday means eat no fish and no beef.
R2	Yes	Eats fish or poultry and no beef gives low cholesterol.
R6	Yes	Low cholesterol gives healthy.
R1	No	No clue on egg.
R3	No	Person eats poultry, so condition is false.
R5	No	Not told the person is vegetarian.
R8	No	Eating veal is unknown.

Forward Chaining Exercise: A, B, C, E are true

Rule	Condition	Conclusion
Rule 1	X, B, E true	Y true
Rule 2	Y, D true	Z true
Rule 3	A true	X true
Rule 4	Y, C true	W true

Step	Fired rule	Result
Start	A, B, C, E true	Facts available.
1	Rule 3	X becomes true.
2	Rule 1	Y becomes true.
3	Rule 4	W becomes true.
End	Rule 2 fails because D is not true.	System returns W.

Backward Chaining

Backward chaining is goal-driven. It is best when we want to find the reason after something has happened. The expert system starts with a goal and tries to find evidence to prove it. If evidence is not found, backtracking is used.

Forward Chaining vs Backward Chaining

Feature	Forward chaining	Backward chaining
Starting point	Known facts/data	Goal/hypothesis
Direction	Facts -> conclusions	Goal -> supporting facts
Main use from slides	Data-driven reasoning	Goal-driven reasoning
Stop condition	Problem solved or no rule fires	Goal proven or no rules can prove subgoal

Backward Chaining - Simplest Process

Step	What to do
1	To prove goal G, first check if G is already a fact.
2	If G is a fact, G is proven and stop.
3	If not, find a rule that concludes G.
4	Try to prove each condition of that rule.
5	G is true if all required premises are true.

Backward Chaining Example: Catch 6:00 Flight

Rule	IF	THEN
1	Wake up at 4:00	Pack at 4:30
2	Pack at 4:30	Leave home by 5:00
3	Leave home at 5:00	Park car by 5:15
4	Park car at 5:15	Check in by 5:30
5	Check in by 5:30	Catch 6:00 flight

Backward chaining starts with the goal "catch 6:00 flight", then looks for the rule that can prove it, then keeps moving backward until it reaches the starting premise: wake up at 4:00.

06 ML - First Iteration sem 1 2024 2025 (1).pdf

INTRODUCTION

This chapter gives a first overview of machine learning.

What is Machine Learning?

Machine Learning is a subset of AI that enables systems to learn and improve from data without being explicitly programmed. The key concept is learning from data.

Why Machine Learning?

Application area	Examples from slides
Search	Google search
Ads	Google ads
Recommendation	Netflix, YouTube
Voice assistants	Siri, Alexa
Computer vision	Face recognition
NLP	Google Translate, ChatGPT

When Should We Use ML?

Condition	Simple meaning
A pattern exists	The data has a repeatable pattern.
We cannot pin it down mathematically	Exact hand-written rules are hard.
We have a representative data set	We have examples that show the problem well.

Core Concepts of Machine Learning

Concept	Simple meaning
Data	The foundation of ML.
Algorithms	The engine of learning.
Models	The result of learning.
Training set	Data used to train the model and let it learn patterns.
Test set	Data used to evaluate performance on unseen data.

Types of Machine Learning

Type	Simple meaning	Source examples
Supervised learning	Learning with labeled data.	Image recognition, spam detection
Unsupervised learning	Finding patterns in data without labels.	Customer segmentation, anomaly detection
Reinforcement learning	Learning by trial and error using feedback from the environment.	Self-driving cars, AlphaGo

Machine Learning Tasks

Task	What it does	Type named in slides
Classification	Puts data into predefined classes.	Supervised learning
Regression	Predicts continuous values.	Supervised learning

Task	What it does	Type named in slides
Clustering	Groups similar data points.	Unsupervised learning
Association	Discovers rules in large data.	Unsupervised learning

Machine Learning Techniques

Technique	Simple meaning
Neural networks	Mimic the human brain.
Decision trees	Flowchart-like tree structures.
Support vector machines	Find the best boundary that separates classes.
Ensemble methods	Combine predictions from multiple models.

Challenges in Machine Learning

Challenge type	Challenge	Simple meaning
Technical	Data quality	Bad data can hurt learning.
Technical	Overfitting	Model learns training details too much, including noise.
Technical	Underfitting	Model is too simple to learn the real pattern.
Technical	Resources	ML may need resources.
Ethical	Bias and fairness	Output may be unfair.
Ethical	Privacy concerns	Data may involve private information.
Ethical	Future of work and automation	Jobs and work may change.
Ethical	Mental health	Human well-being may be affected.
Ethical	Malicious use of ML	ML can be used harmfully.

07 Traditional ML - sem 2 2025 2026 Supervised learning 1 (2).pdf

TRADITIONAL ML - SUPERVISED LEARNING 1

This chapter covers supervised learning tasks, regression, classification, linear regression, loss functions, gradient descent, overfitting, and underfitting.

Content

Source page(s)	Topic covered
1-5	Supervised learning overview and tasks
6-9	Classification and income prediction examples
10-16	Regression, features, training/test datasets, learning algorithms
17-24	Linear regression, OLS, cost/loss, optimization
25-28, 37	Overfitting and underfitting
31-36	Gradient descent

Supervised Learning

Supervised learning starts with training examples that have correct labels. The algorithm learns how to map input data to specific outputs.

Tasks of Supervised Learning

Task	What it predicts	Example from source
Classification	A category/label.	Handwritten digit label 0-9.
Regression	A continuous value.	Annual income from years of education.

Classification Example: Handwritten Digits

Step	Simple meaning
Model sees many labeled images.	Example: image of "5" -> label 5.
Model learns patterns.	It notices image features linked to labels.
Model predicts new images.	It guesses the number for an unseen handwritten image.

Predicting Annual Income

The slides compare a rigid rule-based model with machine learning. A hand-written rule might say every extra year of higher education increases annual income by $5,000. Machine learning instead learns the relationship from labeled data.

Approach	How it works	Source issue
Rules-based model	Human writes fixed rules.	Rules become complicated as more factors are added.
Supervised learning	Machine learns relationship from X and Y examples.	Goal is to predict Y accurately for new X.

Regression: Predicting a Continuous Value

Regression predicts a continuous target variable Y. Continuous means the value can move smoothly without gaps, such as weight or height.

Term	Meaning
X	Input data/features.

Term	Meaning
Y	Target output we want to predict.
Feature	Useful attribute, such as years of education or job title.
Numerical feature	Number-based feature, such as years of work experience.
Categorical feature	Category-based feature, such as job title.

Training and Test Datasets

Dataset	What it contains	Purpose
Training dataset	Labeled examples with features and target values.	Used to train the model.
Testing dataset	Used during evaluation.	Tests whether model works on unseen data.

Generalization means the model should not only memorize training data. It should predict accurately on new data.

Importance of Training and Testing

If training/testing is used	If it is skipped
Helps check whether the model generalizes to new data.	Model may look better than it really is.
Helps detect overfitting.	Real-world accuracy may suffer.

Linear Regression (Ordinary Least Squares)

Linear regression is used in the source to predict income Y from years of education X. The goal is to learn a linear model that predicts a new Y for an unseen X with as little error as possible.

The slide uses the form Y = mX + c, also described with parameters $\beta 0$ and $\beta 1$.

Linear regression part	Simple meaning
Slope $\beta 1$	How much Y changes when X increases by one unit.
Intercept $\beta 0$	Starting value when X is zero.
Prediction $\hat{y}$	The model's guessed value.

If the slope is 5 and income is measured in thousands of dollars, then one more year of education increases predicted income by 5 units, meaning $5,000.

Cost Function / Loss Function

A cost or loss function measures how inaccurate the model's predictions are. For linear regression, the slide names Mean Squared Error (MSE) as the common cost function.

Goal	Meaning
Define cost function	Measure error.
Minimize cost function	Adjust $\beta 0$ and $\beta 1$ so predictions become more accurate.

Optimization Process

Method	Source meaning
Ordinary Least Squares (OLS)	Directly computes parameters that minimize the cost function.
Gradient descent	Iteratively adjusts parameters to reduce the cost function.

Overfitting

Overfitting happens when a model learns training data too well, including noise and small mistakes.

Overfitting behavior	Result
Works very well on training data.	Looks strong during practice.
Works poorly on new data.	Fails to generalize.
Memorizes instead of understanding.	Like memorizing exact past exam answers, then failing when questions change.

Ways to Combat Overfitting

Method	Simple meaning	Source example
Get more data	More examples make memorization harder.	Use 500-1000 spam emails instead of 50.
Simplify the model	Smaller models are less likely to memorize noise.	Use smaller neural network or logistic regression instead of huge deep network.

Underfitting

Underfitting behavior	Result
Model is too simple.	It cannot learn the real pattern.
Performs poorly on training data.	It did not learn enough.
Performs poorly on new data.	It also fails outside training.

Overfitting vs Underfitting

Concept	Main problem	Performance
Overfitting	Learns too much detail/noise from training data.	Good on training, poor on new data.
Underfitting	Too simple to learn the real pattern.	Poor on training and poor on new data.

Gradient Descent: Learn the Parameters

Gradient descent is used to find the minimum of the model's loss function by repeatedly improving the parameter guesses.

The slide's analogy is walking through a valley blindfolded. You feel which direction slopes downward, take a step, and repeat until the ground is flat. The bottom of the valley is the minimum loss.

Gradient descent idea	Simple meaning
Start with guessed β_0 and β_1	Begin with estimated parameters.
Compute partial derivatives	See how changing each parameter affects loss.
Move opposite the direction that increases loss	Walk downhill.
Stop when loss is minimized	The algorithm has converged.

For β_1, if $dz/d\beta_1$ is negative, increasing β_1 reduces loss. If it is positive, decrease β_1. If it is zero, do not change β_1 because an optimum has been reached.

09_Traditional_ML_Supervised_learning_3_part_1_sem_1_2025_2026.pdf

TRADITIONAL ML - SUPERVISED LEARNING 3

This chapter covers non-parametric learners, k-nearest neighbors, decision trees, random forests, cross-validation, hyperparameter tuning, and ensemble models.

Supervised Learning: A Quick Recap

Supervised learning uses labeled training data. The goal is to predict an output label from input data.

Term	Meaning	Source example
Features X	Input data.	House size, email text.
Label Y	Output to predict.	House price, spam/not spam.

Without labels, the problem is not supervised learning.

Parametric Models and Non-Parametric Models

Feature	Parametric models	Non-parametric models
Function form	Assume a fixed form.	Do not assume a fixed form.
Example equation	$Y = mx + b$.	No predefined equation.
What they learn	Parameters.	Structure directly from data.
Examples	Linear regression, logistic regression.	Decision tree, k-nearest neighbors.
Strength	Simple, fast, easier to interpret.	More flexible, handles complex patterns.
Limitation	Less flexible.	More complex as more data is added.

What Are Non-Parametric Learners?

Non-parametric models do not assume a fixed structure for the function. They learn directly from data, which helps them adapt to complex and nonlinear patterns.

Advantages of Non-Parametric Learners

Advantage	Simple meaning
Handles complex and nonlinear relationships.	Can learn patterns that are not straight lines.
More flexible than parametric models.	Less tied to a fixed formula.
Useful when the true function is unknown or complex.	Helpful when one formula is not enough.

The source example is predicting housing prices using many factors such as size, location, rooms, school proximity, crime rate, and neighborhood appeal.

Why Use k-NN?

k-nearest neighbors predicts by looking at similar examples. For house prices, it can find similar houses and average their prices.

k-NN Recap

k-NN feature	Meaning
Type	Supervised, non-parametric.
Use cases	Classification and regression.

k-NN feature	Meaning
Classification prediction	Majority vote.
Regression prediction	Average.
Strength	No need to define a mathematical function in advance.
Requirement	Need to compute distance between data points.

k-Nearest Neighbors (k-NN)

To classify a mysterious green circle, measure its distance to all known points, select the k closest points, and use majority vote.

k value	Source example result
k = 3	If two nearest points are red and one blue, predict red.
k = 5	Label is based on majority vote among five nearest neighbors.

For continuous variables such as house prices, k-NN takes the average of the k closest values. For categories such as cat vs dog, it uses the mode/most common label.

How to Use k-NN to Predict Housing Prices

Step	What happens
1	Store training data X with features such as zip code, neighborhood, bedrooms, square feet, distance from public transport, and corresponding sale prices Y.
2	Sort training houses by similarity to the house being predicted.
3	Take the mean of the k closest houses as the sale price guess.

Euclidean Distance in k-NN

Euclidean distance measures straight-line distance between points. In 2D, the slide gives:

sqrt((x2 - x1)^2 + (y2 - y1)^2)

This distance helps decide which data points are nearest.

k-NN Example: Predict Label for (3,4) with k = 3

Point	Feature 1	Feature 2	Label
P	1	2	YES
Q	4	6	NO
R	5	7	NO
S	2	3	YES

Point	Distance to (3,4)	Label	Rank
S	1.41	YES	1
Q	2.24	NO	2
P	2.83	YES	3
R	3.61	NO	4

k nearest points	Labels
S, Q, P	YES, NO, YES

Final prediction: YES, because YES appears twice and NO appears once.

Choosing k: Tuning Hyperparameters by Cross-Validation

A hyperparameter is a setting chosen before training. It is not learned from the data.

Hyperparameter example	Model
k	k-NN
max_depth	Decision trees
learning_rate	Neural networks

Weights in regression or splits in trees are not hyperparameters because they are learned during training.

Cross-Validation

Step	Meaning
Split data into folds	Example: 5 groups.
Train/test each fold	Train on 4 folds, test on 1.
Repeat	Each fold becomes the test set once.
Try different k values	Example: 1, 3, 5, 7.
Measure performance	Use accuracy or mean squared error.
Average results	Gives a more reliable evaluation.

Higher k Prevents Overfitting

k size	Behavior	Risk
Small k, such as k = 1	Focuses only on nearest neighbor.	Can overfit to random noise.
Larger k	Considers more neighbors.	More general.
Too large k, such as k = N	Looks at all training points.	Too simple and may predict only the majority class.

The slide's animal example: if 90 out of 100 animals are cats and 10 are dogs, k = N = 100 may predict "cat" even for a dog-like test point.

Where to Use k-NN in the Real World

Use case	Why k-NN fits
Fraud detection	Can update quickly with new training examples.
Housing price prediction	Nearby houses can be similar in price.
Missing data imputation	Mean or mode of nearby points can fill missing values.

Random Forest and Decision Tree

Random Forest: Many Trees, One Strong Forest

Random forest is an ensemble of decision trees. Each tree trains on a random data subset. Each split considers a random subset of features. Classification uses majority voting, while regression uses averaging.

Random Forest Benefits and Trade-Off

Benefit / trade-off	Meaning
Reduces overfitting	Many trees help avoid depending too much on one tree.
Robust with noisy or large datasets	Handles difficult data better.
Helps identify important features	Shows which inputs matter.

Benefit / trade-off	Meaning
Less interpretable	Harder to explain than one tree.
Slower to train	Takes more time/resources.

Decision Tree vs Random Forest

Feature	Decision Tree	Random Forest
Number of models	Single model.	Ensemble of trees.
Interpretability	Easy to interpret.	Harder to interpret.
Overfitting	Prone to overfitting.	More accurate and less overfitting.
Speed/resources	Simpler.	Slower and more resource-intensive.

When to Use Decision Tree vs Random Forest

Situation	Use Decision Tree	Use Random Forest
Need simple model	Yes.	Less suitable.
Need clear interpretation	Yes.	Less suitable.
Can accept some overfitting risk	Yes.	Less necessary.
Want better accuracy and robustness	Less suitable.	Yes.
Large or noisy dataset	Less suitable.	Yes.
Interpretability less critical	Less suitable.	Yes.

10 Unsupervised learning.pdf

UNSUPERVISED LEARNING

This chapter covers clustering, dimensionality reduction, k-means clustering, and principal component analysis.

Introduction - Why do we need Unsupervised Learning?

Unsupervised learning helps when we do not have labels or predefined categories. It finds hidden patterns, groups similar data, and reduces complexity.

Main Unsupervised Learning Tasks

Task	Simple meaning
Clustering	Group data by similarity.
Dimensionality reduction	Compress data while keeping useful structure.

Examples

Example from slides	How unsupervised learning helps
Advertising platform	Segments people into similar groups for relevant ads.
Airbnb	Groups housing listings into neighborhoods.
Data science team	Reduces dimensions to simplify modeling and reduce file size.

Unlike supervised learning, evaluation is harder because there are no predefined labels. The quality depends on whether the patterns are meaningful for the task.

Clustering

Clustering groups similar data points together. The source example is Acxiom's Personicx system, which categorizes U.S. households into 70 clusters within 21 life-stage groups.

Personicx cluster idea	Source example
Starting Out	Young people beginning careers or education.
Top Wealth	Wealthy, established professionals.

Advertisers can use clusters to target ads, such as gym memberships to active groups or luxury products to top wealth groups.

K-Means Clustering

K-means divides data into k clusters.

k value	Effect
Larger k	Smaller, more detailed groups.
Smaller k	Broader groups.

Student performance example:

k setting	Possible grouping
Larger k, such as 10 groups	Small categories like 90-92%, 93-95%.
Smaller k, such as 2 groups	High Performers and Low Performers.

A centroid is the central point of a cluster, representing the average position of points in that group. Points closest to a centroid are assigned to that centroid's cluster.

K-Means Clustering Steps

Step	What happens
1. Initialize centroids	Choose k centroids at random.
2. Assign data points	Assign each point to the nearest centroid using distance.
3. Update centroids	Recalculate each centroid as the average position of its cluster points.
4. Repeat until stable	Repeat assigning and updating until centroids no longer change much.
5. Stable clusters	The clusters become well-defined and stable.

The slides also say k-means can group handwritten digit images by finding patterns in pixel values.

Dimensionality Reduction

Dimensionality reduction is like compression. It reduces data complexity while keeping as much important structure as possible.

Problem	Simple meaning
Too many features	Example: 128 x 128 image has 16,384 pixels/dimensions.
Hard to analyze or visualize	Too much information makes work difficult.
Noise/repetition	Some features may not add useful information.

Principal Component Analysis (PCA)

PCA is presented as a common dimensionality reduction technique.

Source analogies:

Analogy	PCA meaning
Library with thousands of books	PCA groups similar books into categories, so we manage fewer categories instead of many books.
Weather factors	PCA finds a simpler set of factors when variables such as temperature, humidity, sunlight, and wind overlap.

The supplementary slide also introduces spaces and bases, showing that points can be described using a different basis while still making the math work.

11_Neural_Networks_and_Deep_Learning_may_2026_sem_2_2025_2026.pdf

NEURAL NETWORKS AND DEEP LEARNING

This chapter explains deep neural networks, image classification, brain inspiration, neurons, feature learning, layers of abstraction, software packages, CNNs, RNNs, and applications.

Content

Source page(s)	Topic covered
1-4	Deep learning and the function f
5-6	Image classification and gradient descent training
7-8	Deep learning history and enabling factors
9-14	Biological neurons and artificial neurons
15-19, 29	Feature learning and layers of abstraction
20-23	Interpretability, software packages, CNNs, RNNs
24-28	Deep learning applications
25	DNN vs CNN vs RNN

Deep Learning

In machine learning, f is the function the model tries to learn. It maps an input to an output.

Input	Output	Meaning of f
Image	Label or class	Maps pixels to the correct image class.
Sentence	Sentiment or translation	Maps text to meaning or translated text.
Game state	Best move	Maps game situation to action.

The real world is messy, so f can be complicated. The slides mention natural language, vision problems, and games as difficult cases.

!!! EXAM TOPIC: Neural Networks !!!

A neural network is a computational model inspired by biological neurons. It is used to recognize patterns, classify data, and make predictions.

Image Classification

The source example is a deep neural network classifying an image such as a lion.

Network part	What it does in simple language
Input layer	Receives raw image data. Each pixel is treated as a feature.
Hidden layer 1	Detects basic features such as edges or lines.
Hidden layer 2	Combines basic features into shapes or patterns.
Hidden layer 3	Finds complex parts such as ears, nose, or mane.
Output layer	Gives class probabilities, such as Lion 90%, Dog 5%, Cat 3%.

How Hidden Layers Learn

Operation named in slide	Simple meaning
Matrix multiplication	Combines previous layer outputs with weights.
Activation functions	Add non-linearity so the network can learn complex patterns.
Weights and biases	Values adjusted during training to reduce errors.

Training a Deep Neural Network with Gradient Descent

Step	Simple meaning
Start with random weights.	The model begins with rough guesses.
Predict output.	Example: it guesses "Lion."
Calculate loss.	Compare prediction with the correct label.
Adjust weights.	Use gradient descent to reduce future error.
Repeat many times.	With many examples, the model learns patterns.

Gradient descent is described as the backbone of learning in neural networks.

Where Deep Learning Does Well, and Some History

Period / factor	Source detail
1940s-1960s	Neural networks historically referred to as cybernetics.
1980s-1990s	Referred to as connectionism.
Around 2006	Neural networks became deeper and deep learning came into vogue.
Compute	Moore's Law, GPUs, ASICs.
Data	Data in a nice form, such as ImageNet.
Algorithms	Ideas such as backprop, CNN, LSTM.
Infrastructure	Linux, TCP/IP, Git, ROS, PR2, AWS, AMT, TensorFlow.

Neuron

The slides say the human brain has around 10 billion neurons, each connected on average to 10,000 other neurons. Neurons receive signals through synapses. When sufficiently activated, neurons "fire" by sending electrical signals to other neurons.

Biological Neuron vs Artificial Neural Network Idea

Biological part	Simple role	AI connection in slides
Dendrites	Receive signals.	Inputs.
Soma / cell body	Processes information.	Processing in artificial neuron.
Axon	Carries output.	Output.
Synapses	Points of connection between neurons.	Modeled by adjustable weights.

Neural Network Structure

Layer / component	Simple meaning
Input layer	Receives data.
Hidden layers	Extract features and patterns.
Output layer	Produces final result.
Inputs x	Data values entering the neuron.

Layer / component	Simple meaning
Weights w	Importance values for inputs.
Bias b	Extra adjustable value.
Activation function	Decides the neuron output.

Single Neuron Calculation

Source values:

Item	Values
Inputs	$x1=2$, $x2=1$, $x3=3$, $x4=1$
Weights	$w1=0.3$, $w2=0.1$, $w3=0.4$, $w4=0.2$
Bias	$b=-1.2$

Calculation from the food freshness example:

Step	Working
Weighted sum	$z = (0.3 \times 2) + (0.1 \times 1) + (0.4 \times 3) + (0.2 \times 1) - 1.2$
Value	$z = 0.6 + 0.1 + 1.2 + 0.2 - 1.2 = 0.9$
Activation rule	If $z >= 0$, output = 1. If $z < 0$, output = 0.
Result	Since $z = 0.9$, output = 1.
Food freshness status	Fresh.

Neurons, Feature Learning, and Layers of Abstraction

Feature learning means the network learns useful patterns layer by layer. In the visual example, early layers detect basic edges, middle layers detect parts such as eyes/nose/mouth, and higher layers detect a complete face.

Level	What is detected
Low-level features	Edges and basic lines.
Mid-level features	Parts such as eyes and noses.
High-level features	Whole face or complete object.

Why Linear Models Do Not Work

The slides explain that simple template-like models average images in each class. This creates blurry templates and loses details such as orientation, color, and position.

Linear/template problem	Simple meaning
No abstraction	Cannot understand different shapes or angles.
Blurry details	Important features disappear when images are averaged.
Not flexible	Cannot handle small differences, such as left vs right.

Deep neural networks are better here because they learn layers of features instead of averaging everything into one blurry template.

DNN

A deep neural network processes data hierarchically:

Layer stage	What happens
Input layer	Raw pixel data enters.

Layer stage	What happens
Early hidden layers	Detect simple features such as diagonal lines or edges.
Middle hidden layers	Combine features into object parts, such as face-like shapes.
Deeper layers	Assemble parts into complete objects.
Output layer	Gives final classification such as face or cat.

Interpretability in DNNs

Point	Simple meaning
Interpretability	Understanding how a model makes decisions.
DNN challenge	DNNs work well but can be hard to explain because they learn complex patterns through many layers.
Why it matters	In critical areas such as healthcare, people need to know why a decision was made.
Source conclusion	DNNs are powerful but not always interpretable; visualization tools and explainable AI help make decisions clearer.

Deep Learning Software Packages

The source lists TensorFlow, Torch, PyTorch, Caffe, Theano, and more. The point is that people rarely need to implement every part of neural networks from scratch.

Convolutional Neural Networks (CNNs)

CNNs are designed for image input and are effective for computer vision. They are also used in deep reinforcement learning and mimic how biological visual systems process visual input.

Recurrent Neural Networks (RNNs)

RNNs have built-in memory and suit language problems. They are also important in reinforcement learning because they help an agent keep track of what happened before, even when not everything is visible at once.

DNN VS CNN VS RNN

Model	Main idea	Best for	Limitation / special note
Deep Neural Network (DNN)	General-purpose neural network with many hidden layers and fully connected layers.	Tabular data, basic classification, regression.	Not good at image patterns or time sequences.
Convolutional Neural Network (CNN)	Specialized for images and spatial data using convolution layers.	Image classification, facial recognition, object detection.	Captures spatial features in pictures.
Recurrent Neural Network (RNN)	Specialized for sequential data and uses memory/loops.	Time series, language, speech, music.	Good for patterns over time, such as predicting next word.

Deep Learning Applications

Application	Source example
Drug discovery	Predicting molecule bioactivity.
Photo/video tagging	Face and object recognition.
Search	Powering Google search results.
Language	Google Translate and natural language generation/understanding.
Robotics	Mars rover Curiosity selecting inspection-worthy soil targets.
Question answering	Facebook neural network with short-term memory answering Lord of the Rings plot questions.
Self-driving cars	Road signs, lanes, obstacles.
Art generation	Neural style mimics an artist's style and remixes another image.

chapt 12 Generative AI-sem 1 2024 2025.pdf

GENERATIVE AI

This chapter covers generative AI, large language models, natural language processing, business applications, responsible AI, and discriminative versus generative models.

Introduction to Generative AI

Generative AI is a subset of AI focused on creating new content such as text, images, and music. It enables creativity and automation.

Capability	Source wording
Generates text	Creates written content.
Generates images	Creates visual content.
Generates code	Creates programming content.
Enhances efficiency	Helps different industries work faster.
Powered by LLMs	Uses advanced models to understand natural language.

Large Language Models (LLMs)

LLMs are advanced AI models trained on large text datasets. They use transformers for deep learning.

LLM feature	Simple meaning
Language mastery	Learns grammar, patterns, and meaning.
Applications	Chatbots, content tools, virtual assistants.
Continual evolution	Research keeps expanding capabilities.

Natural Language Processing (NLP)

NLP is AI technology for understanding and generating human language. It goes beyond recognizing words by analyzing meaning and context.

NLP application	Source example
Customer service	Chatbots.
Social media	Sentiment analysis.
Reporting	Automated report generation.

Business Applications of Generative AI and LLMs

Business area	Applications from source
Marketing and sales	Personalized content creation, lead generation automation.
Customer service	24/7 chatbots and email responses.
Operations	Automates reporting and optimizes processes.
HR	Candidate screening and onboarding.
R&D	Idea generation and data synthesis.

Ethical Considerations and Responsible AI

Concern	Simple meaning
Bias mitigation	Reduce potential bias in AI outputs.

Concern	Simple meaning
Transparency	Be open about how AI is used and what data it was trained on.
Accountability	Set clear responsibility for AI-generated content.
Human-in-the-loop	Keep human review and oversight for quality and accuracy.

Discriminative versus Generative

Discriminative models	Generative models
Focus on classification.	Create new content.
Example: identify objects in a picture.	Example: generate a non-existing face.
Question: "Who is this in the picture?"	Request: "Generate a new face."

Generative AI Overview

Generative AI is trained on patterns and structures in input data, then generates outputs based on those patterns. The slide mentions ChatGPT, Bing Chat, Bard, DALL-E, and Midjourney. It also names transformer architecture as the backbone of modern natural language processing models, including BERT and GPT.

Resources

Resource listed in slides
Introduction to Generative AI from Google Cloud Skills Boost
Generative AI for Everyone from deeplearning.ai on Coursera
Introduction to Generative AI from Google Cloud Tech channel on YouTube
Introduction to large language models from Google Cloud Tech on YouTube

Chap13-Other AI Approaches.pdf

CHAPTER 13 - Other AI approaches - Fuzzy Logic

This chapter focuses on uncertainty and fuzzy logic.

Content

Source page(s)	Topic covered
1-4	Introduction and uncertainty
5-7	Fuzzy logic definition and vague language examples
8	History
9-13	Crisp logic versus fuzzy logic and temperature examples
14-15	Human-like reasoning and fuzzy logic abilities
16-19	Fuzzy rules and linguistic variables
20-24	Air-conditioner, shower, and washing machine examples
25-27	Applications in consumer products, medicine, and information systems

Introduction

The world is uncertain and not always clear. The slides describe uncertainty as lack of exact knowledge that would let us reach a perfectly reliable conclusion.

Forms of Uncertainty

Form	Source example	Simple meaning
Imprecise measurements	Exact temperature may vary by tool or condition.	Measurements may not match perfectly.
Imprecise definitions	Whether a leader is "good" is subjective.	Words can depend on opinion.
Imprecise knowledge	Where is the pit?	We may not know exact facts.
Uncertain inferences	Rash after gardening probably means poison ivy.	A conclusion can be likely but not guaranteed.

Information may be imperfect because it is inconsistent, incomplete, unsure, or all three.

!!! EXAM TOPIC: Fuzzy Logic !!!

Fuzzy logic was introduced by Lotfi Zadeh in 1965. It is a problem-solving method that uses human-language rules and handles uncertainty, vague criteria, and unclear values.

The key idea: not everything is only yes/no or true/false. Some things have degrees. For example, "cold" depends on what someone means by cold. A temperature might feel cold to one person but not to another.

Fuzzy Logic Core Idea

Idea	Simple explanation
Fuzziness	Something is not clear enough.
Fuzzy set	A set with fuzzy boundaries.
Degrees/scales	Things can be partly true, partly matching, or partly belonging.
Common sense	Experts often solve problems using approximate human reasoning.

Why Fuzzy Logic is Needed

Some questions have exact yes/no answers, such as "Do you need this book?" But many natural-language statements are vague:

Vague statement from slides	Why it is fuzzy
He is quite tall.	"Quite tall" has no single exact boundary.
The student is intelligent.	Intelligence can be judged by degree.
Today is a very hot day.	"Very hot" depends on a scale and context.

Fuzzy logic helps computers handle these human-like vague terms.

History

Year	Source milestone
1965	Fuzzy Sets, Lotfi Zadeh seminar
1966	Fuzzy Logic, P. Marinos, Bell Labs
1972	Fuzzy Measure, M. Sugeno, TIT
1974	Fuzzy Logic Control, E.H. Mamdani
1980	Control of Cement Kiln, F.L. Smidt, Denmark
1987	Sendai Subway Train Experiment, Hitachi
1988	Stock Trading Expert System, Yamaichi
1989	LIFE, Lab for International Fuzzy Engineering

Differences between Fuzzy Logic and Crisp Logic

Crisp Logic	Fuzzy Logic
Precise properties.	Imprecise properties.
Full membership.	Partial membership.
YES or NO.	YES moving gradually toward NO.
TRUE or FALSE.	TRUE moving gradually toward FALSE.
1 or 0.	1 moving gradually toward 0.
Crisp sets.	Fuzzy sets.
"She is 18 years old."	"She is about 18 years old."
"Man 1.6m tall."	"Man about 1.6m tall."

Degree of Membership: "Tall" Man Example

In crisp logic, a numeric height returns only yes or no. In fuzzy logic, a height can belong to the idea of "tall" by a degree.

Height (cm)	Crisp value	Fuzzy value
208	1	1.00
205	1	1.00
198	1	0.98
181	1	0.82
179	0	0.78
172	0	0.24
167	0	0.15
158	0	0.06
155	0	0.01
152	0	0.00

Simple exam reading: crisp logic suddenly changes from 1 to 0 at a boundary. Fuzzy logic changes gradually. A person at 179 cm can still be somewhat tall, even if crisp logic says 0.

Boolean Logic vs Fuzzy Logic for Temperature

Boolean / crisp temperature	Fuzzy temperature
Uses two values.	Uses continuous values.
Temperature is either Hot or Cold.	Temperature can be Extremely Cold, Cold, Quite Cold, Quite Hot, Hot, Extremely Hot.
Sharp boundary.	Smooth shift between meanings.

How Fuzzy Logic Resembles Human Intelligence

Human-like ability	Simple meaning
Handles imprecision and uncertainty.	Can work when facts are not perfectly clear.
Clustering and classification.	Divides situations into parts.
Ranking importance and alternatives.	Focuses on parts with different levels of importance.
Combining parts into a whole.	Builds an integrated decision.

Fuzzy Logic is Able To

Ability / benefit	Simple meaning
Represent vague language naturally.	Handles words like cold, hot, tall, medium.
Enrich, not replace, crisp sets.	Adds gradual meaning instead of removing exact logic.
Allow flexible engineering design.	Helps design systems that handle real-world variation.
Improve model performance.	Source examples: save power consumption, increase lifespan.
Implement simply and often work.	Can be practical and effective.

Fuzzy Rule

A fuzzy rule has the form:

IF part	THEN part
x is A	y is B

In the slides, x and y are linguistic variables, while A and B are linguistic values determined by fuzzy sets.

Linguistic Variable

A linguistic variable is a variable whose values are words or sentences in natural or artificial language.

Linguistic variable	Possible linguistic values from source
Speed	slow, fast, very fast
Driving_speed	slow, medium, fast
Stop_distance	short, medium, long

It is a mathematical way to represent semantic concepts with several terms and degrees of membership.

Fuzzy Rules: Driving Example

Rule	Meaning
IF driving_speed is fast THEN stop_distance is long	Faster driving needs longer stopping distance.

Rule	Meaning
IF driving_speed is slow THEN stop_distance is short	Slower driving needs shorter stopping distance.

The source says driving_speed can range from 0 to 220 km/h and include fuzzy sets such as slow, medium, and fast. Stop_distance can range from 0 to 300m and include short, medium, and long.

Fuzzy Rules: Project and Tip Examples

Conditions	Result
project_duration short AND project_staffing medium AND project_funding inadequate	risk high
project_duration long AND project_staffing large AND project_funding adequate	risk low
project_duration short AND project_staffing large AND project_funding adequate	risk medium
service excellent OR food delicious	tip generous

Air-Conditioner Example

Problem: set a room temperature so it is not too hot or too cold, including when there are many or very few students in the room.

Temperature condition	Fan speed action
IF temperature is cold	set fan_speed to zero
IF temperature is cool	set fan_speed to low
IF temperature is warm	set fan_speed to medium
IF temperature is hot	set fan_speed to high

The membership function has fuzzy temperature sets: Cold, Cool, Warm, Hot. It expresses temperature changes smoothly and naturally.

Bathroom Shower Fuzzy Rules

Water volume	Temperature setting
Full	Hot
Half	Warm
Quarter	Cold

Washing Machine Fuzzy Rules

Load weight	Water amount
Heavy	Full / maximum
Not_so_heavy	Three_quarter
Not_so_light	Half
Light	Quarter / minimum
Medium	Regular

Fuzzy Logic Applications

Fuzzy logic decision making uses fuzzy set operations, if-then-else statements, and logical operators. It resembles human decision making because it works from approximate data and can find precise solutions.

Consumer product / area	Source examples
Home devices	Electrical shower unit, air conditioner, washing machines, refrigerators, television, rice cooker

Consumer product / area	Source examples
Vehicles	Brake control
Medicine	Diagnosis-related decision making
Information systems	Information retrieval and database management

Fuzzy Decision Making in Medicine

Medical diagnosis aspect	Simple meaning
Relative importance of symptoms	Some symptoms matter more than others.
Different disease stages	Symptoms may vary by stage.
Relations between diseases	Diseases can relate to each other.
Hypothesis formation	Forming possible explanations.
Preliminary diagnosis	Early diagnosis stage.
Final diagnosis	Final decision in diagnosis process.

Fuzzy Decision Making in Information System

Benefit	Simple meaning
Soft requests	Allows searches that are not perfectly exact.
Ordering results	Items can more or less satisfy a request.
Imprecise database information	Can handle vague, uncertain, or imprecise information.

AI Subtopic Diagram Companion

Source rule: this companion uses only the concepts already present in the supplied PDFs and the notes created from them. The diagrams are simplified visual versions of those source concepts.

How to use this file: study the short analysis first, then use the diagram to remember the flow, relationship, or structure.

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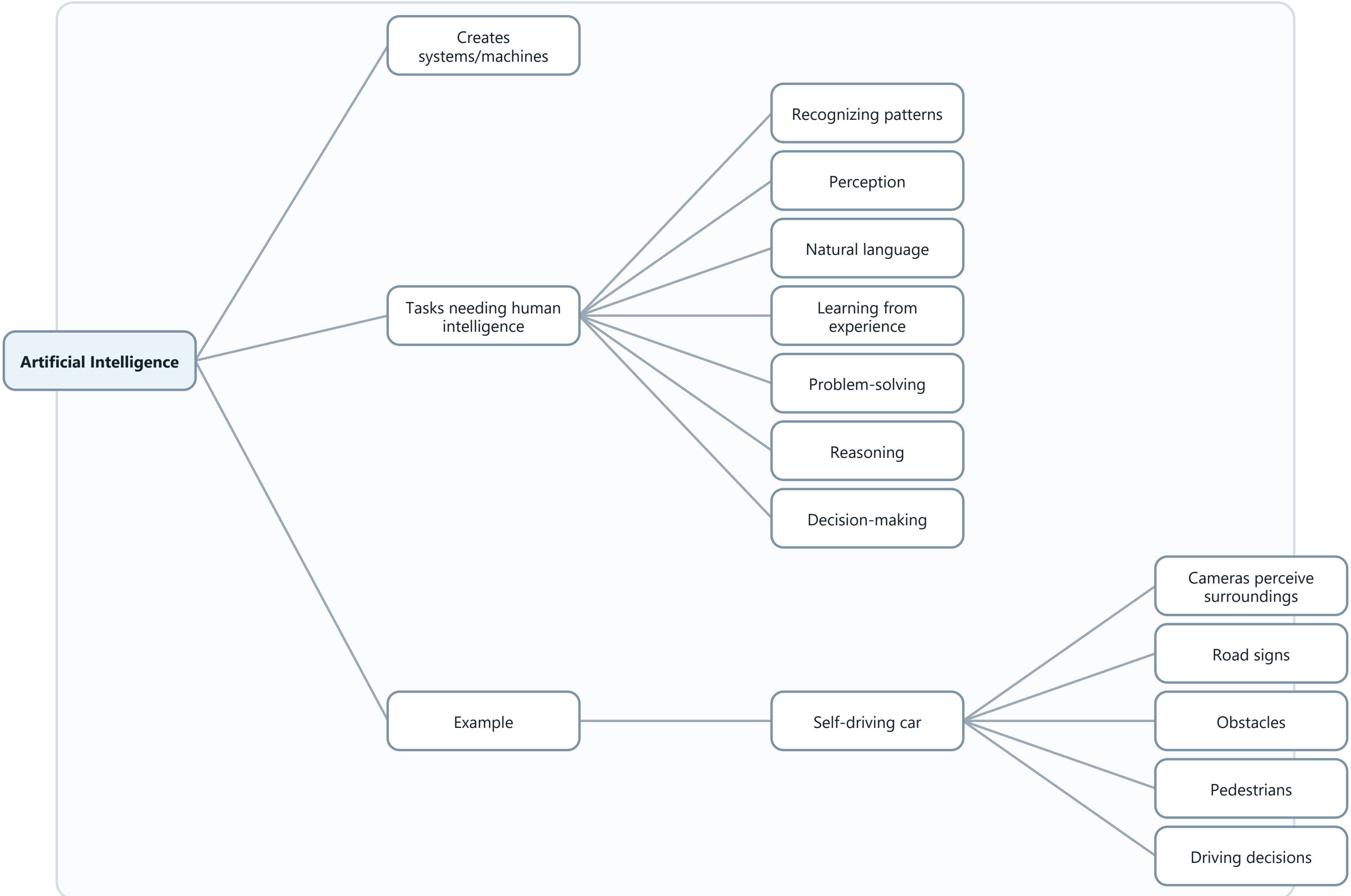
01 Introduction-rrm2024.pdf

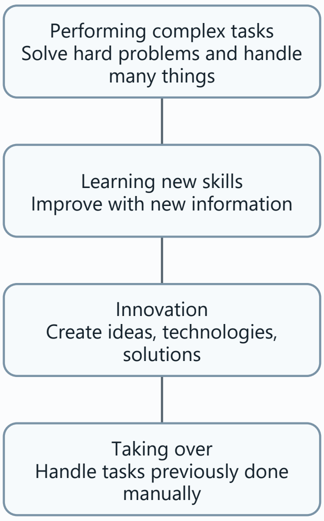
What is Artificial Intelligence?

Diagram needed: Yes. AI is introduced as a broad field with many human-like tasks, so a concept map helps connect the definition to examples.



Intelligence

Diagram needed: Yes. The four levels are easier to remember as a ladder.

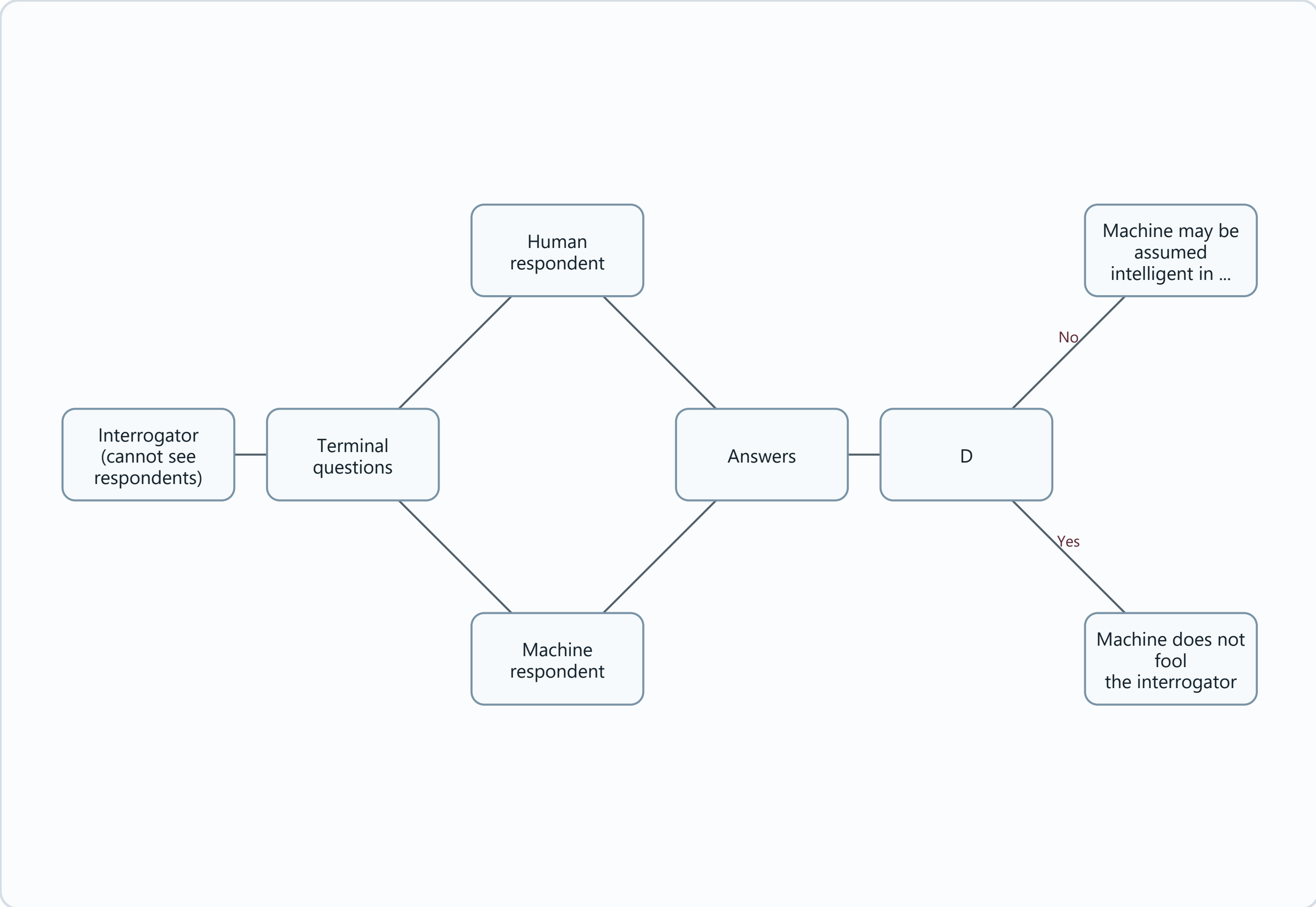


Historical Context

Diagram needed: Yes. AI history is chronological, so a timeline is useful.

Turing Test

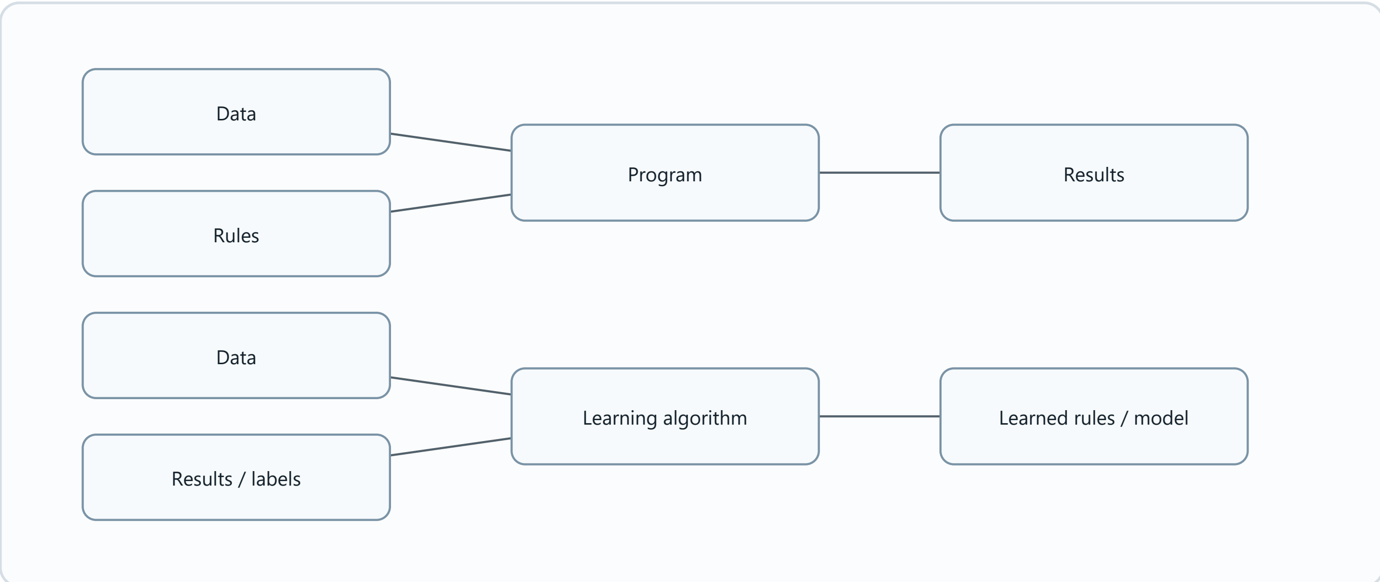
Diagram needed: Yes. The test is a communication setup.



Machine Learning

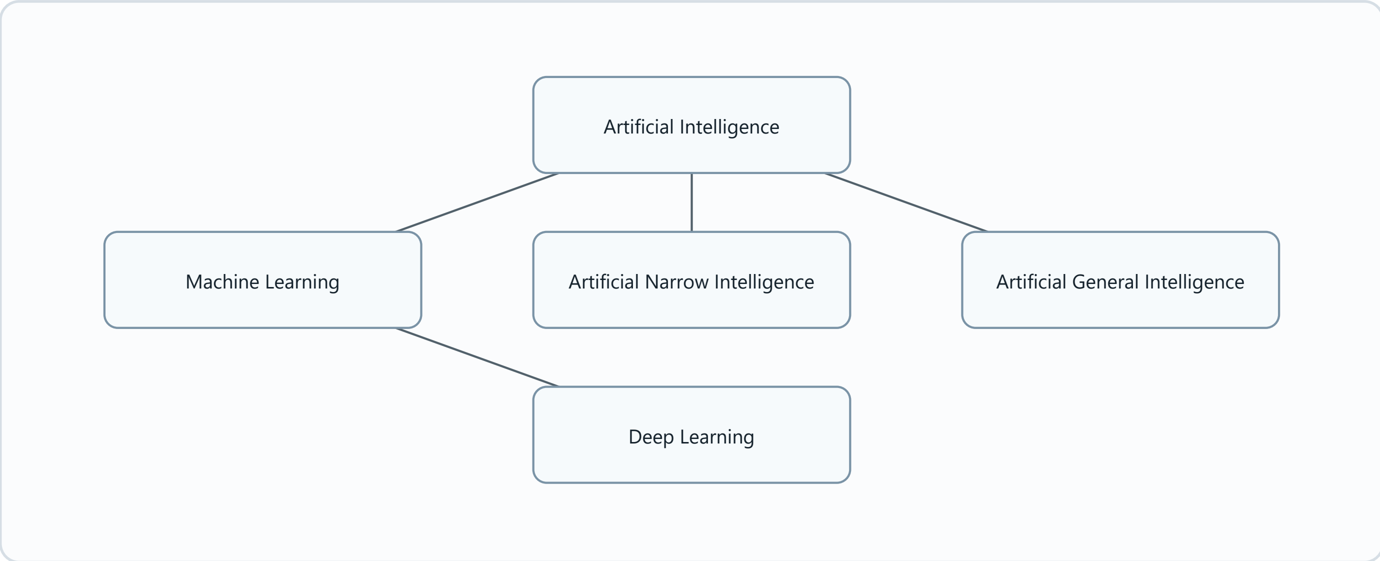
Diagram needed: Yes. The difference between traditional programming and ML is a core comparison.

Traditional programming	Machine learning
Human gives data and rules.	Human gives data and results.
Computer produces results.	Computer learns rules.



Types of AI, Machine Learning, and Deep Learning

Diagram needed: Yes. The slides show AI, ML, and deep learning as nested ideas.



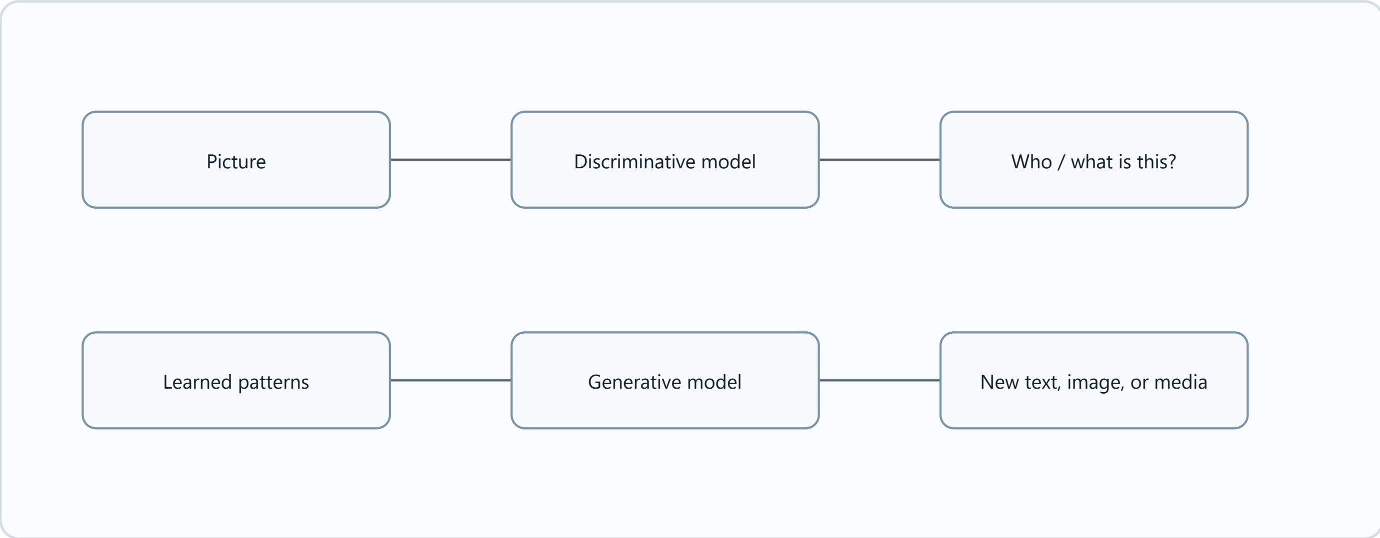
Deep Learning

Diagram needed: Yes. Multiple hidden layers are the main idea.

Discriminative versus Generative

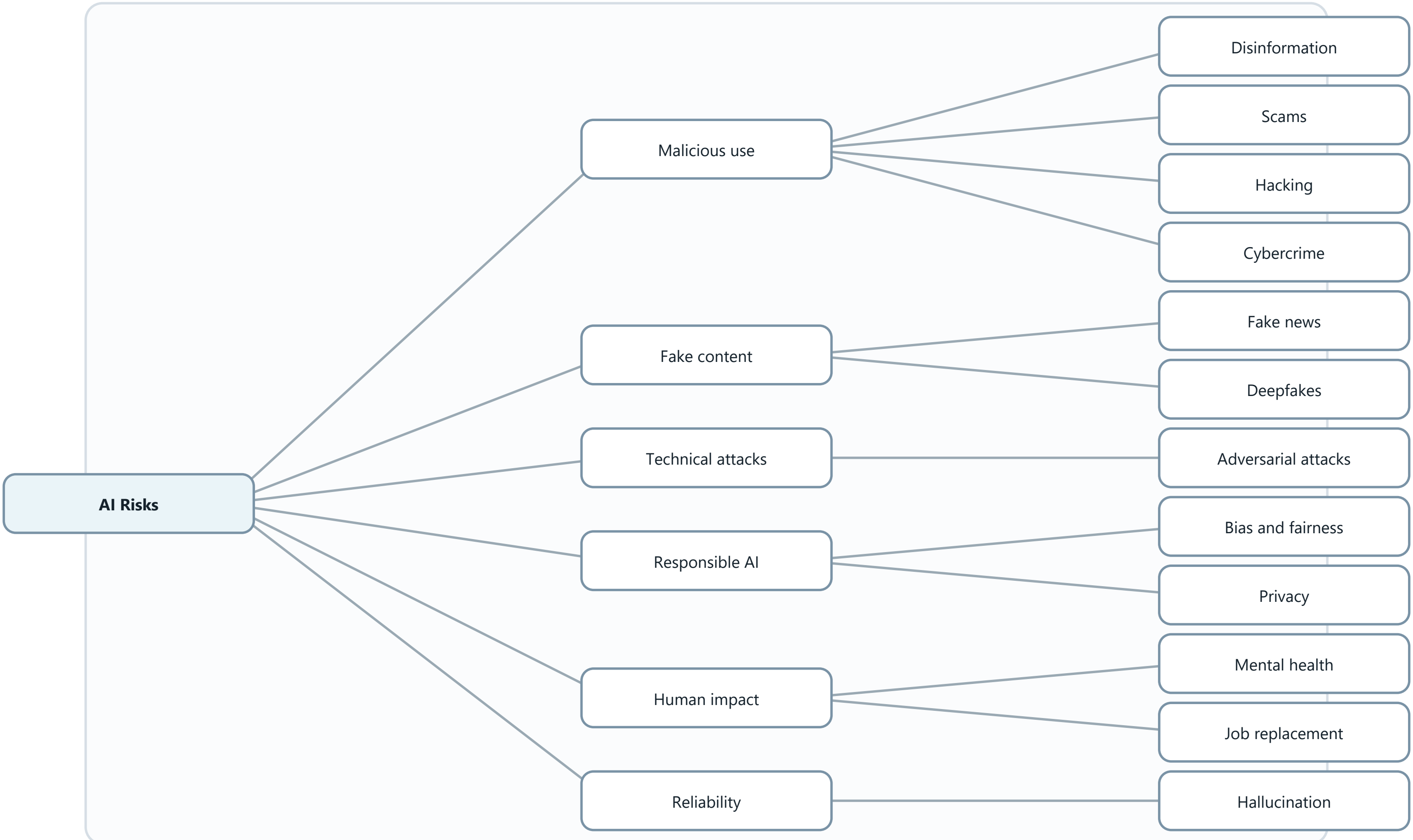
Diagram needed: Yes. The contrast is exam-friendly as a side-by-side flow.

Discriminative	Generative
Recognizes or classifies.	Creates new content.
Example: given a picture, tell who it is.	Example: create a non-existing person's face.



Risks of AI

Diagram needed: Yes. Risks are many, so grouping helps revision.

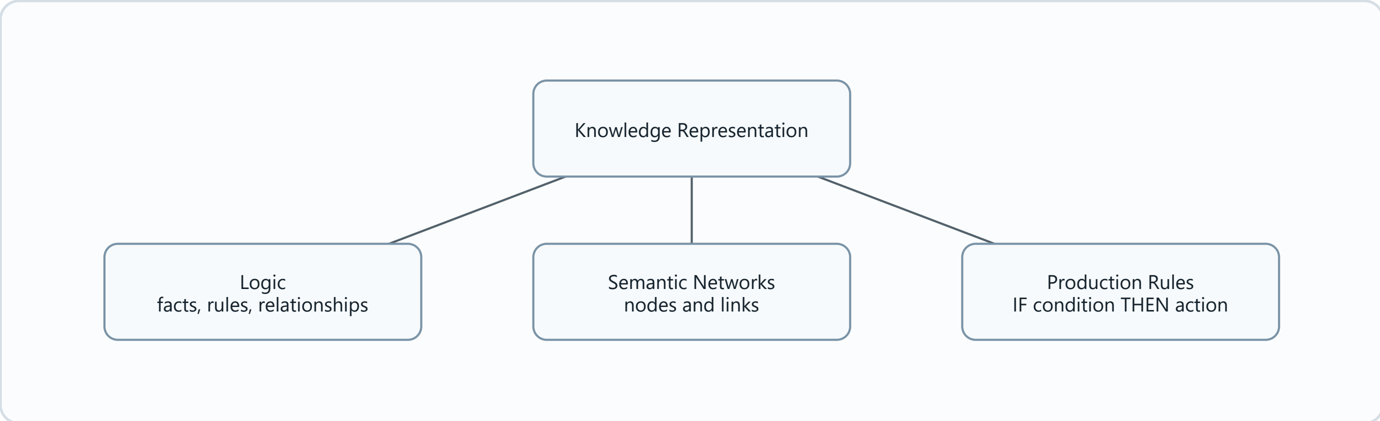


02 Logic and inference- sem 2 202526 (2).pdf

Knowledge Representation

Diagram needed: Yes. The chapter compares logic, semantic networks, and production rules.

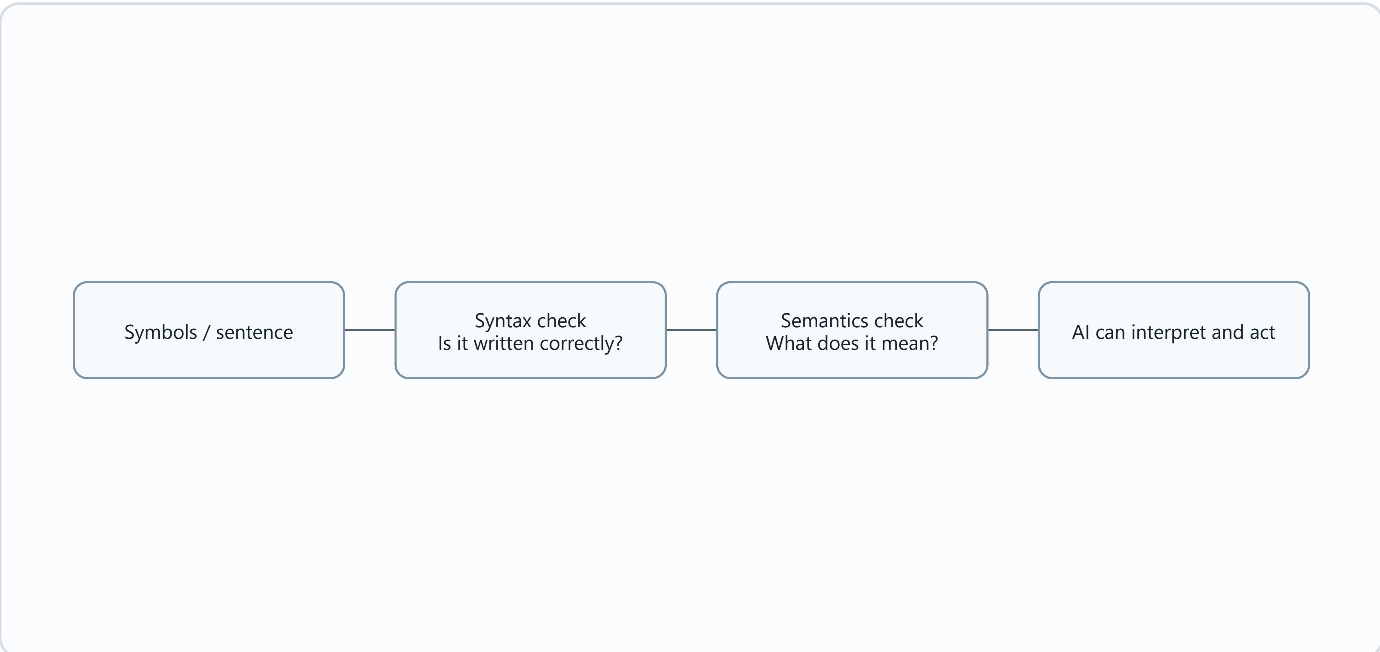
Method	Best visual memory
Logic	Formal symbols and truth.
Semantic networks	Nodes and links.
Production rules	IF condition THEN action.



Syntax and Semantics

Diagram needed: Yes. Students often confuse the two.

Syntax	Semantics
How symbols are legally arranged.	What the symbols mean.

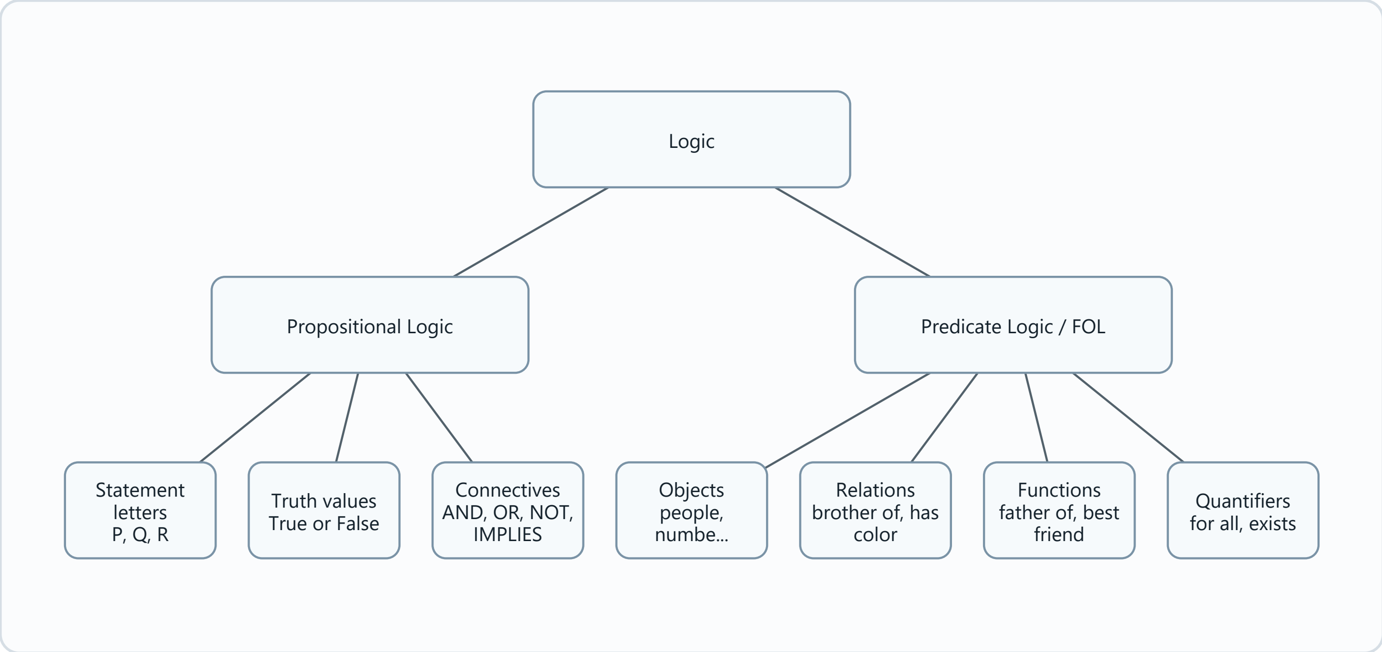


!!! EXAM TOPIC: Predicate and Propositional Logic !!!

Diagram needed: Yes. This is a critical exam comparison.

Propositional logic	Predicate logic / FOL
Whole statements are true or false.	Talks about objects, properties, relationships, and quantities.

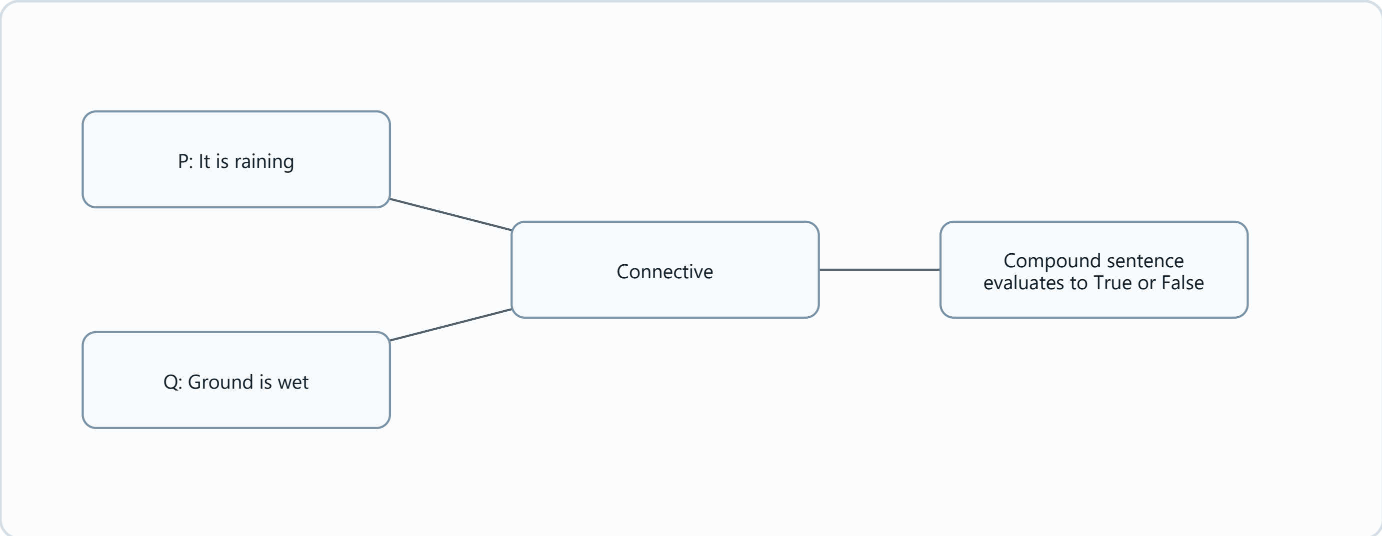
Propositional logic	Predicate logic / FOL
Uses P, Q, R as statements.	Uses predicates such as Brothers(Ravi, Ajay).



Logical Connectives and Truth Values

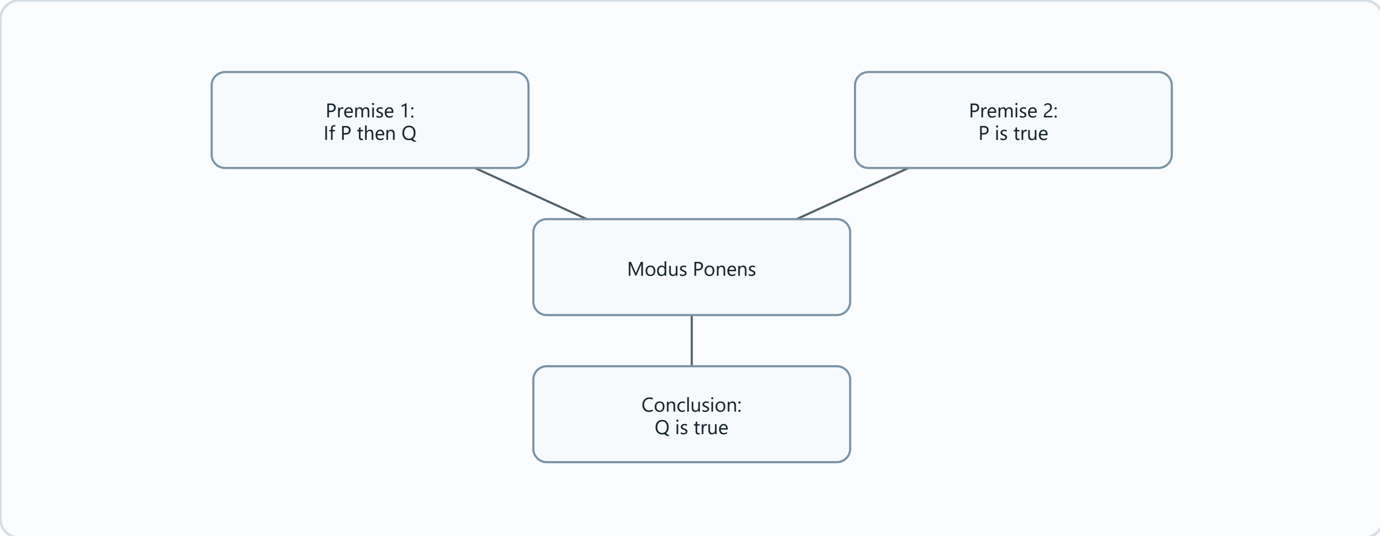
Diagram needed: Yes. Connectives are easier with a compact truth table.

P	Q	P AND Q	P OR Q	P → Q
True	True	True	True	True
True	False	False	True	False
False	True	False	True	True
False	False	False	False	True



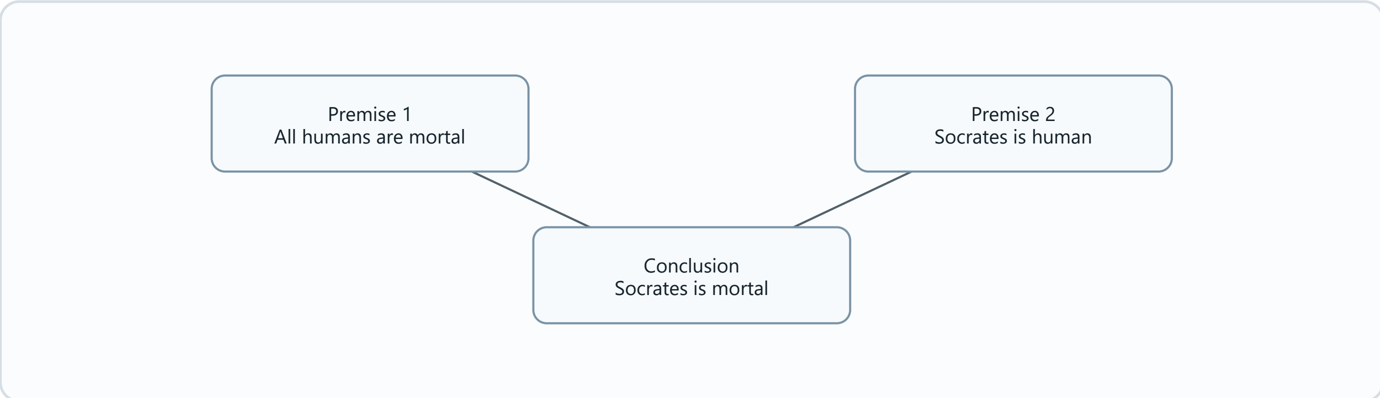
Modus Ponens

Diagram needed: Yes. It is a rule with two premises and one conclusion.



Arguments, Premises, and Conclusions

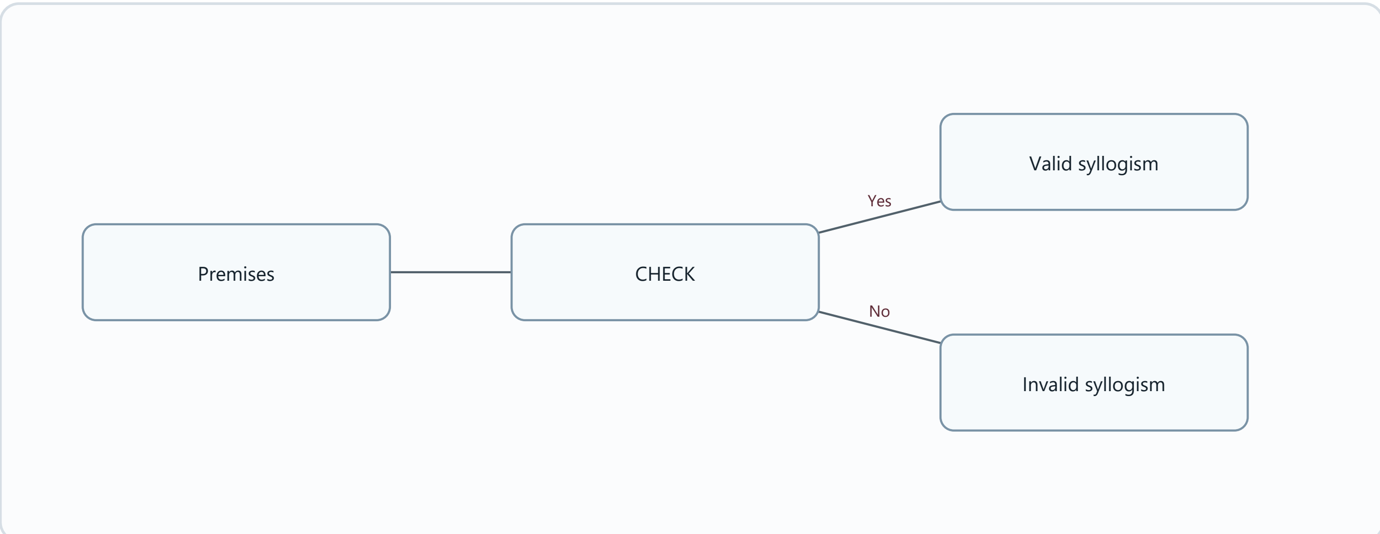
Diagram needed: Yes. The logic flow is simple but important.



Syllogism and Deductive Reasoning

Diagram needed: Yes. Valid and invalid syllogisms are a comparison.

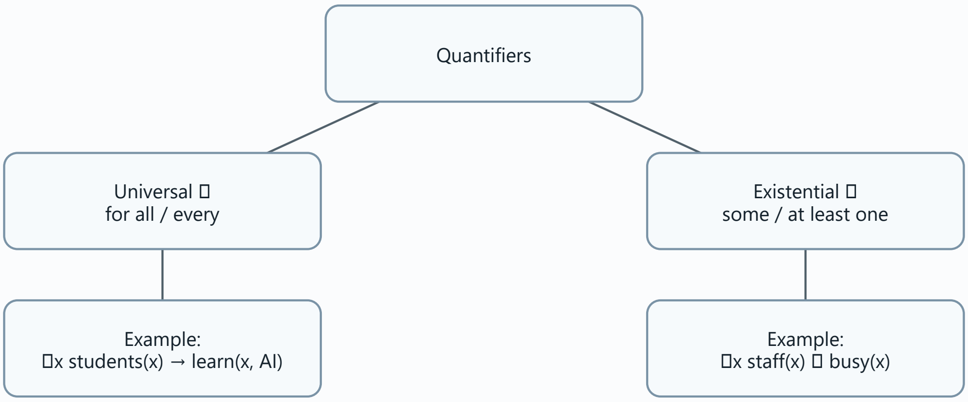
Valid syllogism	Invalid syllogism
Conclusion follows from the premises.	Conclusion does not follow from the premises.



Quantifiers in First-Order Logic

Diagram needed: Yes. The difference between "all" and "some" is central.

Quantifier	Symbol	Main connective from slides
Universal	□	Implication →
Existential	□	AND □



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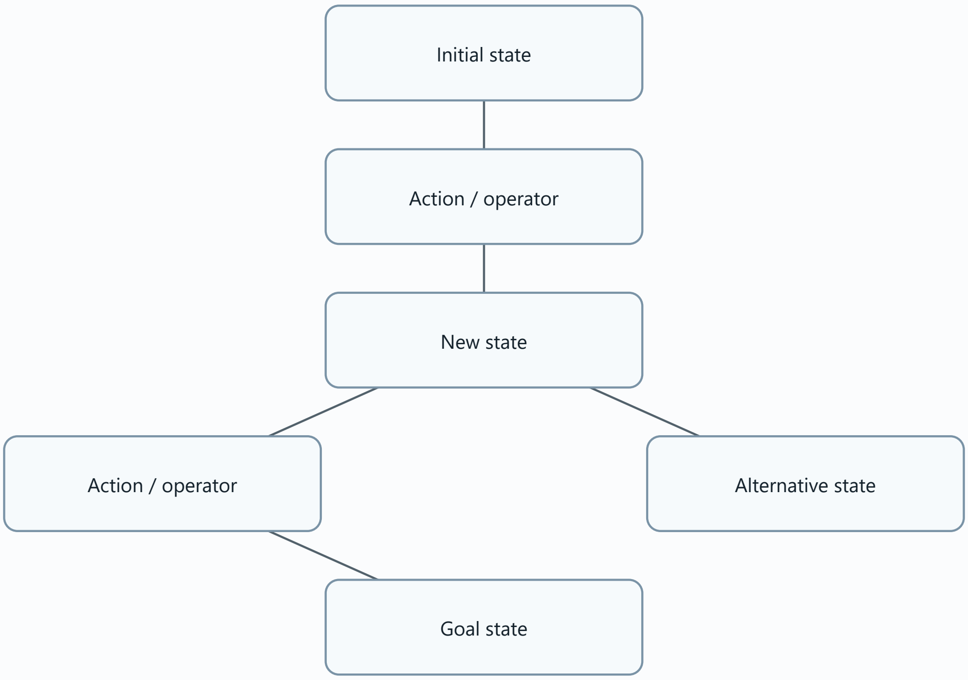
State Search Representation

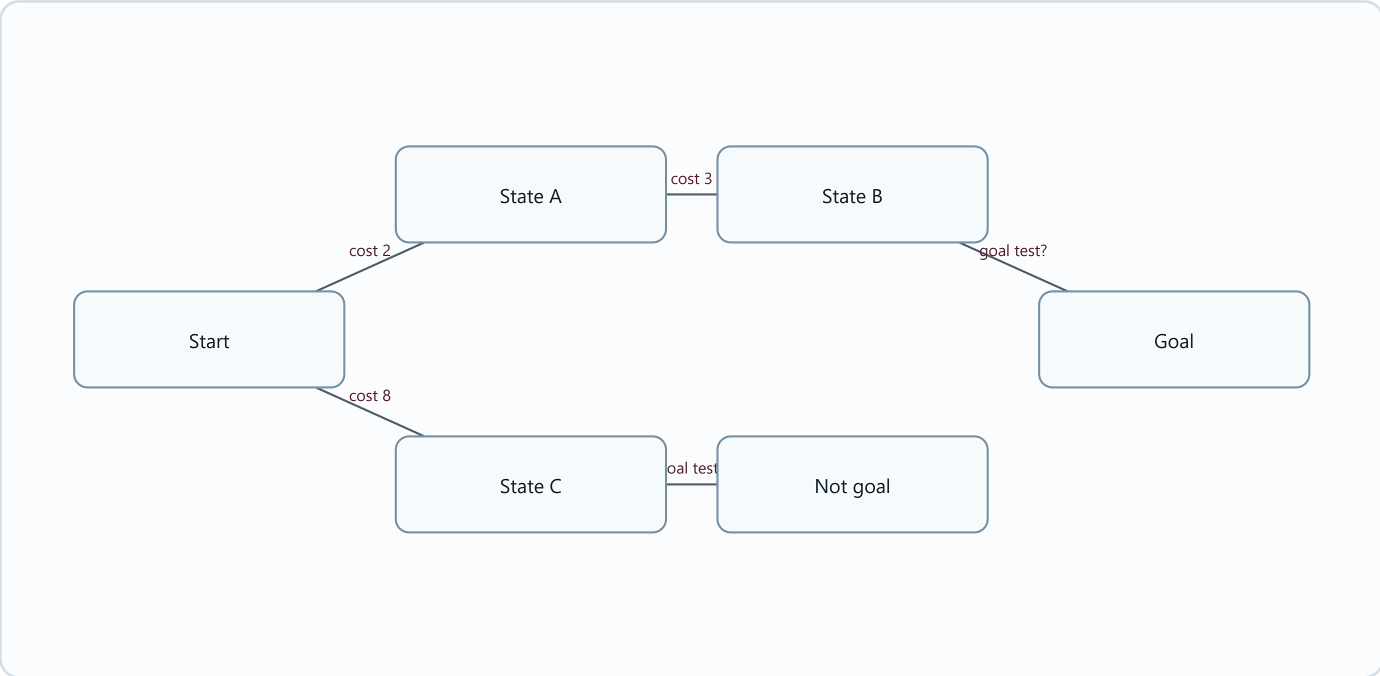
Diagram needed: Yes. A state space is naturally a graph/tree.



Path, Path Cost, and Goal Test

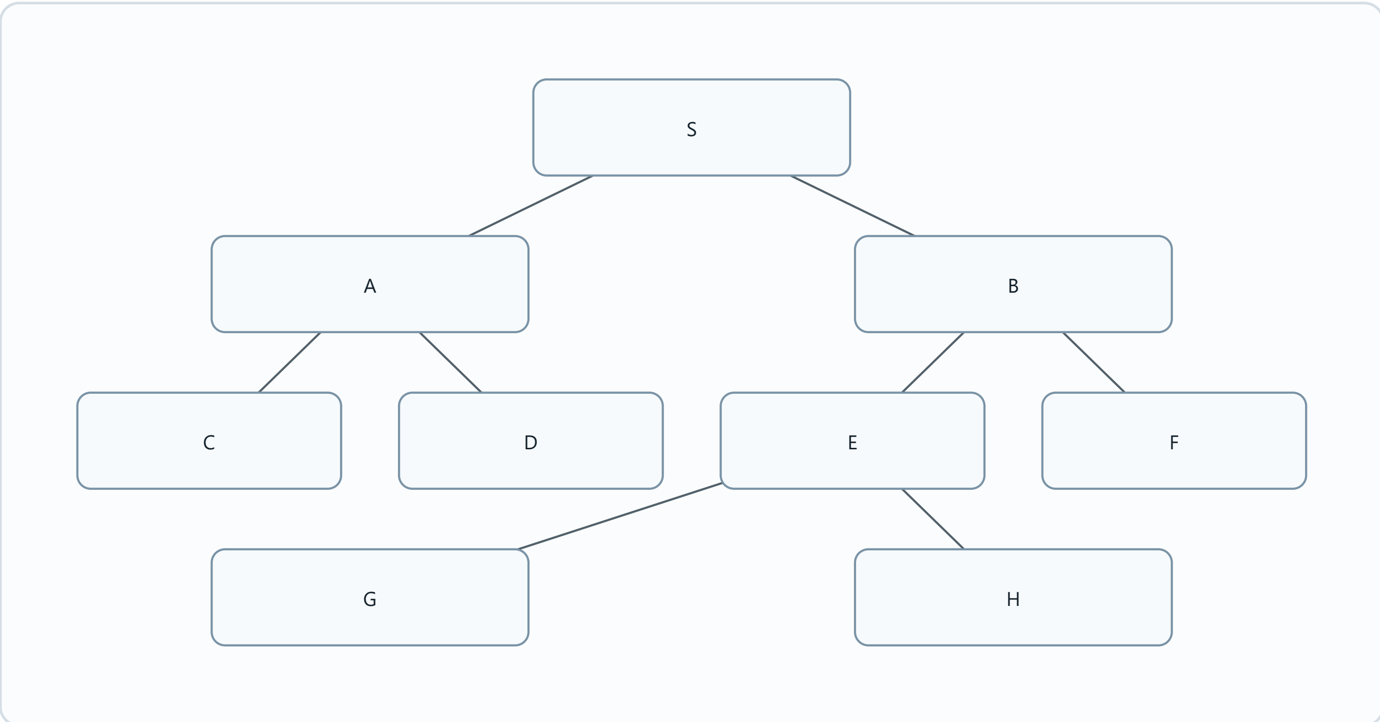
Diagram needed: Yes. These three terms work together in search.

Term	Visual role
Path	Route through states.
Path cost	Number/cost attached to a route.
Goal test	Check whether a state is the goal.



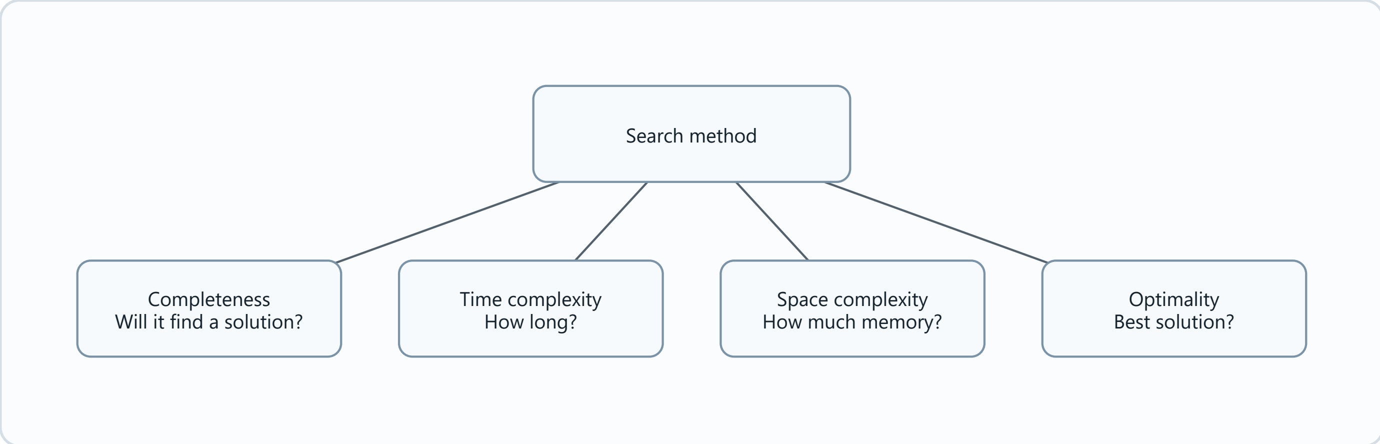
!!! EXAM TOPIC: Search Trees !!!

Diagram needed: Yes. This is a main exam area.



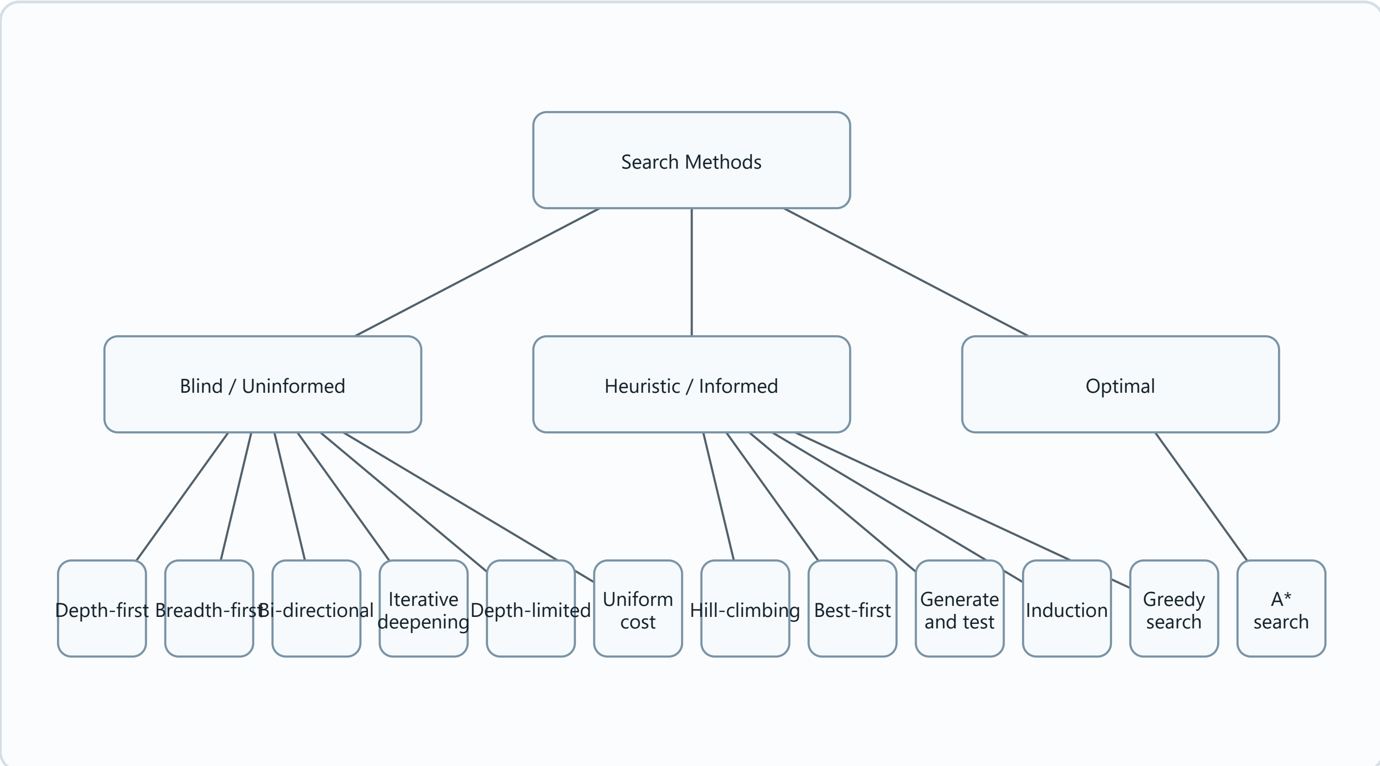
Evaluating a Search

Diagram needed: Yes. The four criteria are a checklist.



Search Methods

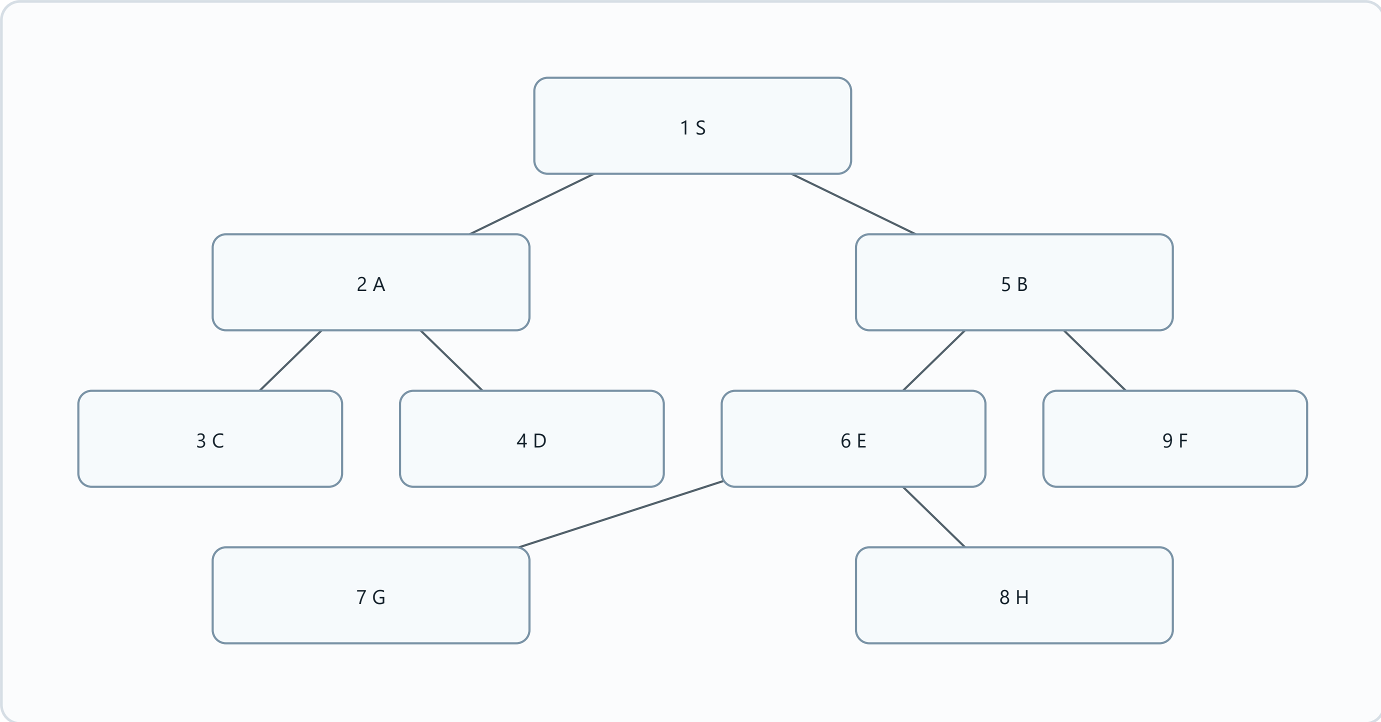
Diagram needed: Yes. The categories prevent confusion.



Depth-First Search (DFS)

Diagram needed: Yes. DFS is a traversal pattern.

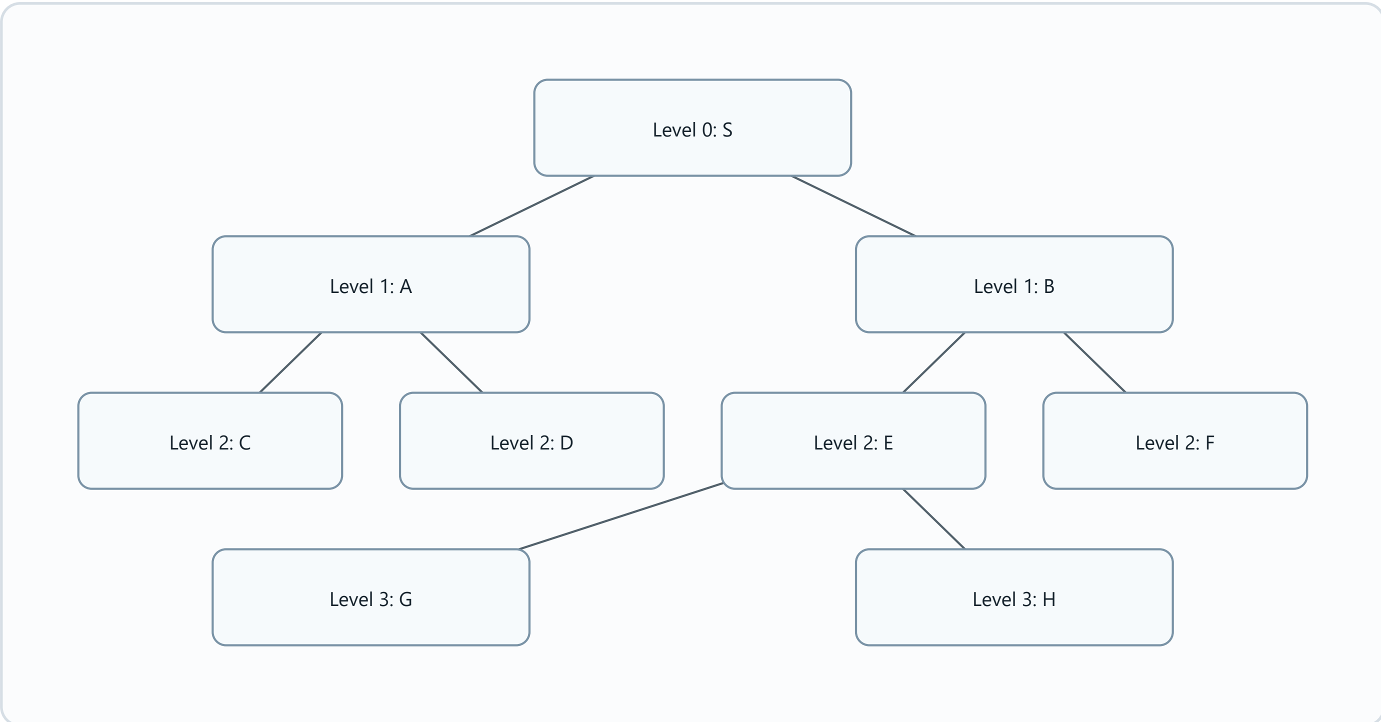
Source DFS order	S, A, C, D, B, E, G, H, F



!!! EXAM TOPIC: Breadth-First Search (BFS) !!!

Diagram needed: Yes. BFS is level-by-level.

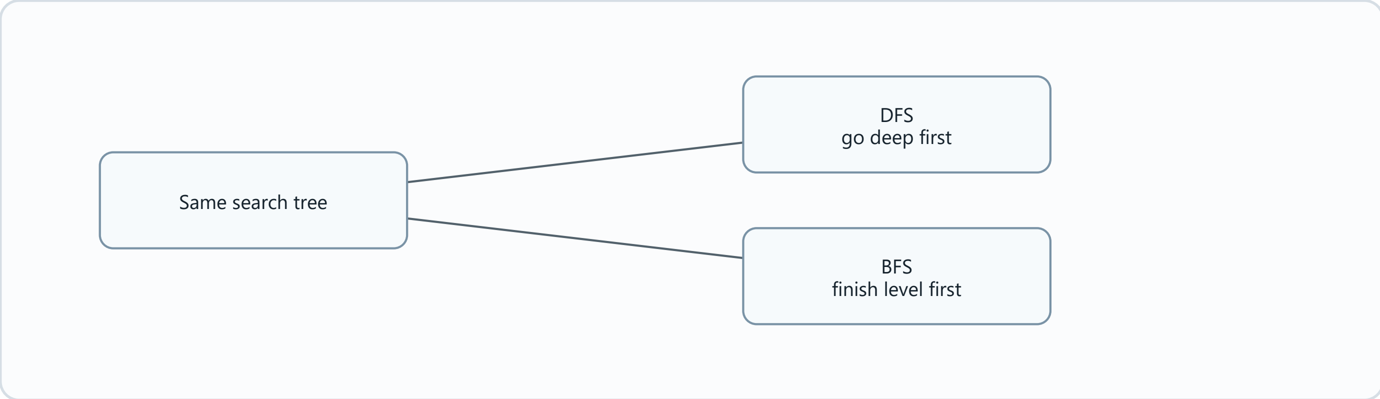
Source BFS order	S, A, B, C, D, E, F, G, H



DFS vs BFS

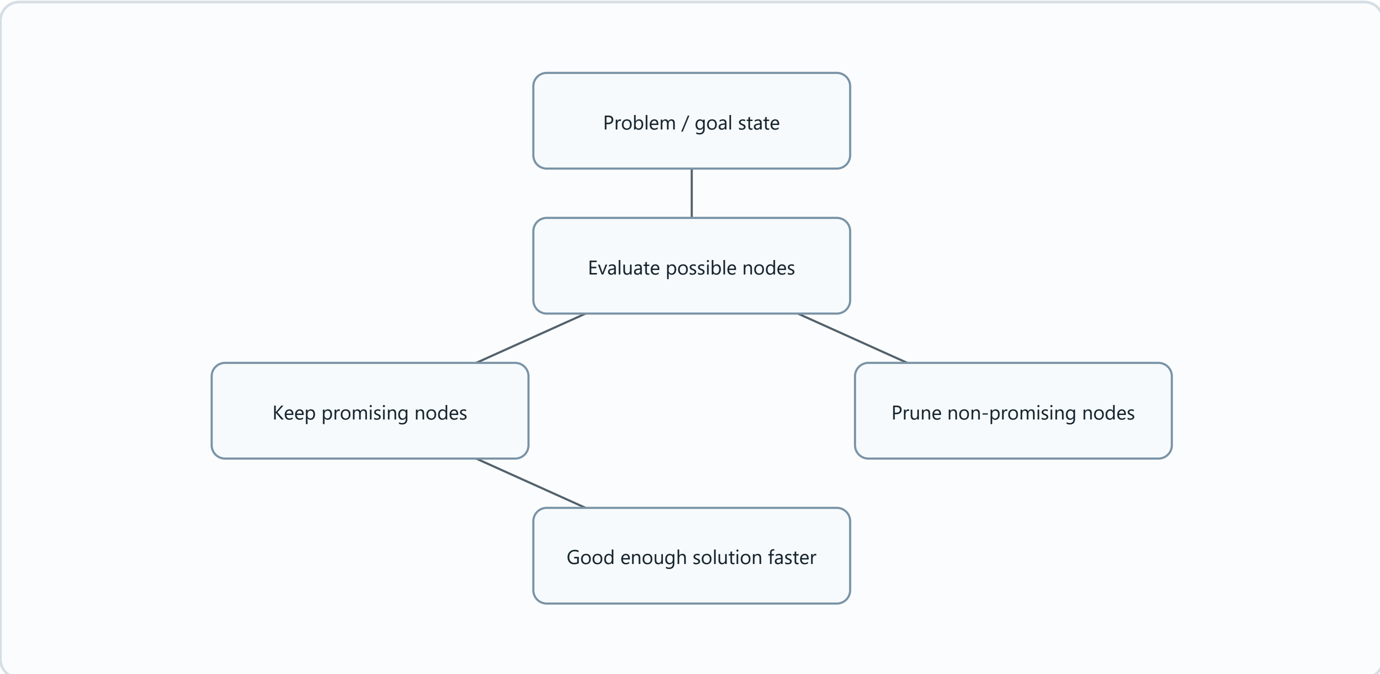
Diagram needed: Yes. This is one of the easiest exam comparisons.

Feature	DFS	BFS
Movement	Branch by branch.	Row by row.
Data behavior in slides	Children added to front of queue.	Children appended to end of queue.
Source order	S A C D B E G H F	S A B C D E F G H



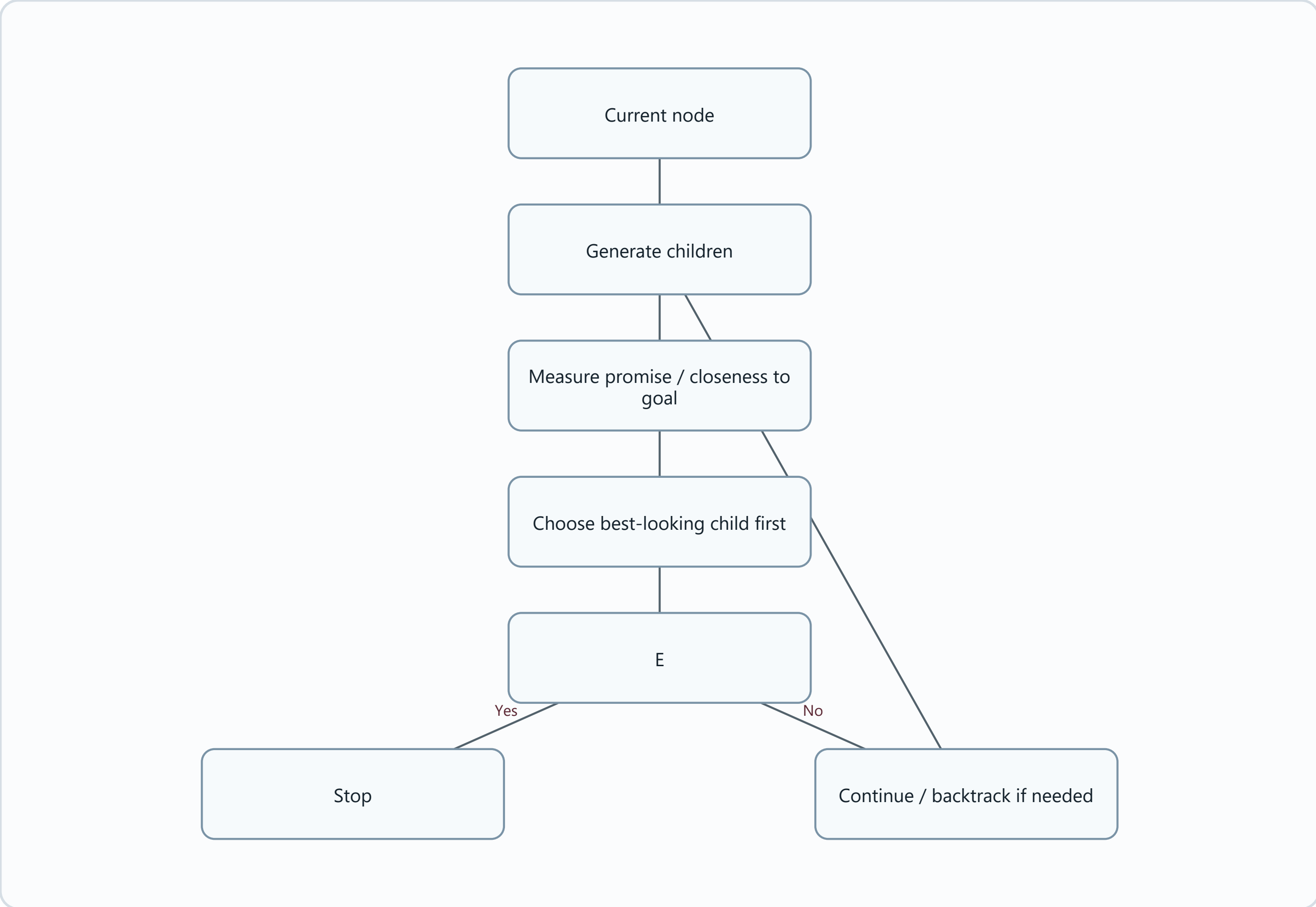
Heuristic Search

Diagram needed: Yes. Heuristic search prunes weak choices.



!!! EXAM TOPIC: Hill-Climbing Search !!!

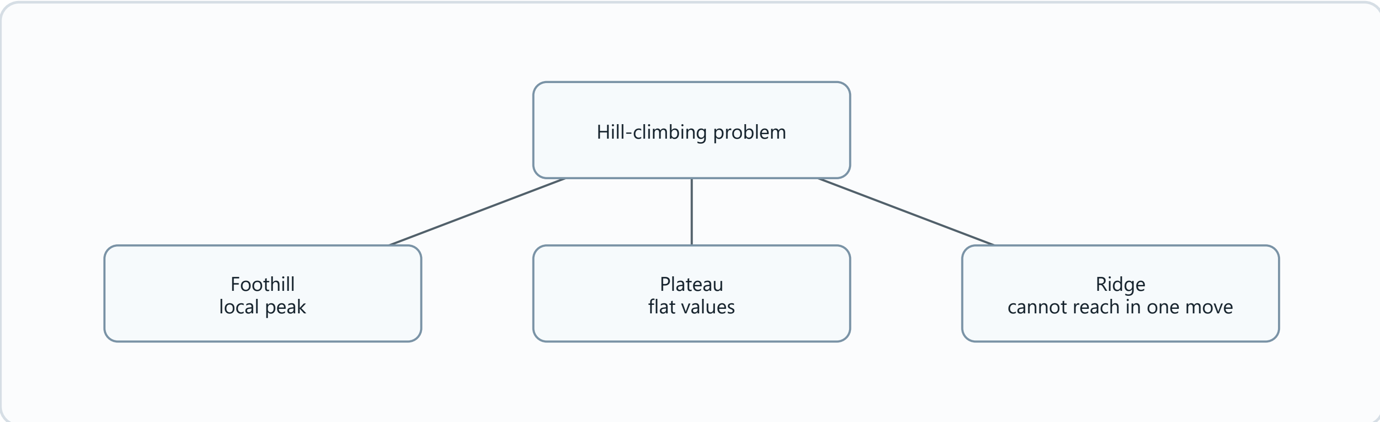
Diagram needed: Yes. Hill-climbing depends on choosing the most promising next node.



Problems with Hill-Climbing

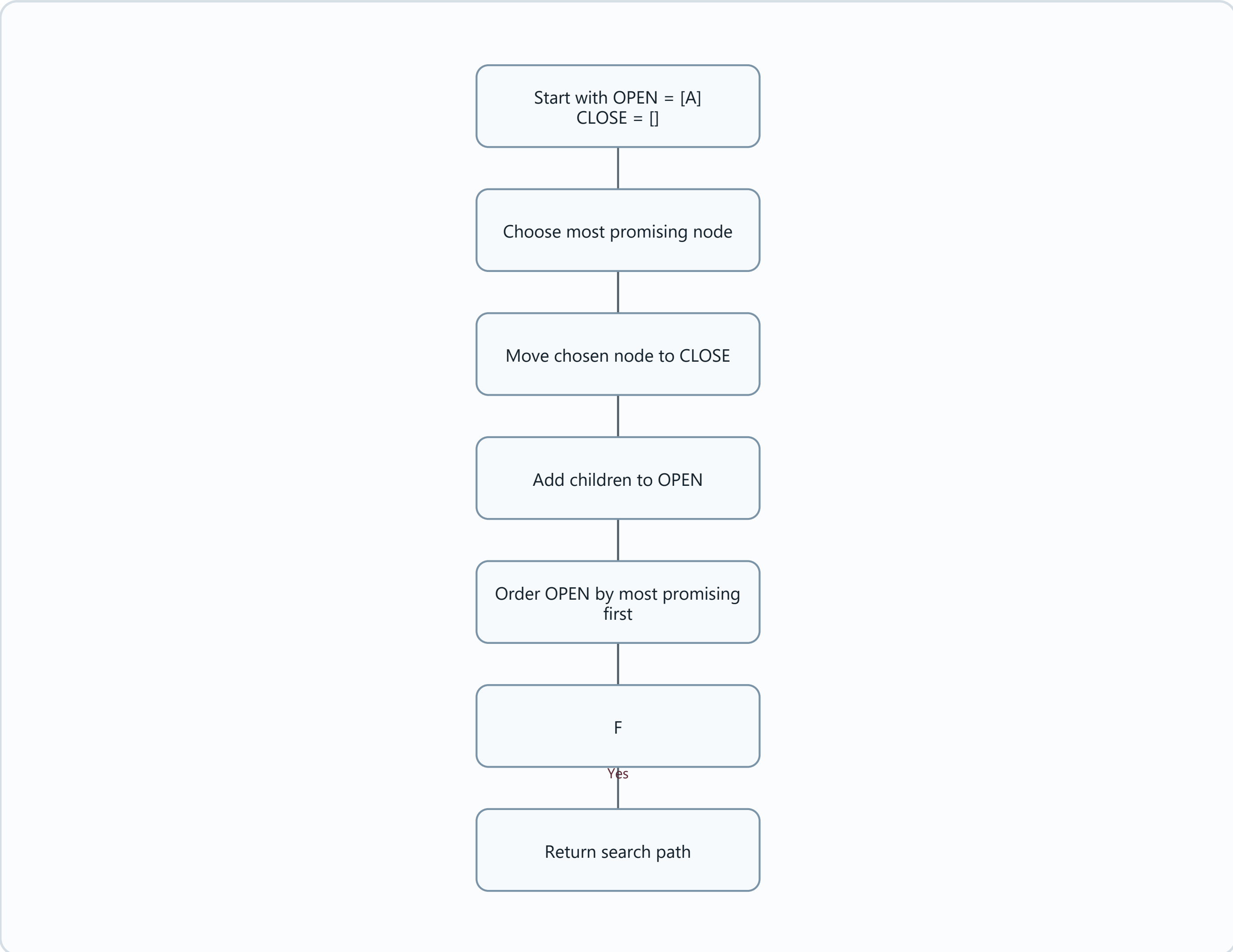
Diagram needed: Yes. The three problems are shape-based.

Problem	Memory image
Foothill	A smaller peak looks best nearby, but a higher peak exists elsewhere.
Plateau	Flat area; no clear best direction.
Ridge	High sloped area not reachable in one move.



Best-First Search

Diagram needed: Yes. The open/closed list example in the slide is process-based.


Start with OPEN = [A]
CLOSE = []
Choose most promising node
Move chosen node to CLOSE
Add children to OPEN
Order OPEN by most promising first
F
Yes
Return search path

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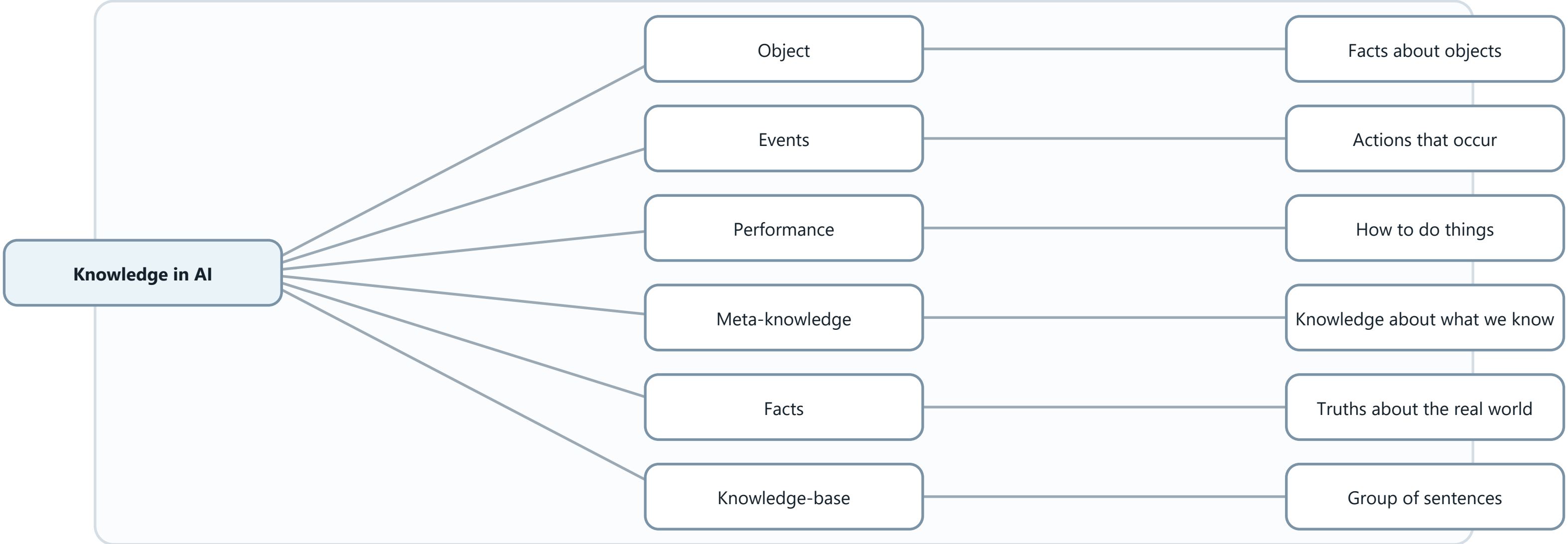
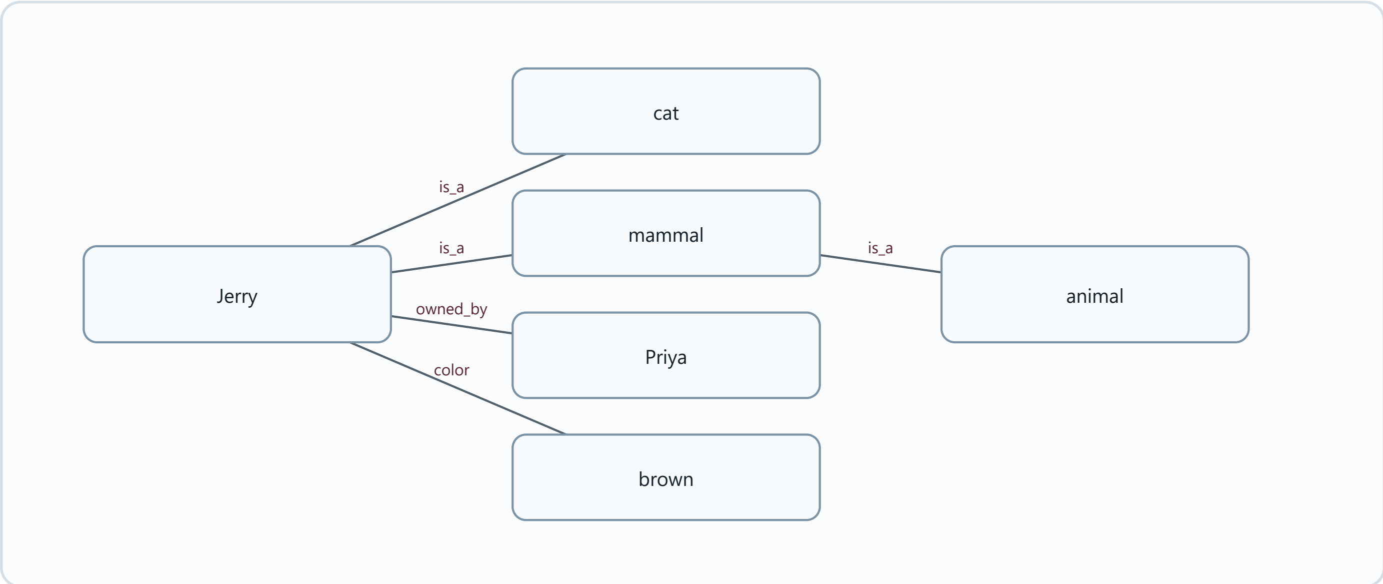
Types of Knowledge

Diagram needed: Yes. The types are easier as a knowledge map.



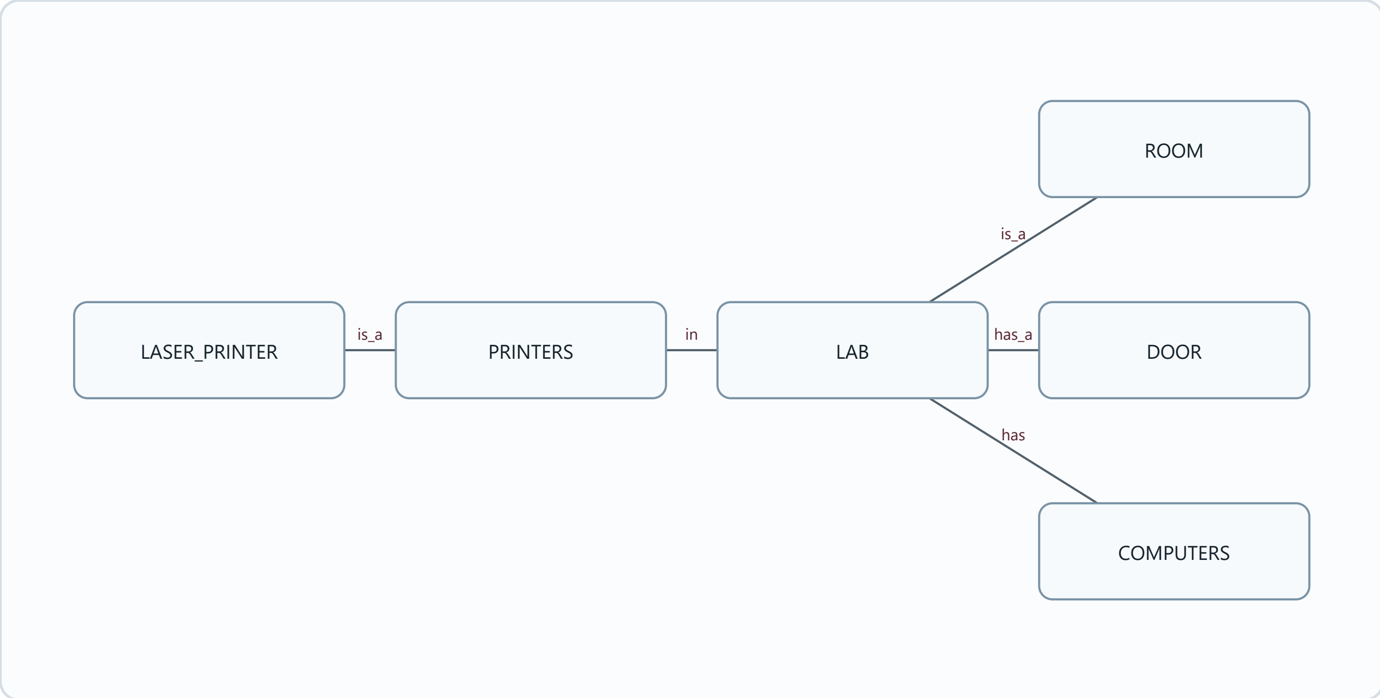
!!! EXAM TOPIC: Semantics / Semantic Networks !!!

Diagram needed: Yes. Semantic networks are diagrams by nature.



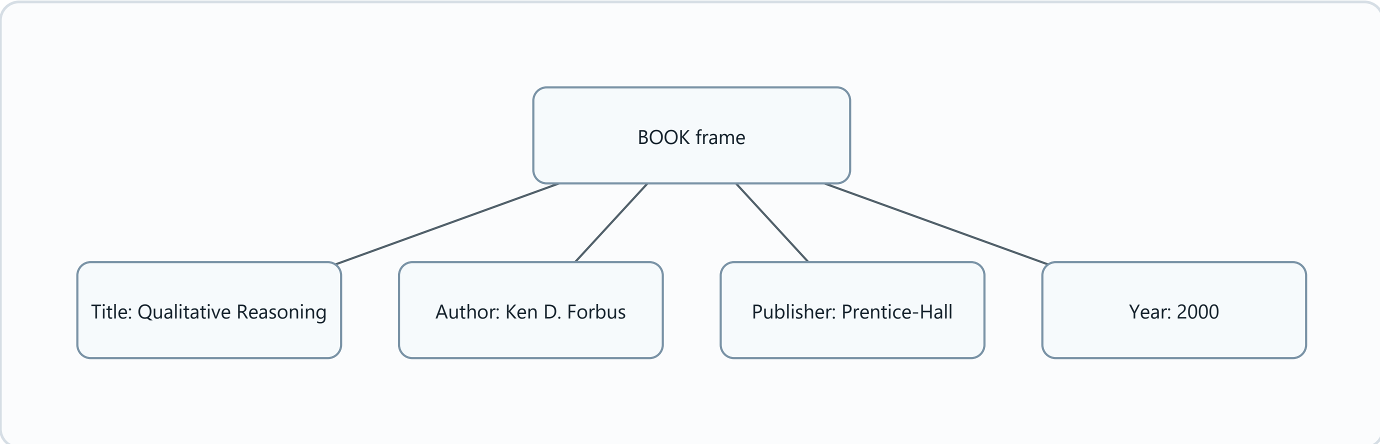
Semantic Network: Lab Example

Diagram needed: Yes. It maps source statements into relationships.



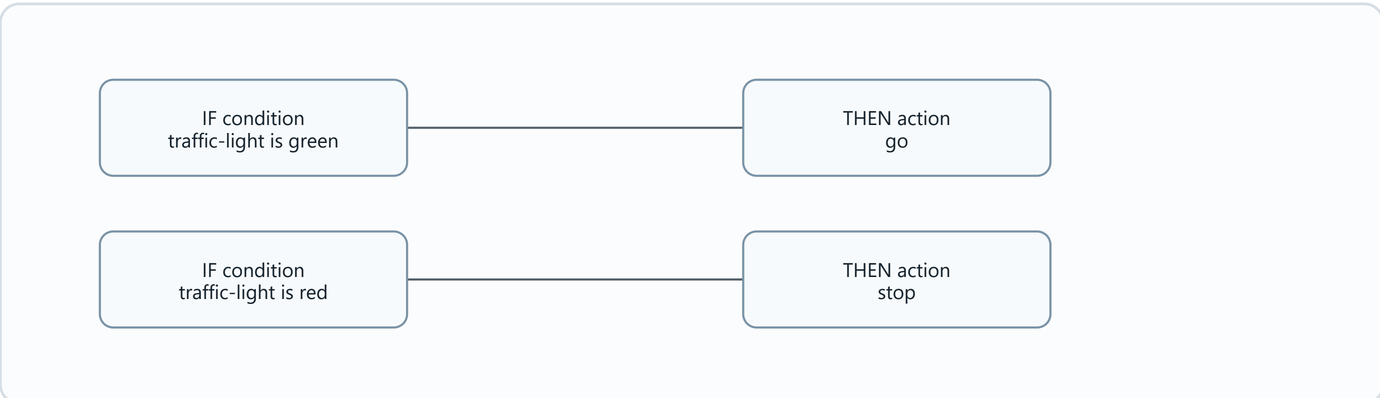
Frames

Diagram needed: Yes. Frames are slot-value structures.



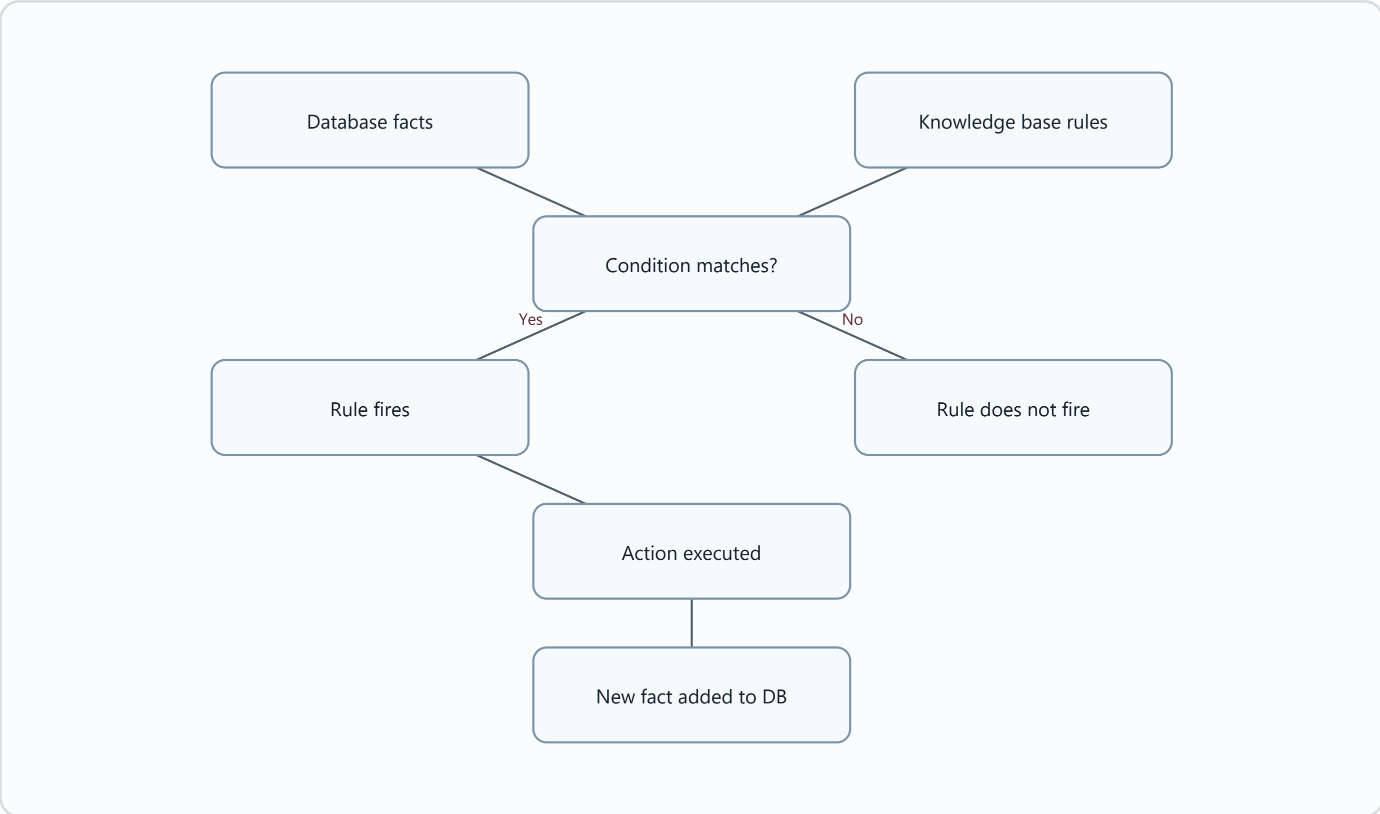
Production Rules

Diagram needed: Yes. The IF-THEN shape is essential.



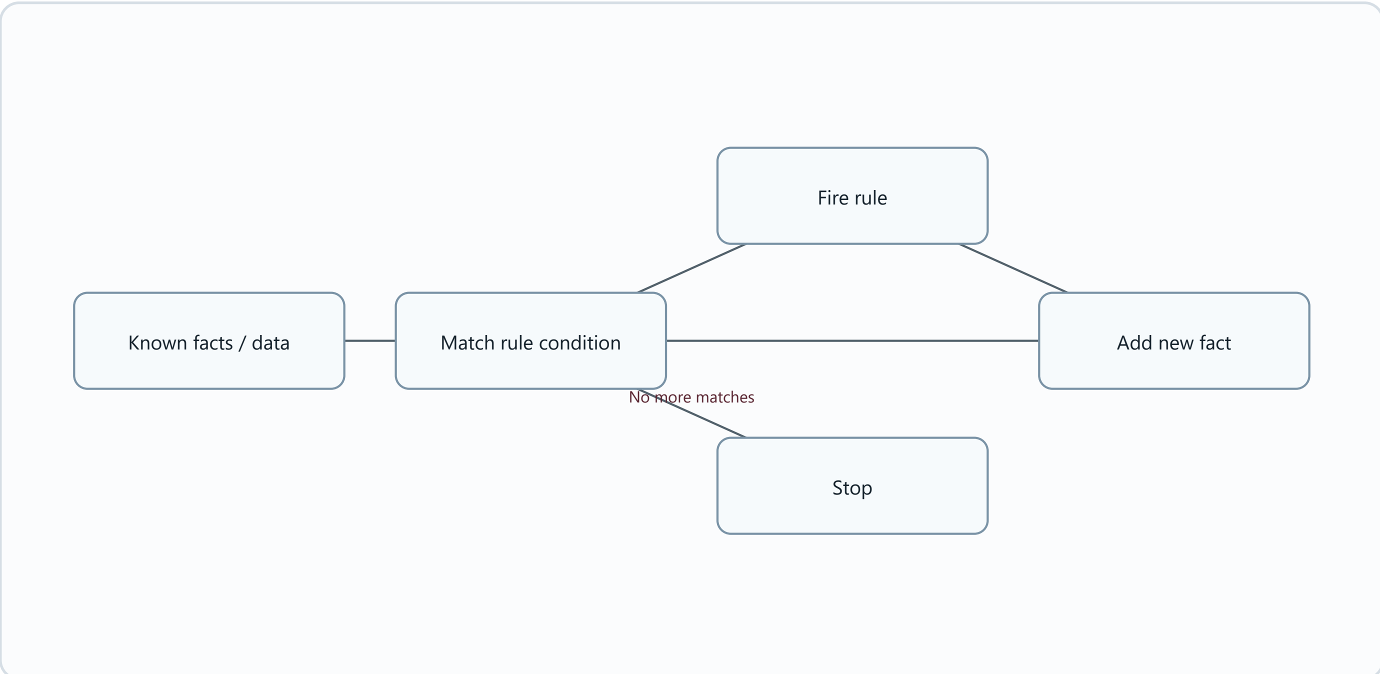
Firing of Rules

Diagram needed: Yes. It shows how facts and rules interact.



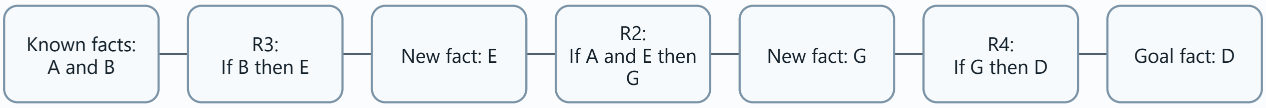
!!! EXAM TOPIC: Forward Chaining !!!

Diagram needed: Yes. Forward chaining is a step-by-step flow.



Forward Chaining Example: A and B prove D

Diagram needed: Yes. The proof chain is linear.

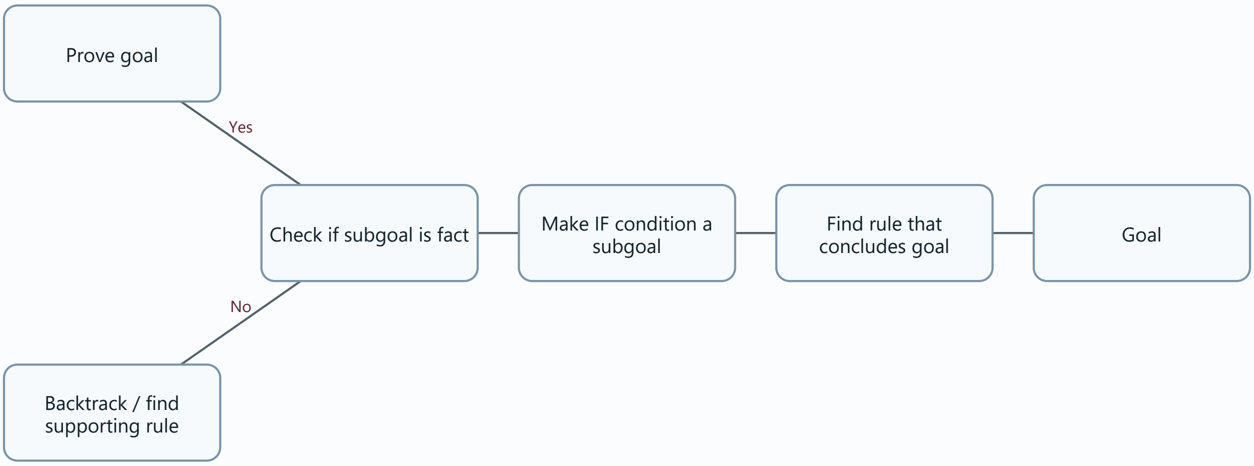


Forward Chaining Health Exercise

Diagram needed: Yes. It makes rule firing order clear.

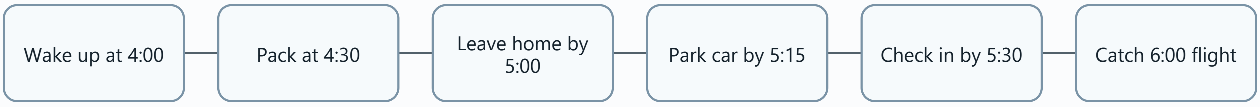
Backward Chaining

Diagram needed: Yes. It is the reverse of forward chaining.



Backward Chaining Flight Example

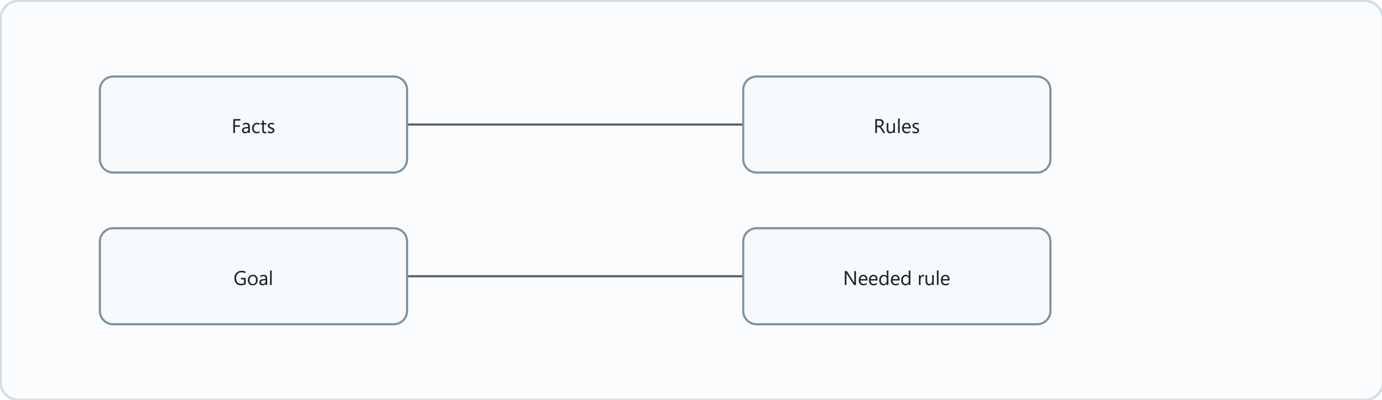
Diagram needed: Yes. The goal is proved by tracing backward.



Forward Chaining vs Backward Chaining

Diagram needed: Yes. The direction is the whole difference.

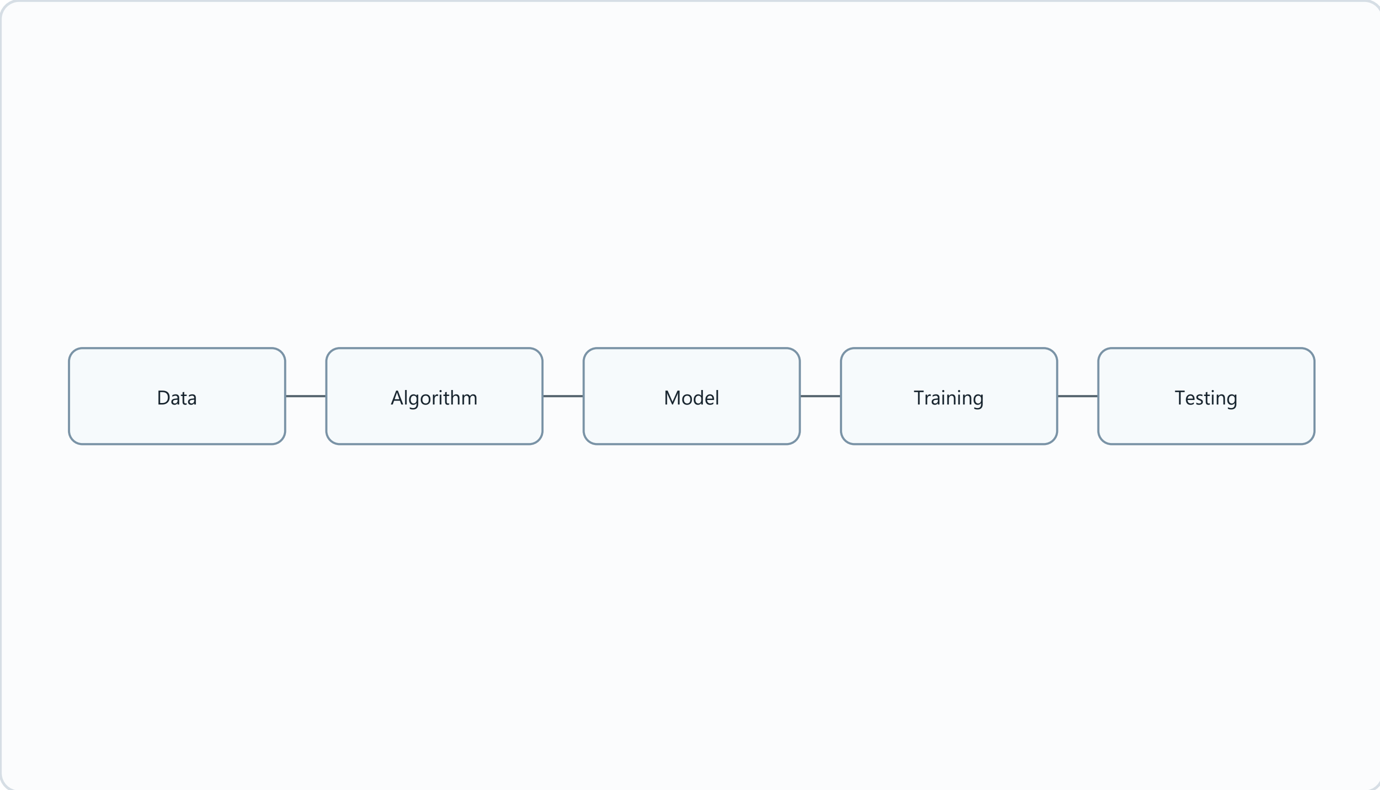
Forward chaining	Backward chaining
Data-driven.	Goal-driven.
Starts from known facts.	Starts from a goal.
Moves toward conclusions.	Searches backward for evidence.



06 ML - First Iteration sem 1 2024 2025 (1).pdf

Core Concepts of Machine Learning

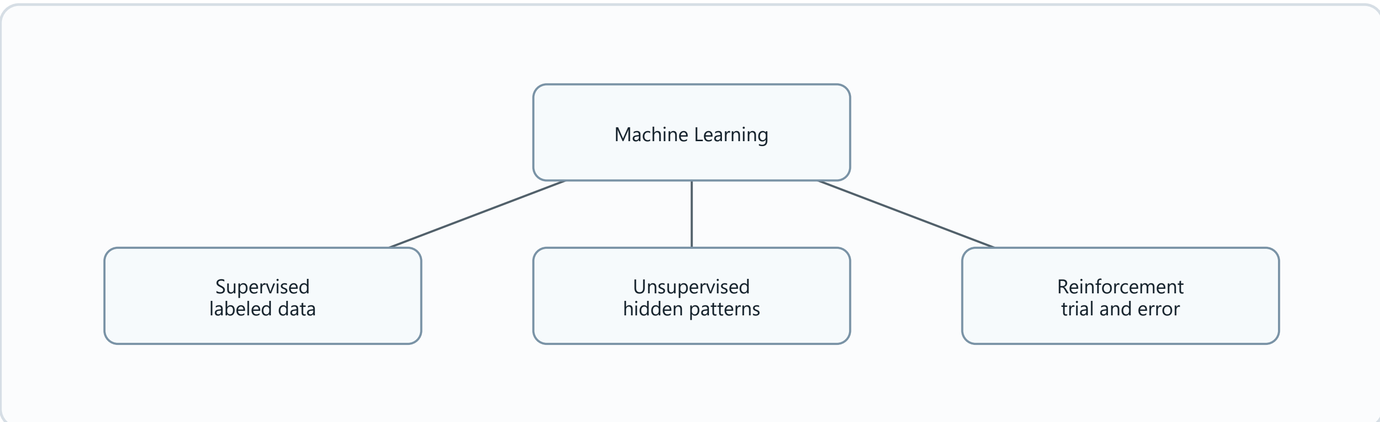
Diagram needed: Yes. The process is a pipeline.



Types of Machine Learning

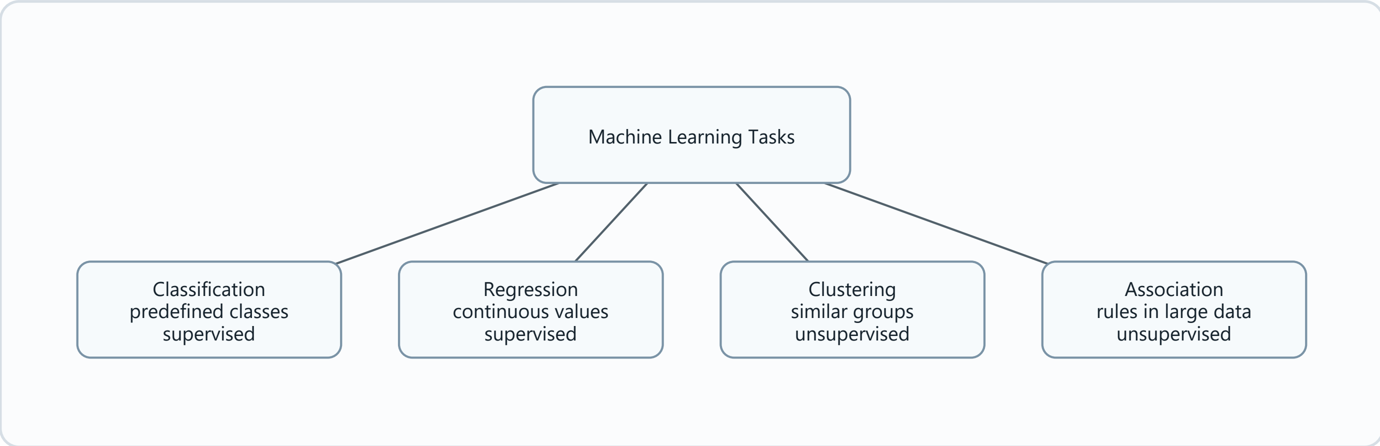
Diagram needed: Yes. The three types are a classification tree.

Type	Key phrase
Supervised	Labeled data.
Unsupervised	No labels; find patterns.
Reinforcement	Trial, error, rewards, penalties.



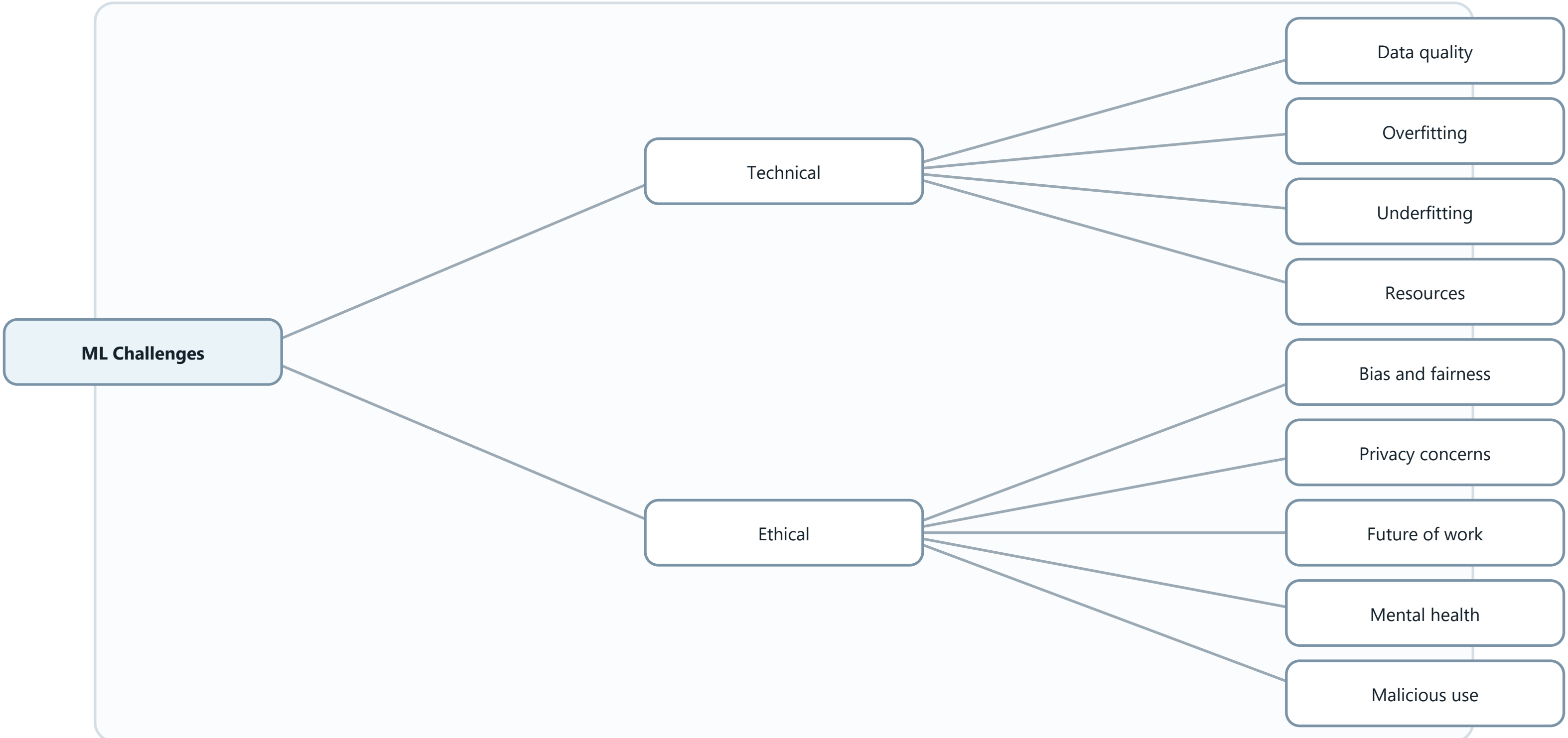
Machine Learning Tasks

Diagram needed: Yes. It shows which tasks belong to which learning type.



Challenges in Machine Learning

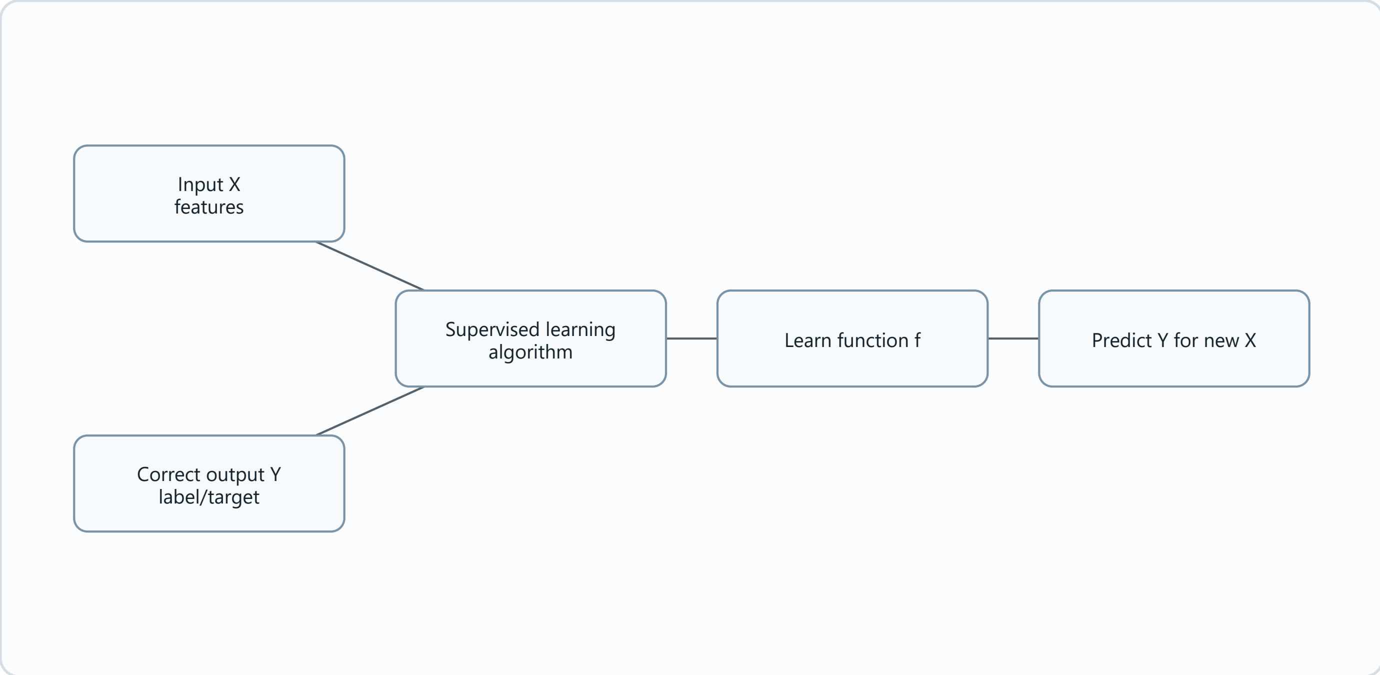
Diagram needed: Yes. Challenges are grouped into technical and ethical.



07 Traditional ML - sem 2 2025 2026 Supervised learning 1 (2).pdf

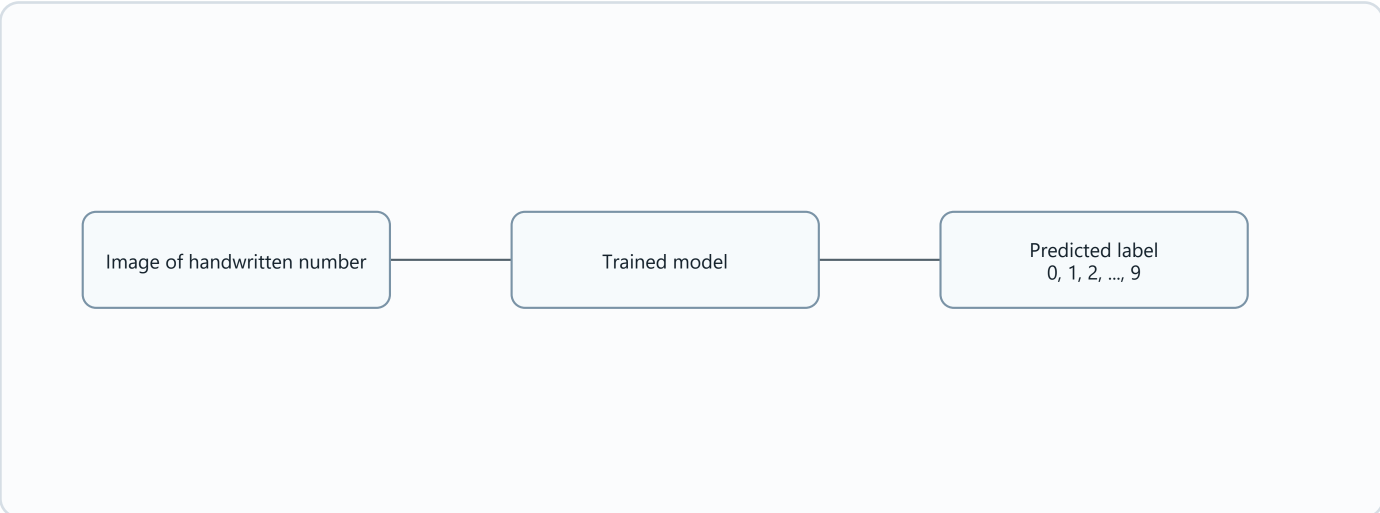
Supervised Learning

Diagram needed: Yes. Supervised learning is input-output learning.



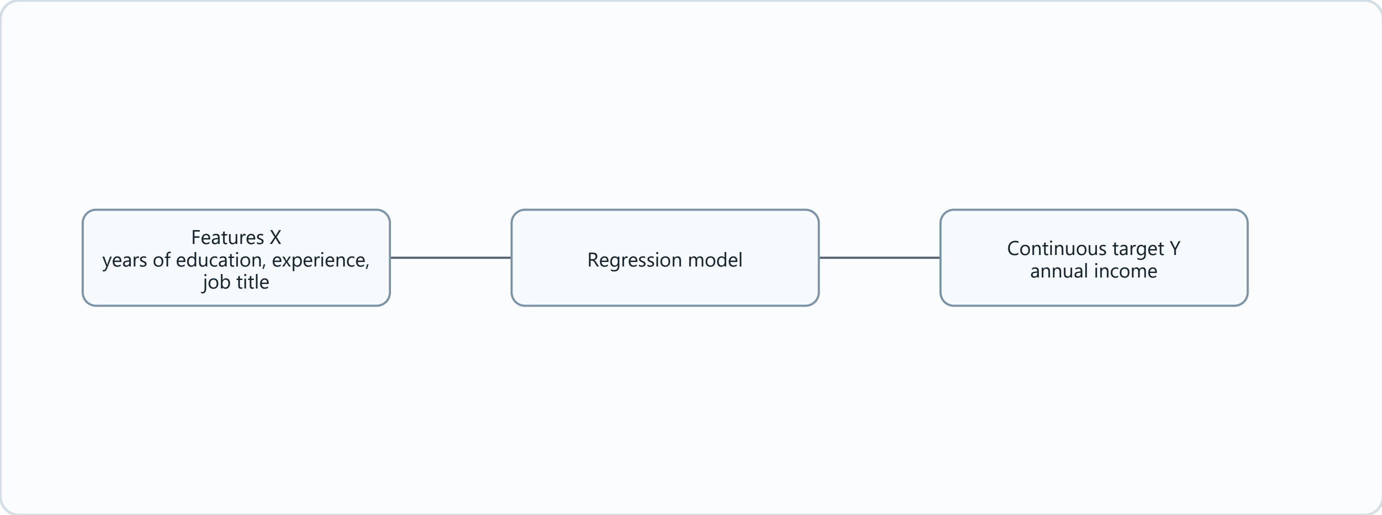
Classification

Diagram needed: Yes. It is a label prediction workflow.



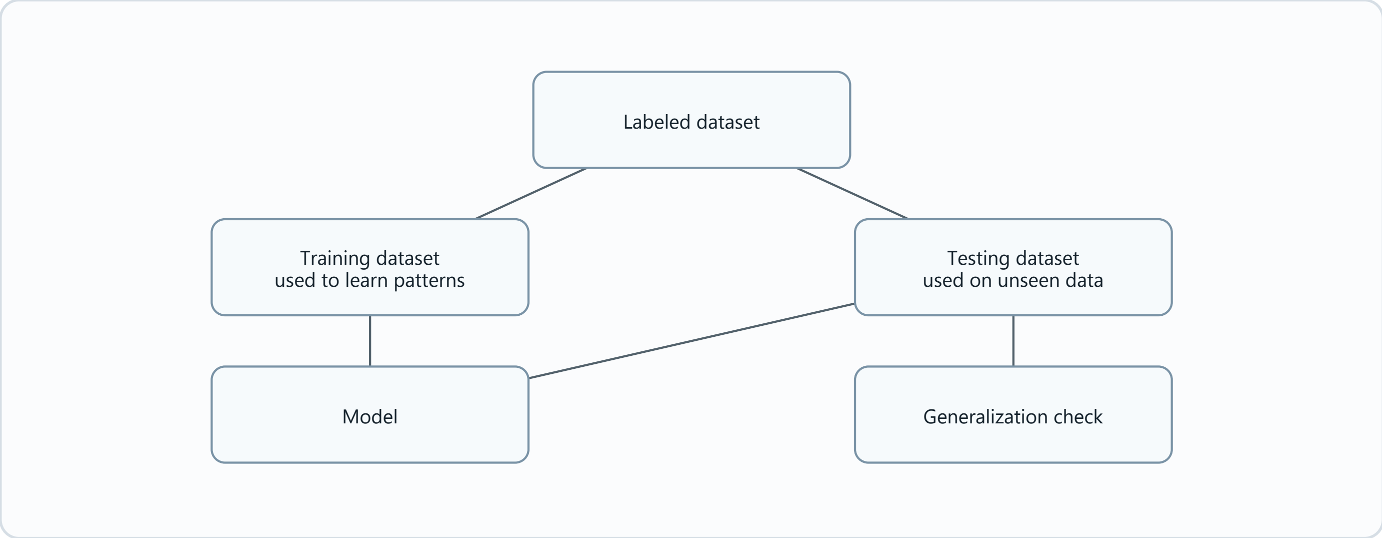
Regression

Diagram needed: Yes. Regression predicts a continuous value.



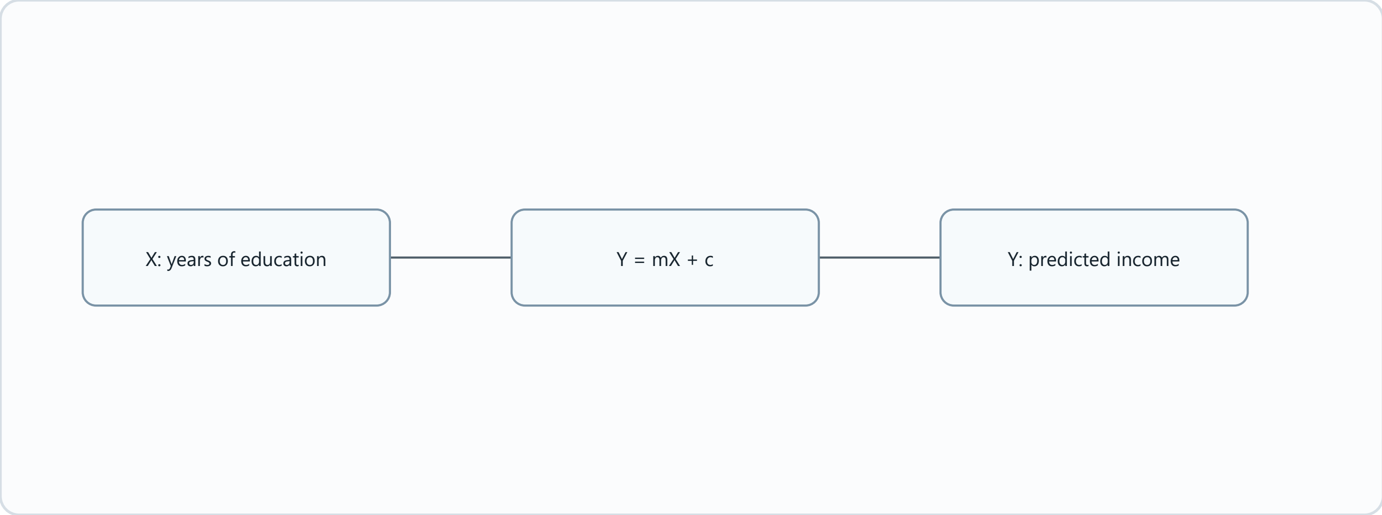
Training and Test Datasets

Diagram needed: Yes. This prevents the common mistake of testing on training data.



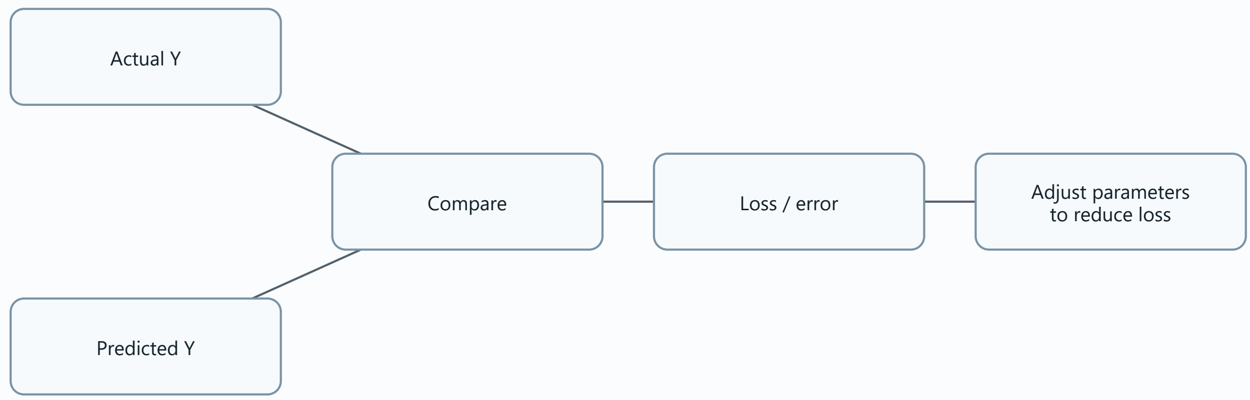
Linear Regression

Diagram needed: Yes. It is a line that maps X to Y.



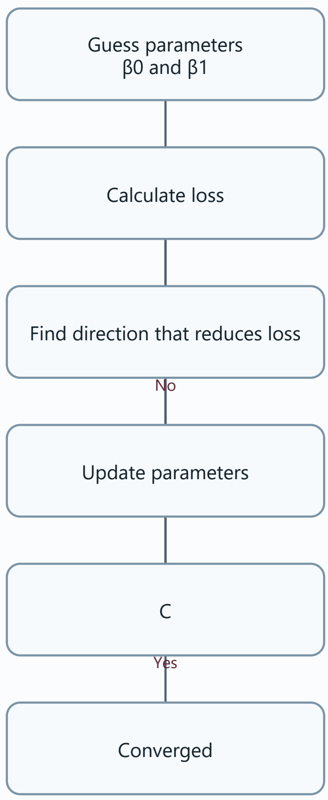
Cost / Loss Function

Diagram needed: Yes. Loss explains why training changes parameters.



Gradient Descent

Diagram needed: Yes. The valley analogy is source-based and highly visual.

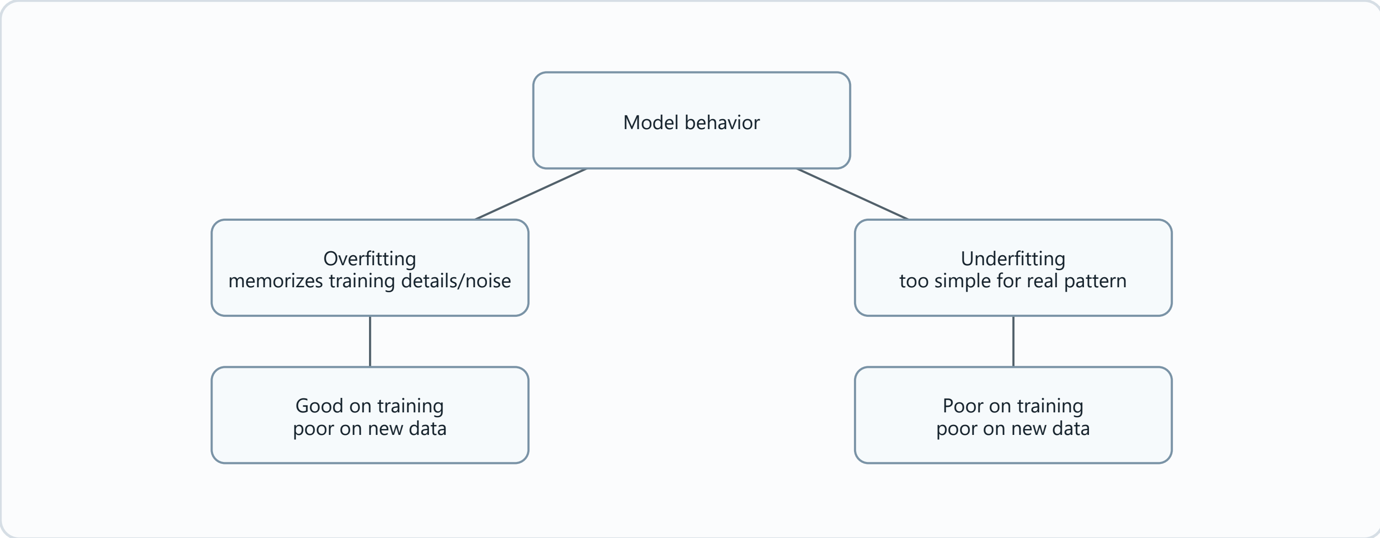


Overfitting and Underfitting

Diagram needed: Yes. These are a key comparison.

Concept	Training performance	New data performance	Simple meaning
Overfitting	Very good.	Poor.	Memorized too much.

Concept	Training performance	New data performance	Simple meaning
Underfitting	Poor.	Poor.	Did not learn enough.

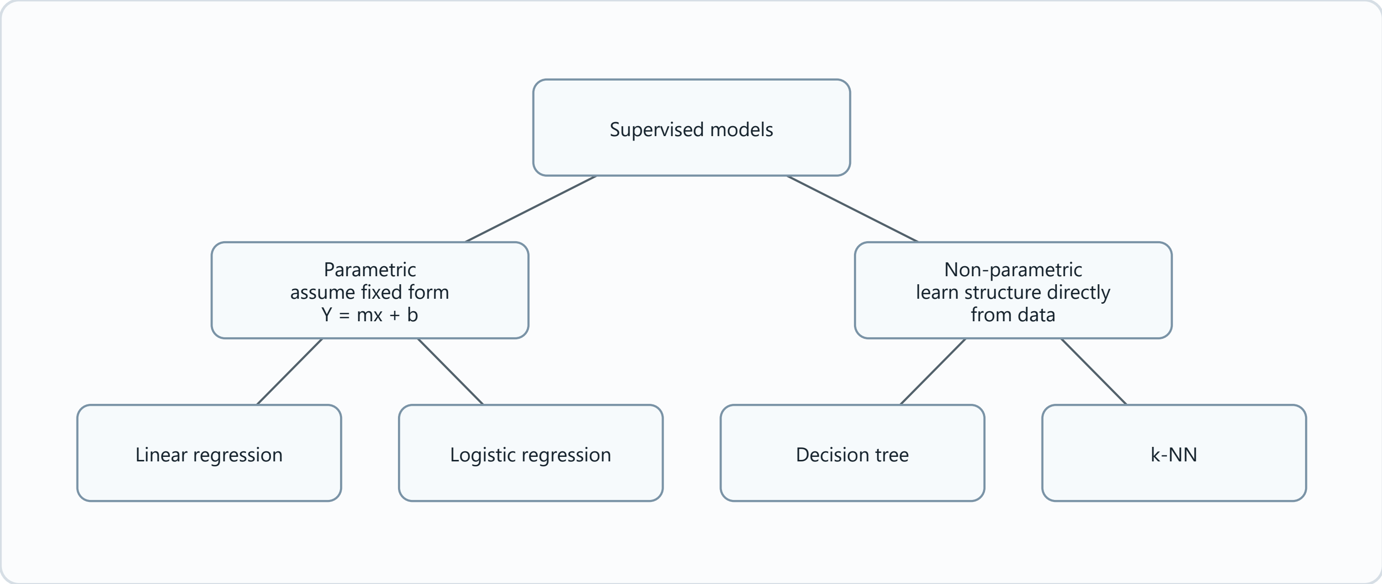


09_Traditional_ML_Supervised_learning_3_part_1_sem_1_2025_2026.pdf

Parametric Models and Non-Parametric Models

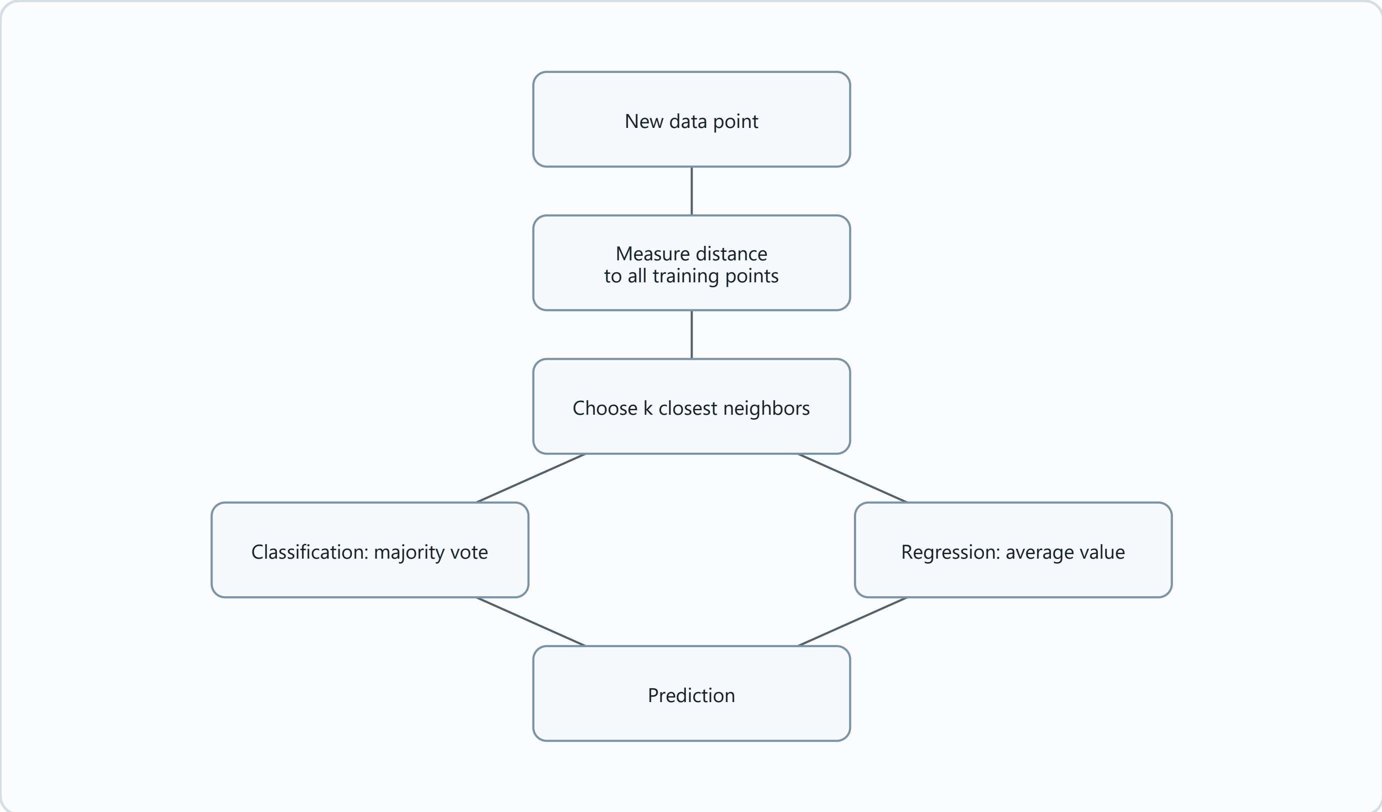
Diagram needed: Yes. The difference is foundational for this chapter.

Parametric	Non-parametric
Fixed function form.	No fixed function form.
Learns parameters.	Learns structure from data.



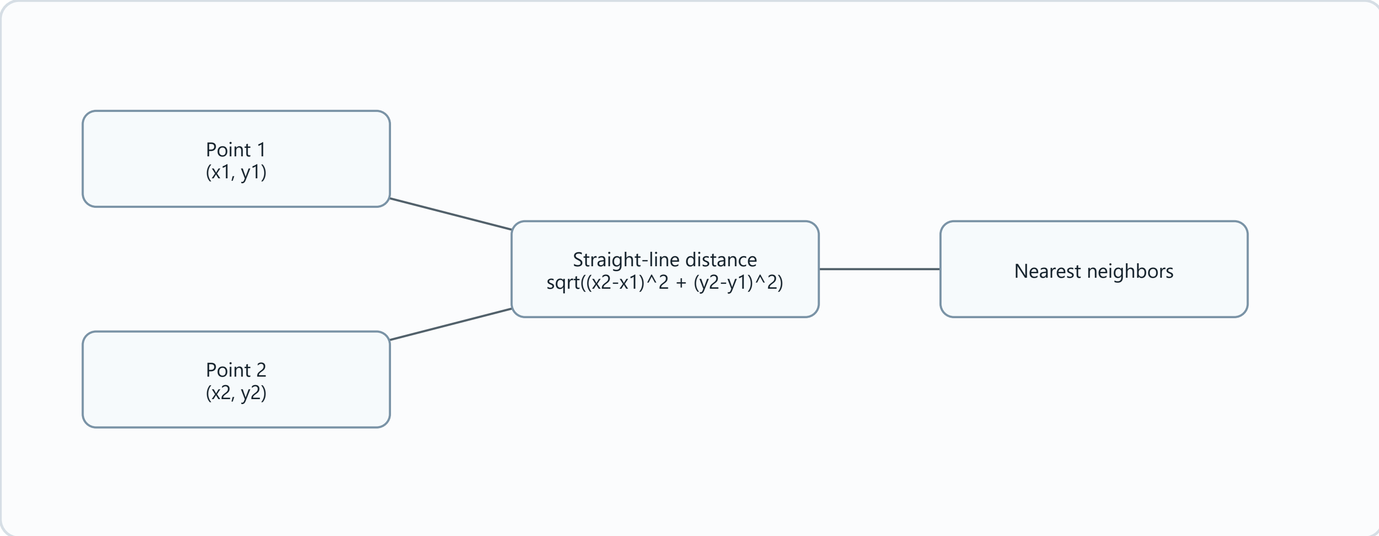
k-Nearest Neighbors (k-NN)

Diagram needed: Yes. k-NN is distance-based.



Euclidean Distance in k-NN

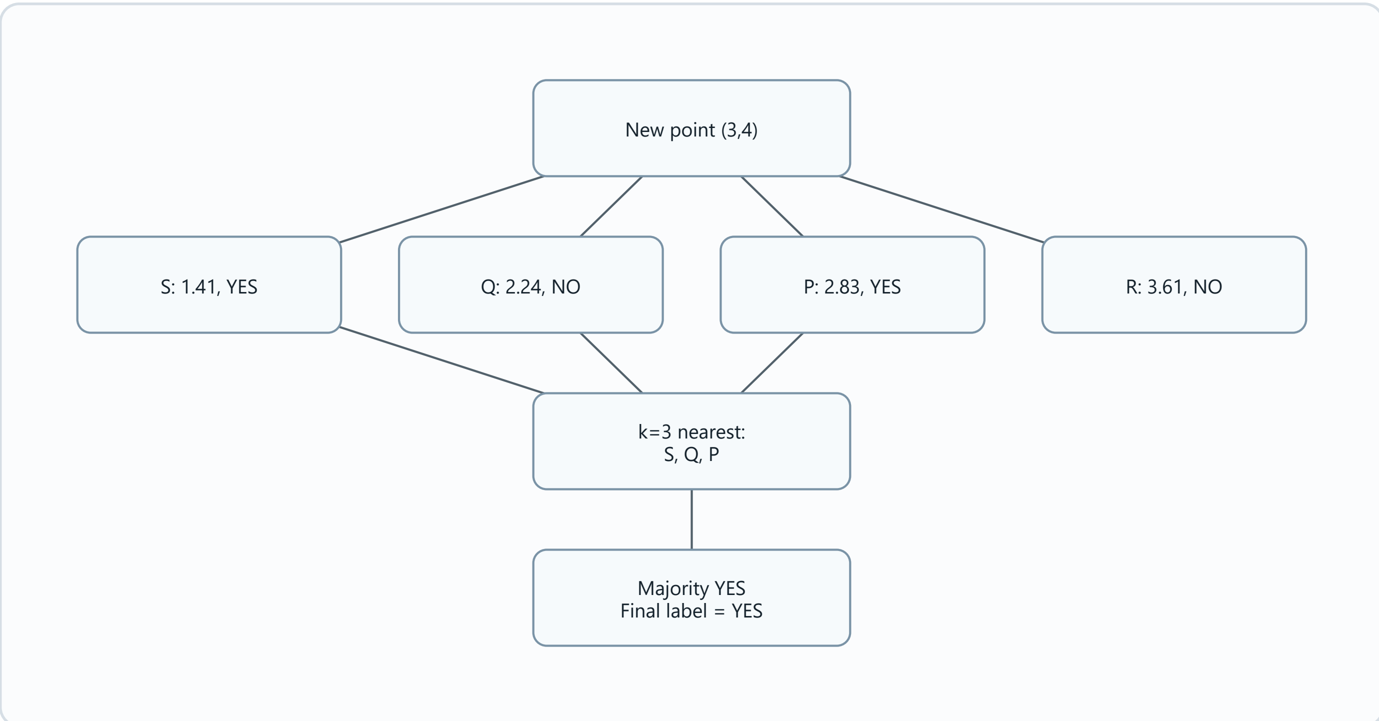
Diagram needed: Yes. The source formula benefits from a point-to-point visual.



k-NN Example for (3,4)

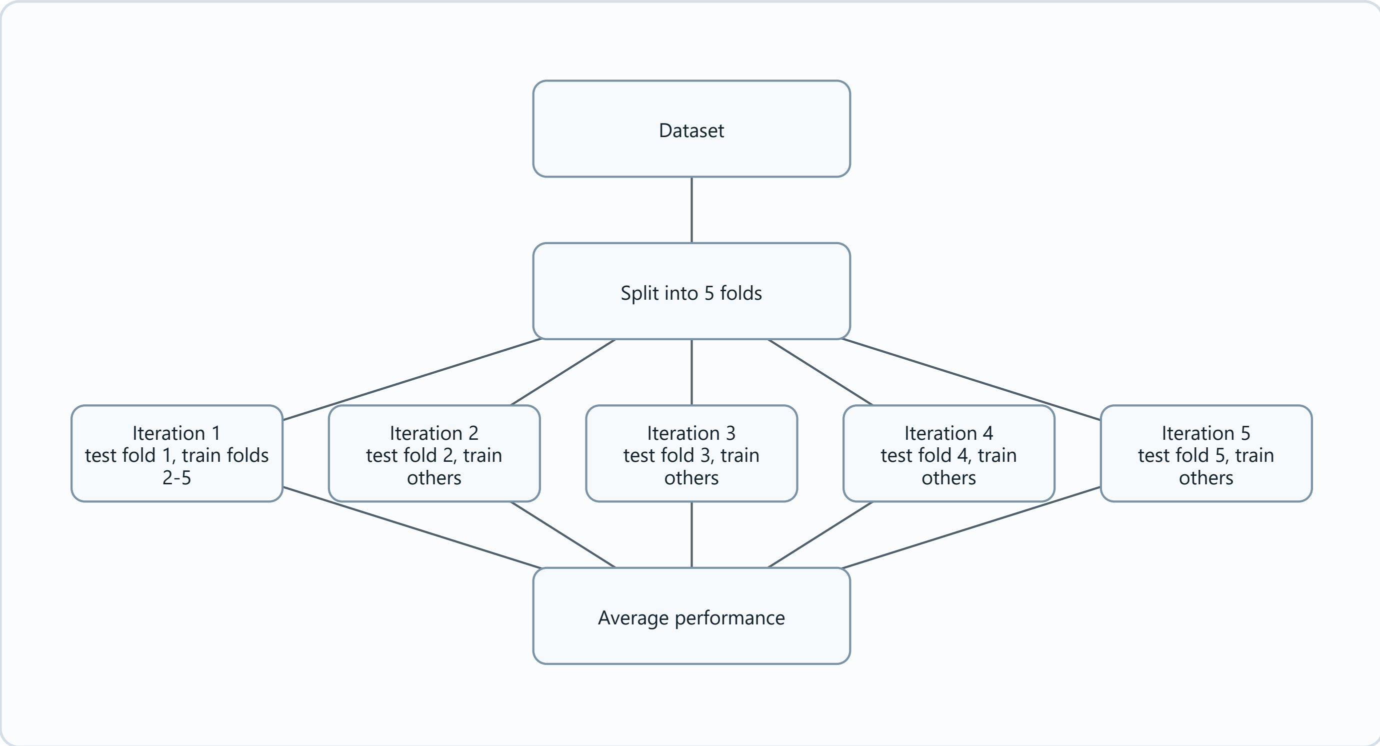
Diagram needed: Yes. The ranking table is the clearest diagram-like format.

Rank	Point	Distance	Label
1	S	1.41	YES
2	Q	2.24	NO
3	P	2.83	YES
4	R	3.61	NO



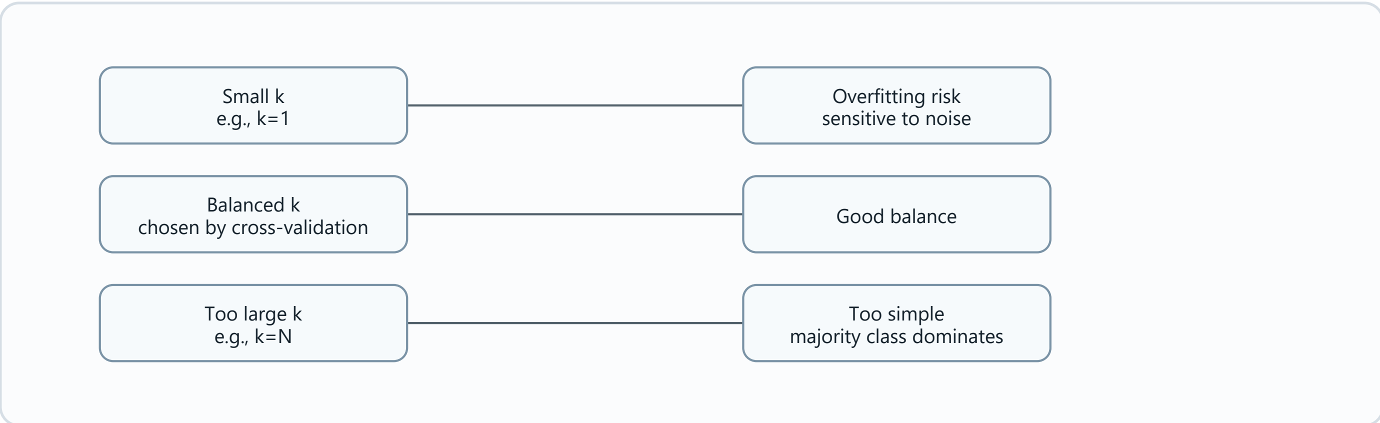
Cross-Validation

Diagram needed: Yes. k-fold cross-validation is naturally a repeated split.



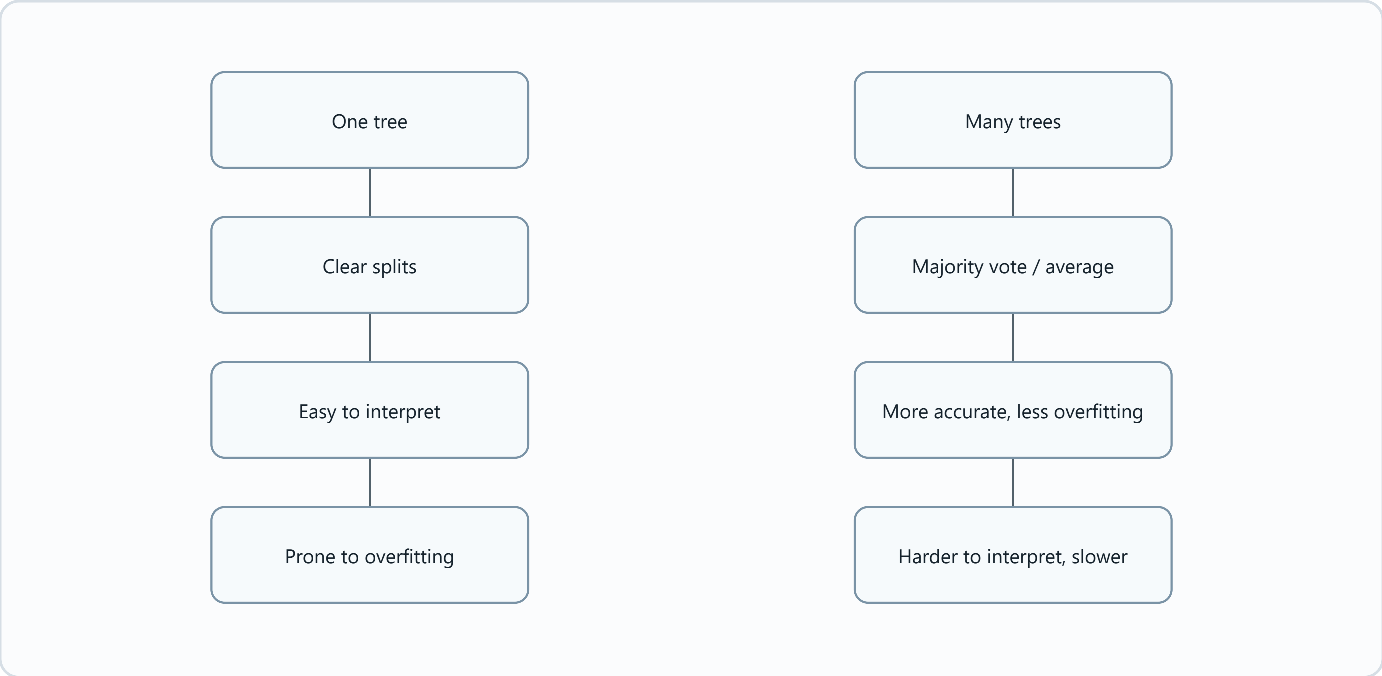
Choosing k

Diagram needed: Yes. The overfitting/oversimplifying balance is visual.



Decision Tree and Random Forest

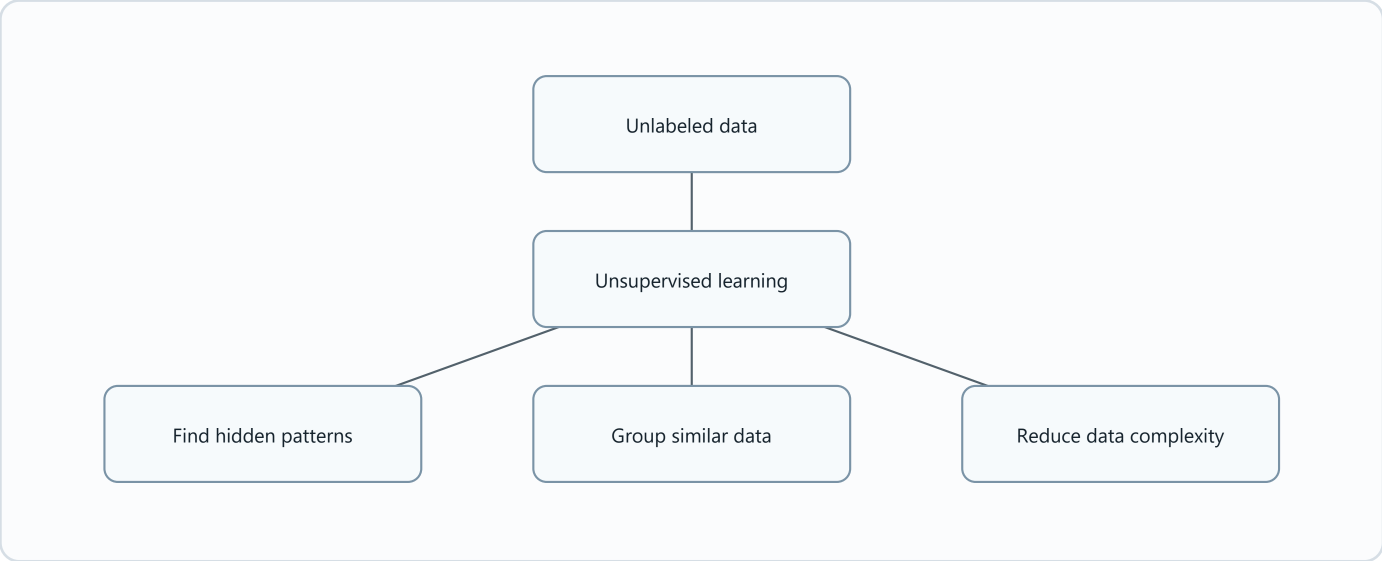
Diagram needed: Yes. The model structures are different.


One tree
Clear splits
Easy to interpret
Prone to overfitting
Many trees
Majority vote / average
More accurate, less overfitting
Harder to interpret, slower

10 Unsupervised learning.pdf

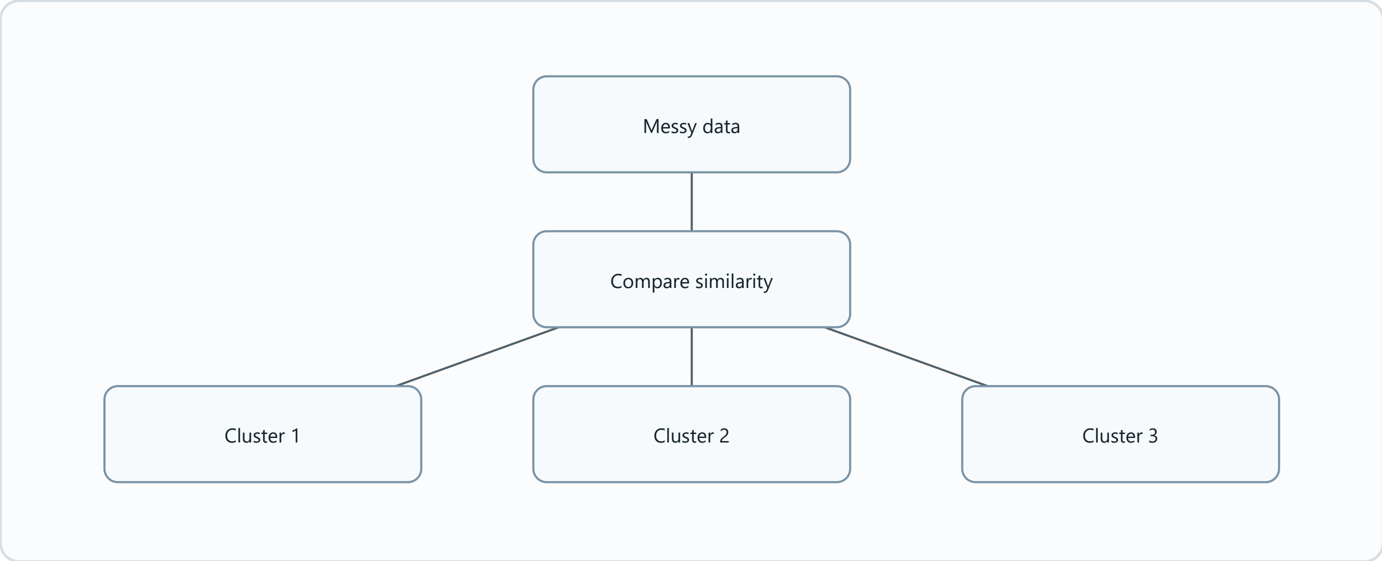
Unsupervised Learning

Diagram needed: Yes. It explains work without labels.



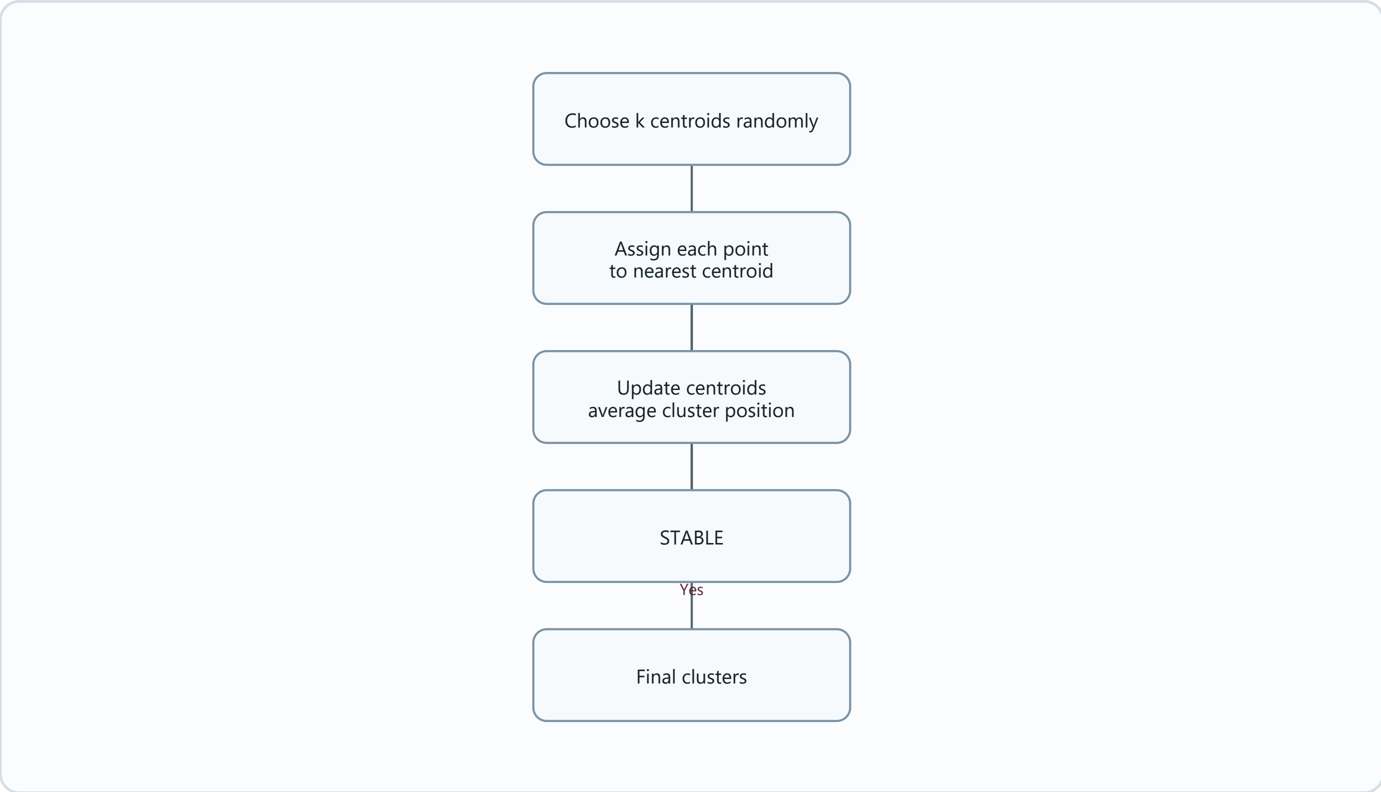
Clustering

Diagram needed: Yes. Clustering means forming groups.



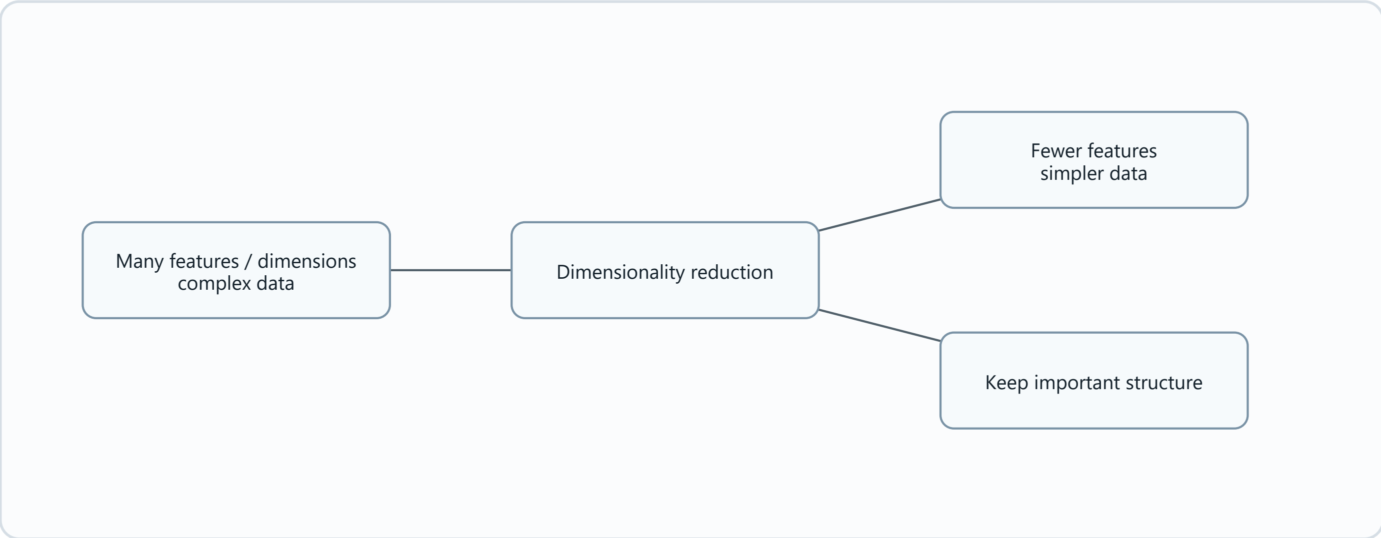
K-Means Clustering

Diagram needed: Yes. The steps are an algorithm loop.



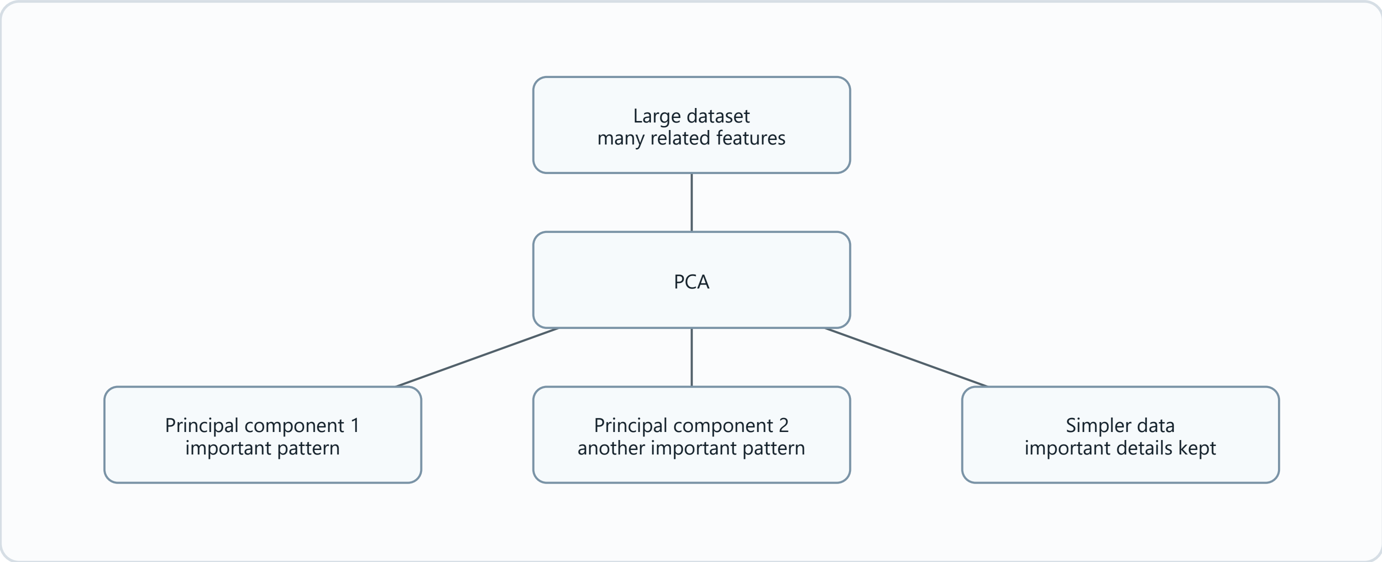
Dimensionality Reduction

Diagram needed: Yes. It is compression while keeping useful structure.



Principal Component Analysis (PCA)

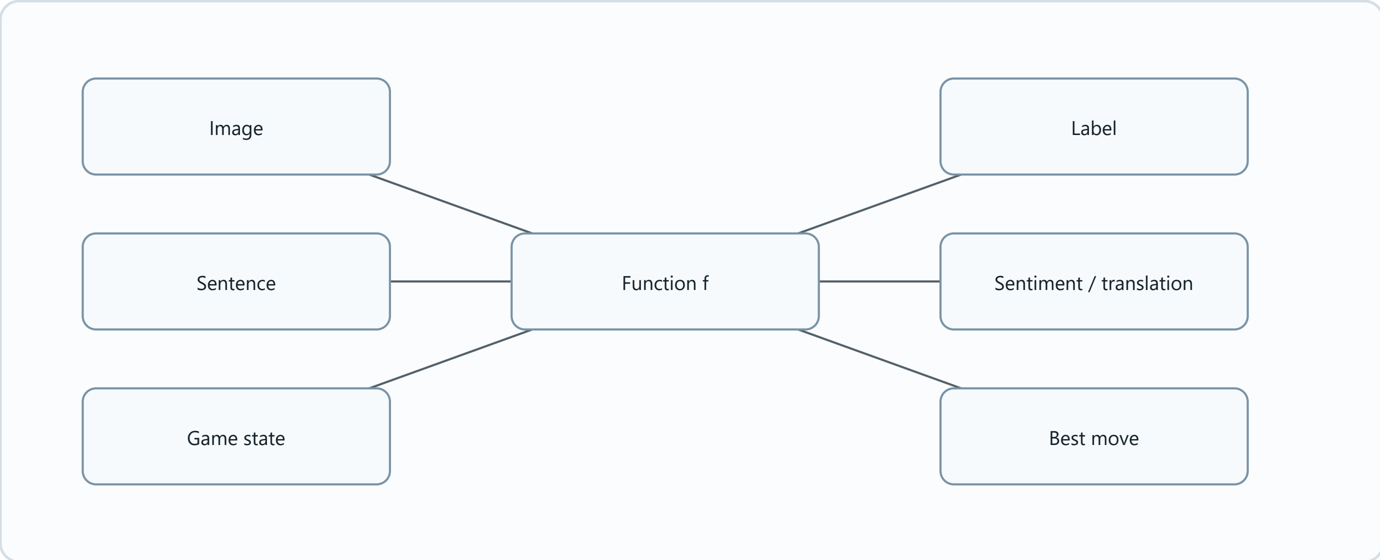
Diagram needed: Yes. The source uses "simplify big data" and "categories/factors" analogies.


Large dataset
many related features
PCA
Principal component 1
important pattern
Principal component 2
another important pattern
Simpler data
important details kept

11_Neural_Networks_and_Deep_Learning_may_2026_sem_2_2025_2026.pdf

Deep Learning Function f

Diagram needed: Yes. The chapter repeatedly says f maps input to output.



!!! EXAM TOPIC: Neural Networks !!!

Diagram needed: Yes. Neural networks are layered structures.

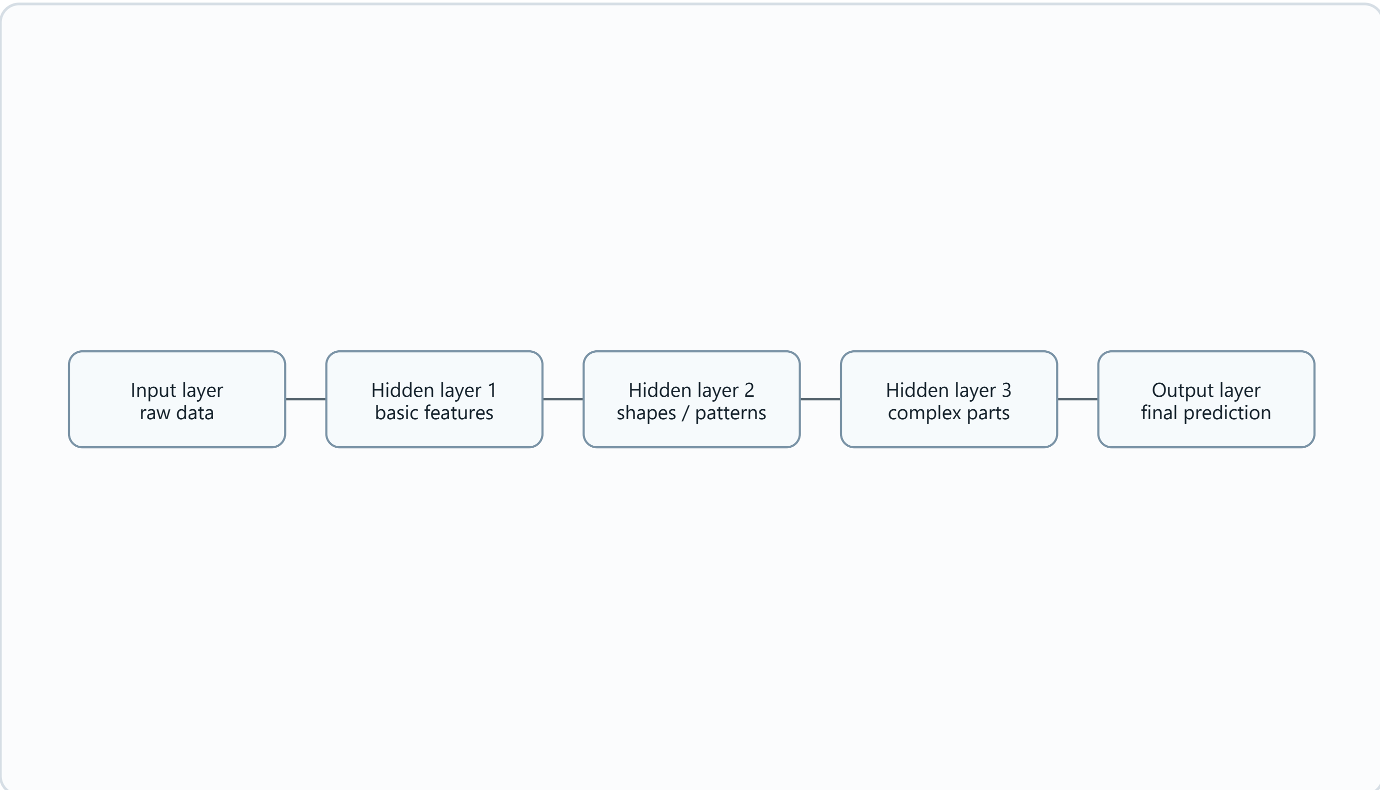
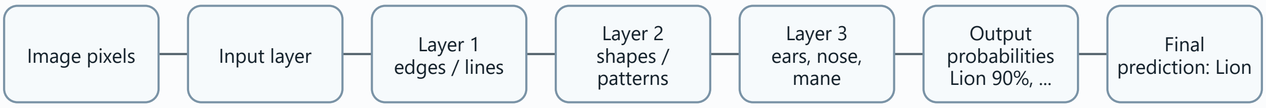


Image Classification

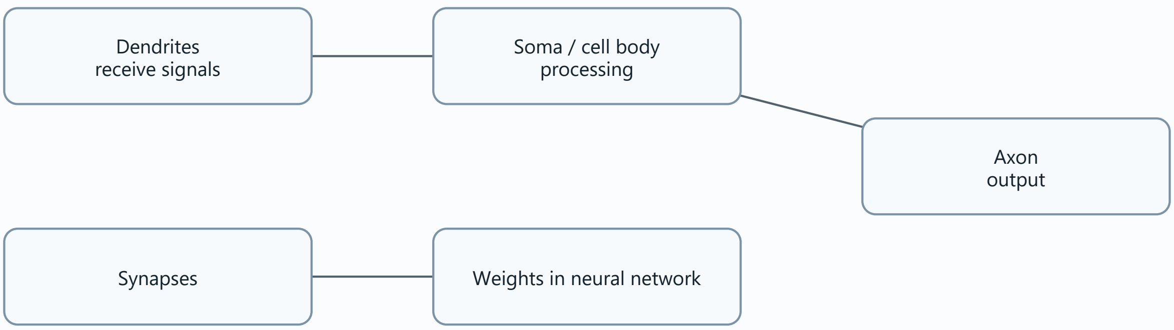
Diagram needed: Yes. The lion example is a full pipeline.



Biological Neuron and Artificial Neuron

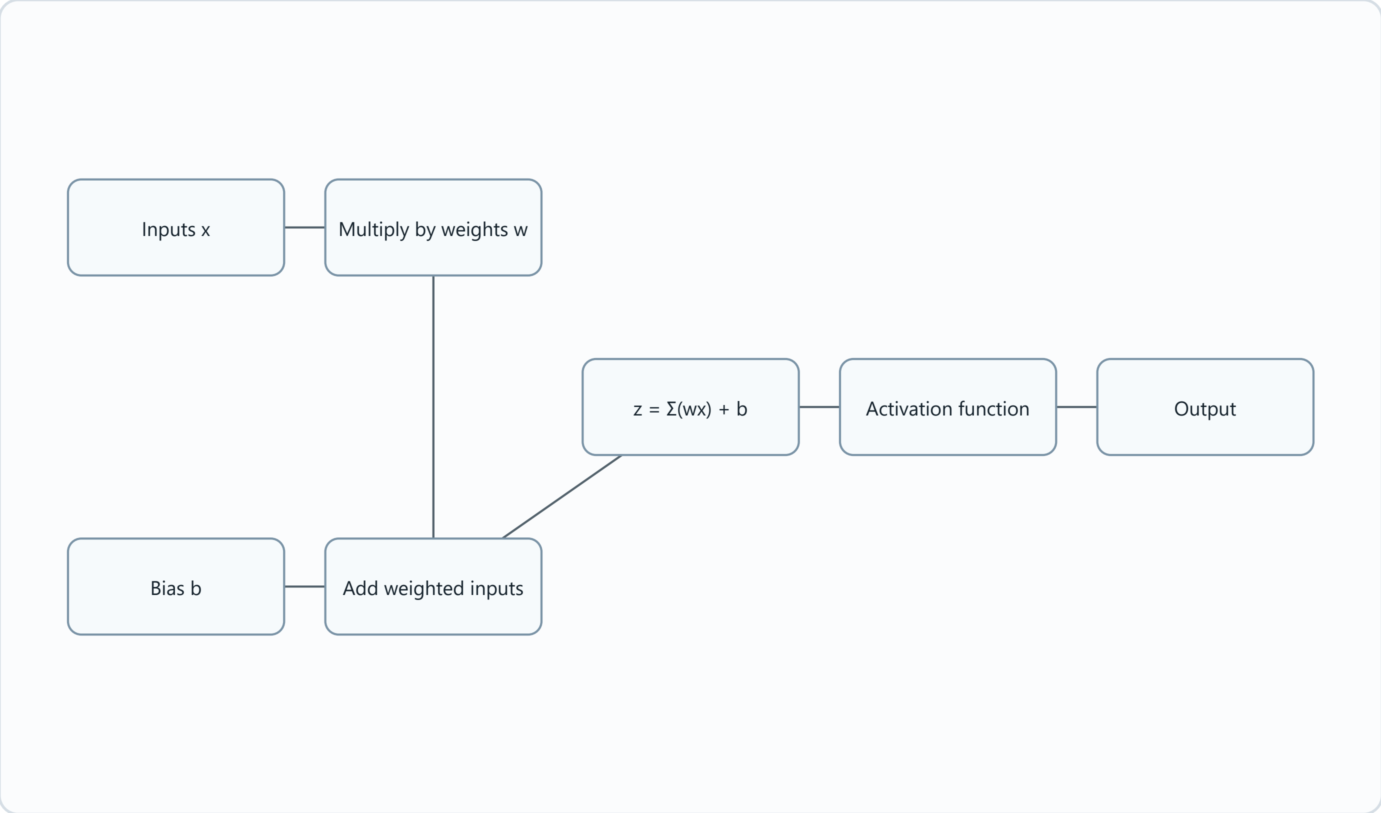
Diagram needed: Yes. The source explicitly maps brain inspiration to AI.

Biological part	Role
Dendrites	Inputs.
Soma / cell body	Processing.
Axon	Output.
Synapses	Connections modeled by weights.



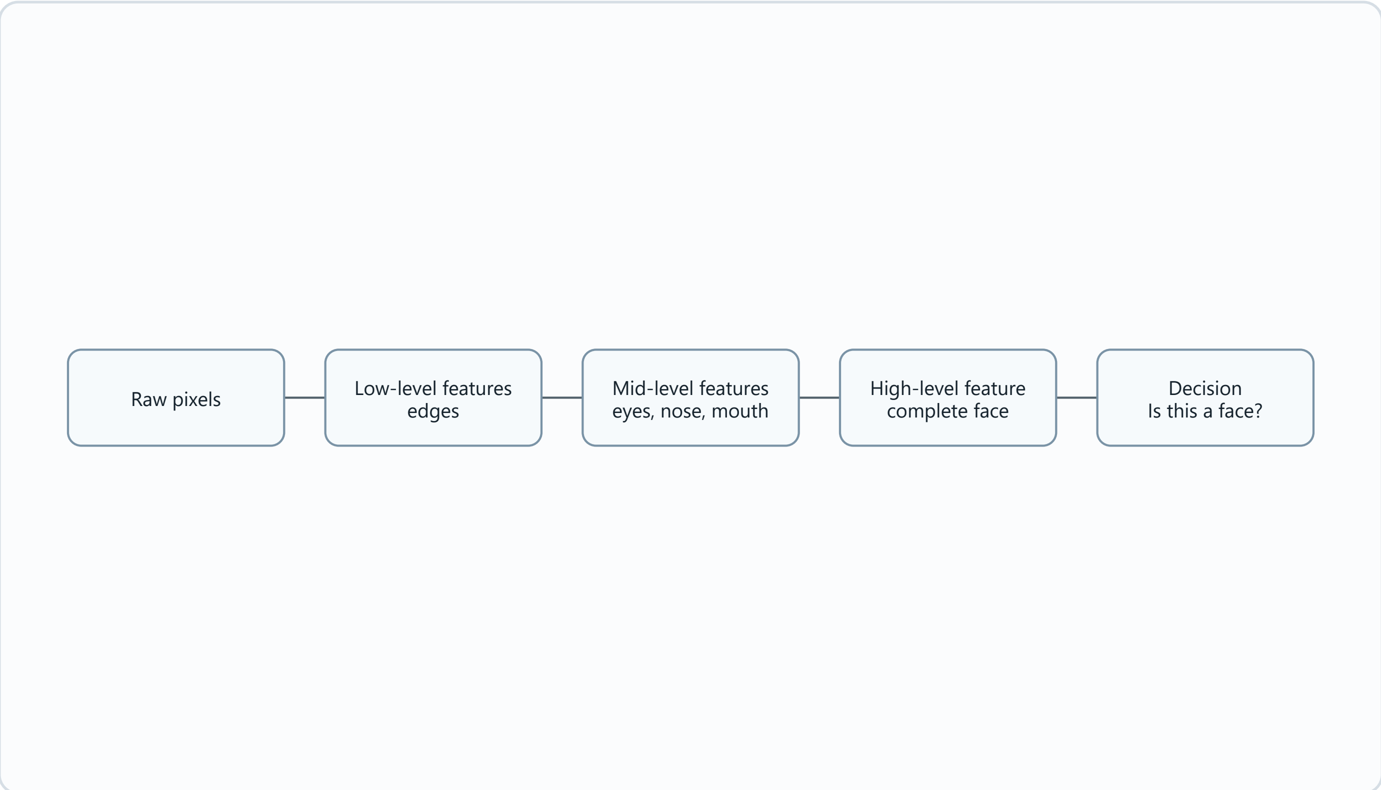
Single Artificial Neuron

Diagram needed: Yes. The weighted sum is a calculation flow.



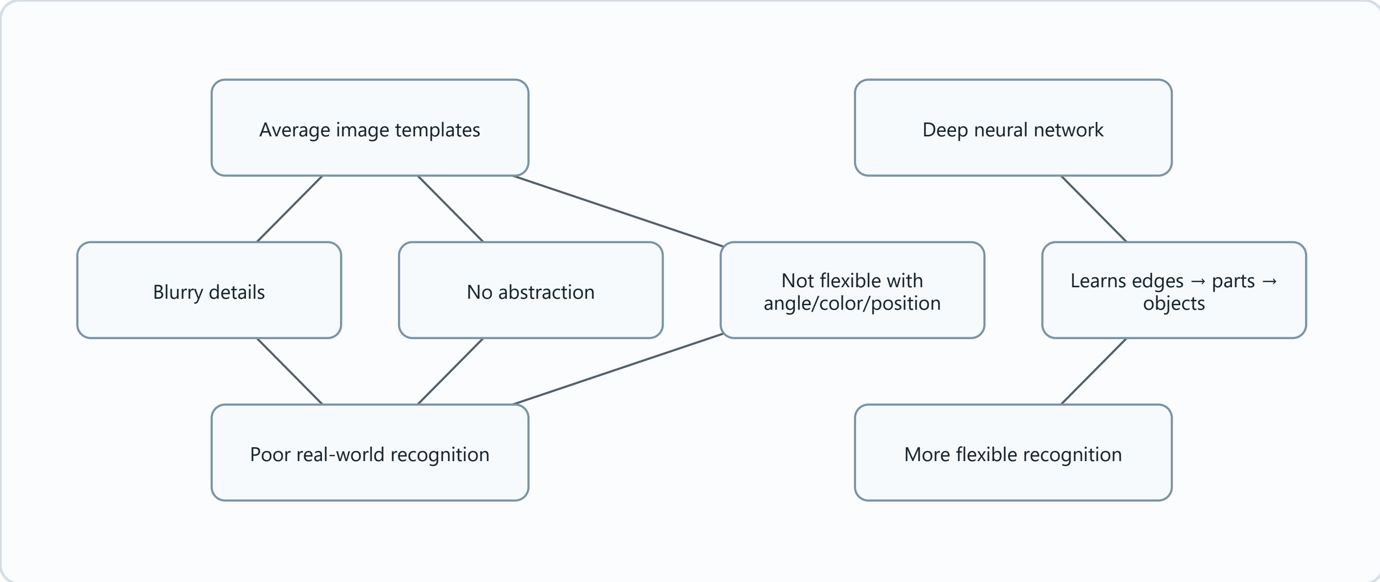
Feature Learning and Layers of Abstraction

Diagram needed: Yes. This is a major neural-network idea.



Why Linear Models Do Not Work

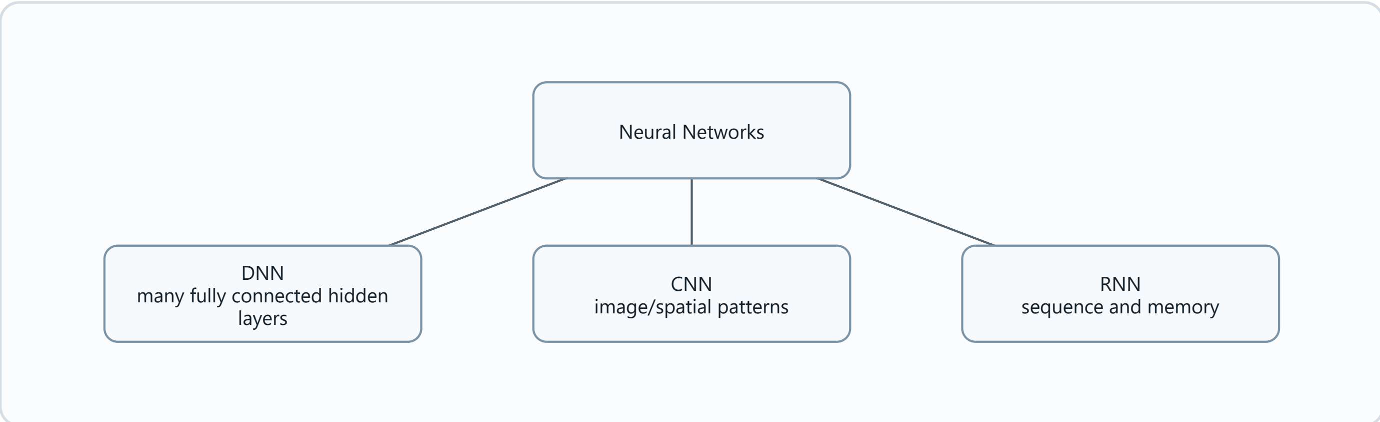
Diagram needed: Yes. It explains why deep networks are needed for images.



DNN vs CNN vs RNN

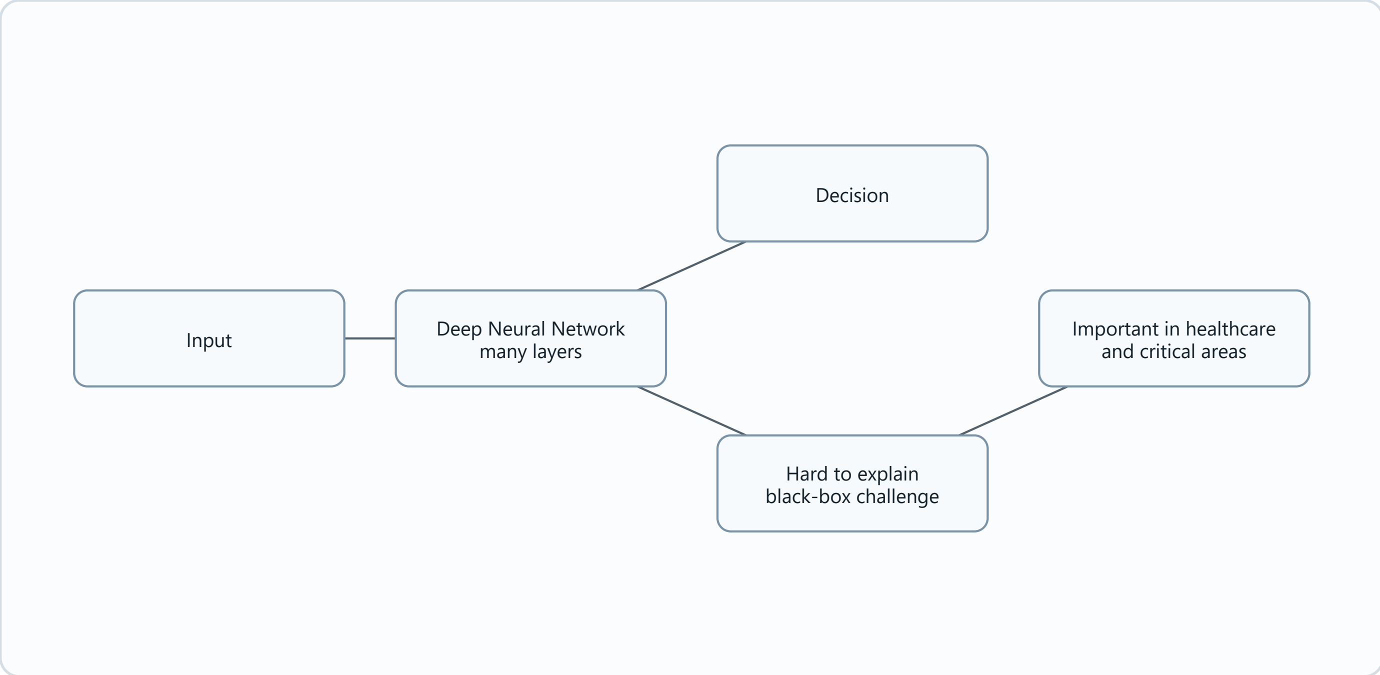
Diagram needed: Yes. This is a direct comparison.

Model	Best for
DNN	General tabular/basic classification/regression.
CNN	Images and spatial data.
RNN	Sequential data such as language, speech, music.



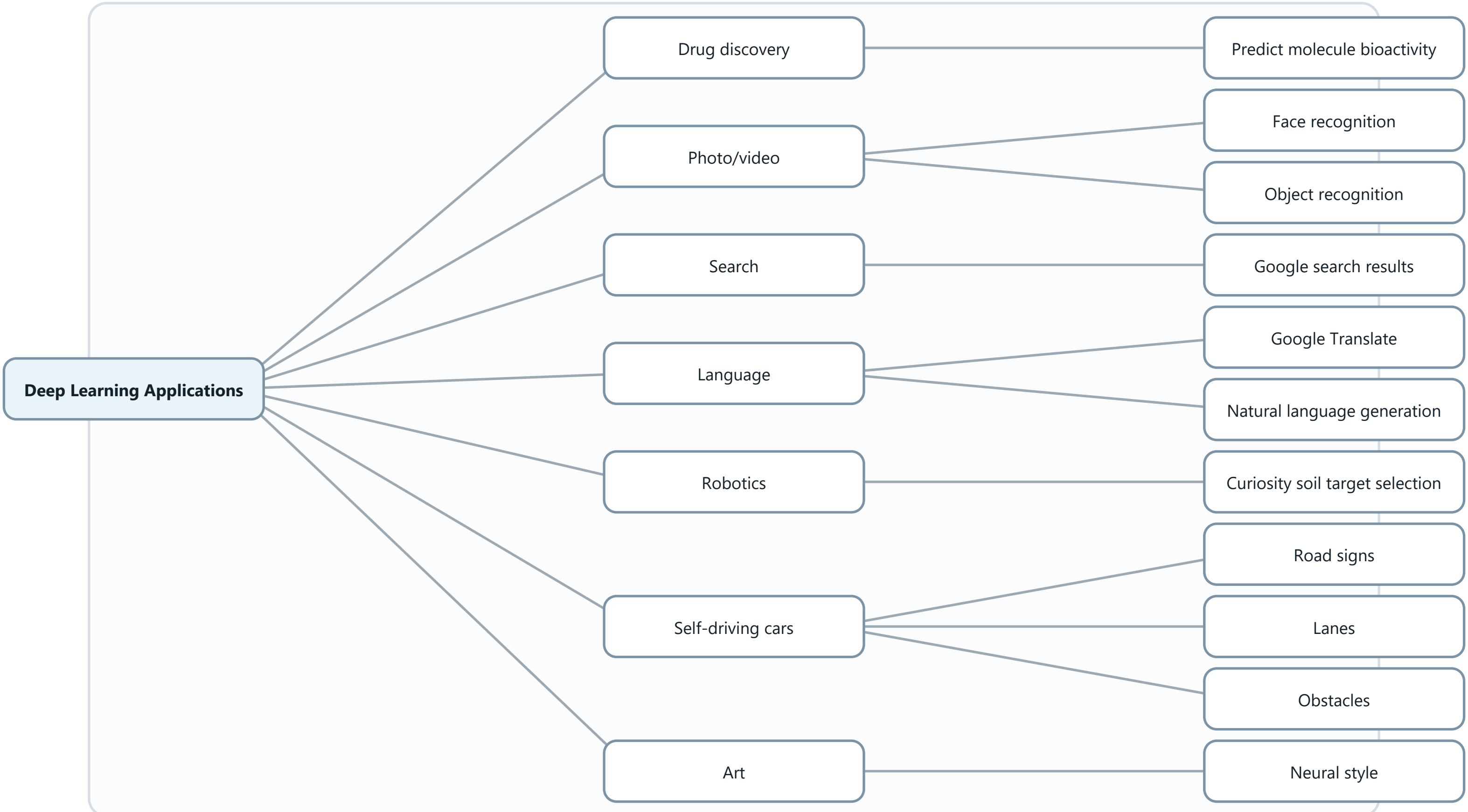
Interpretability in DNNs

Diagram needed: Yes. It explains the black-box problem.



Deep Learning Applications

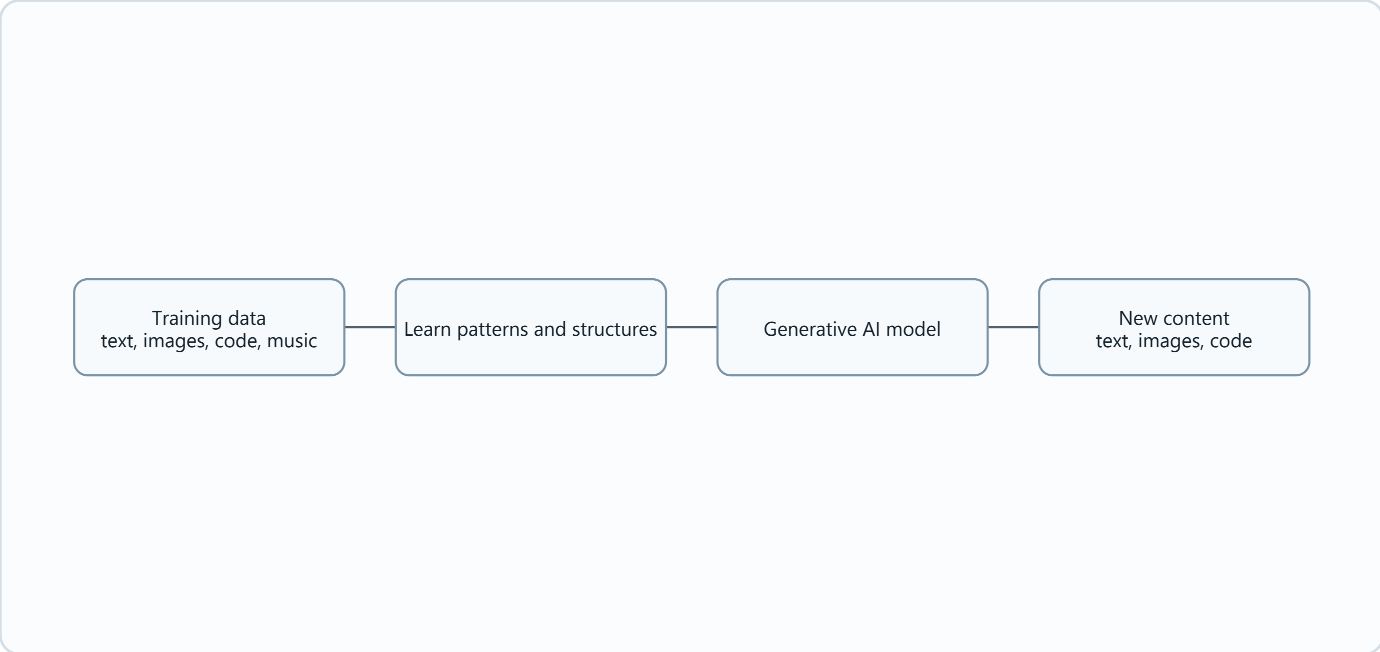
Diagram needed: Yes. The chapter lists many applications.



chapt 12 Generative AI-sem 1 2024 2025.pdf

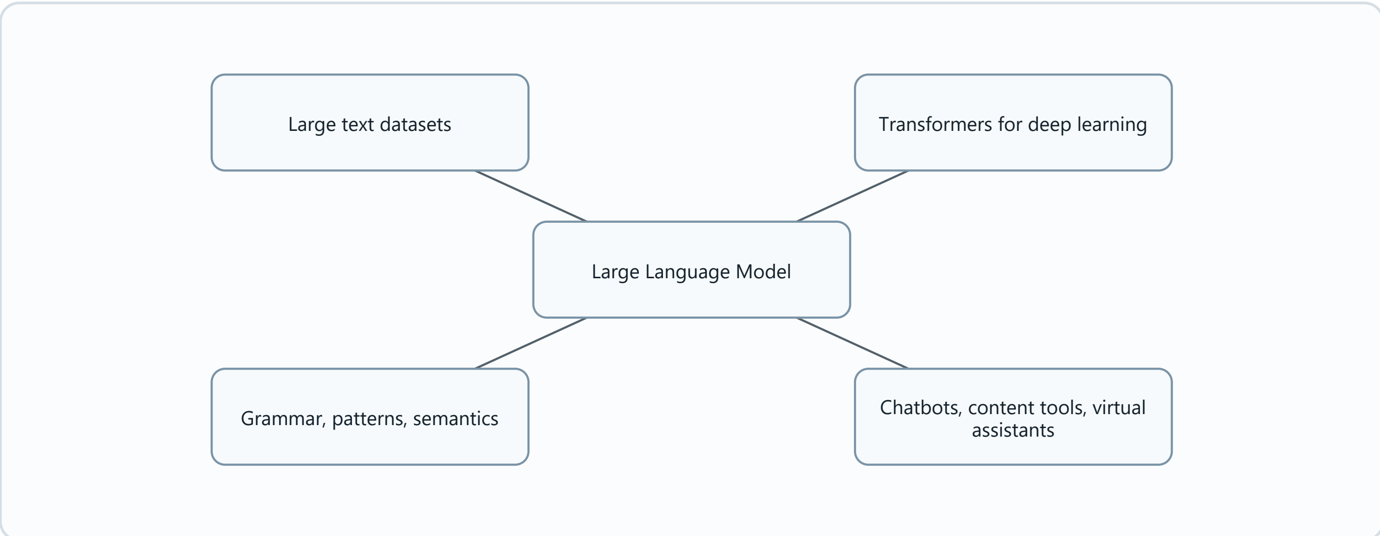
Generative AI

Diagram needed: Yes. The chapter centers on creation from learned patterns.



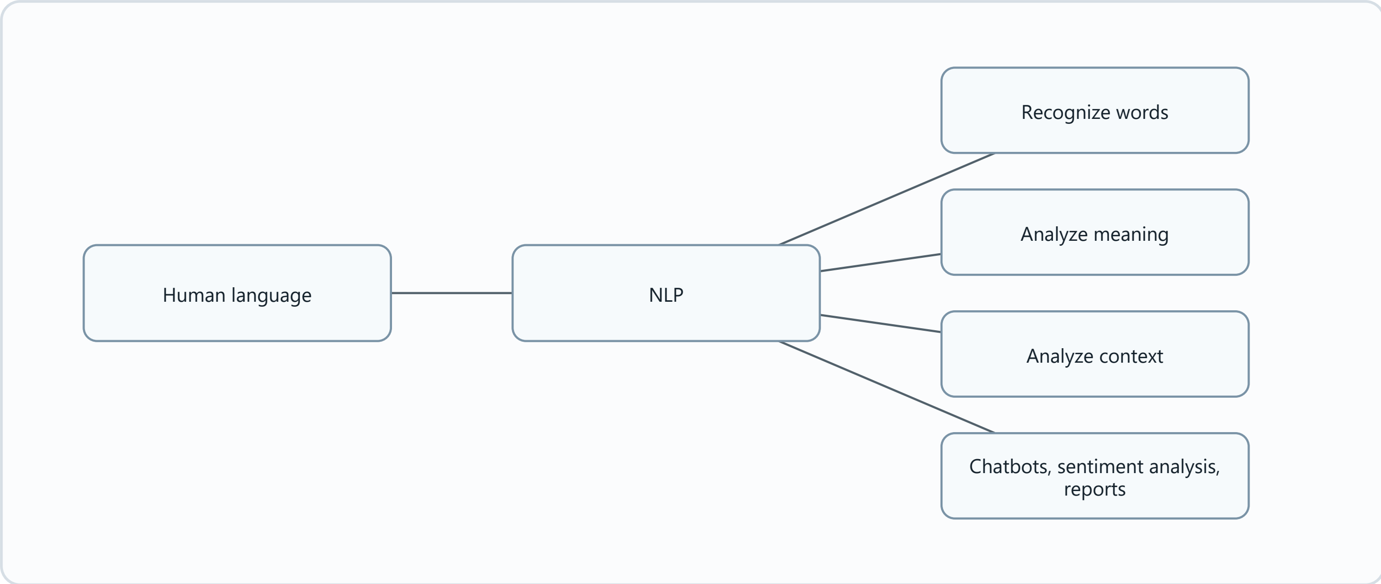
Large Language Models (LLMs)

Diagram needed: Yes. LLMs connect data, transformers, language mastery, and applications.



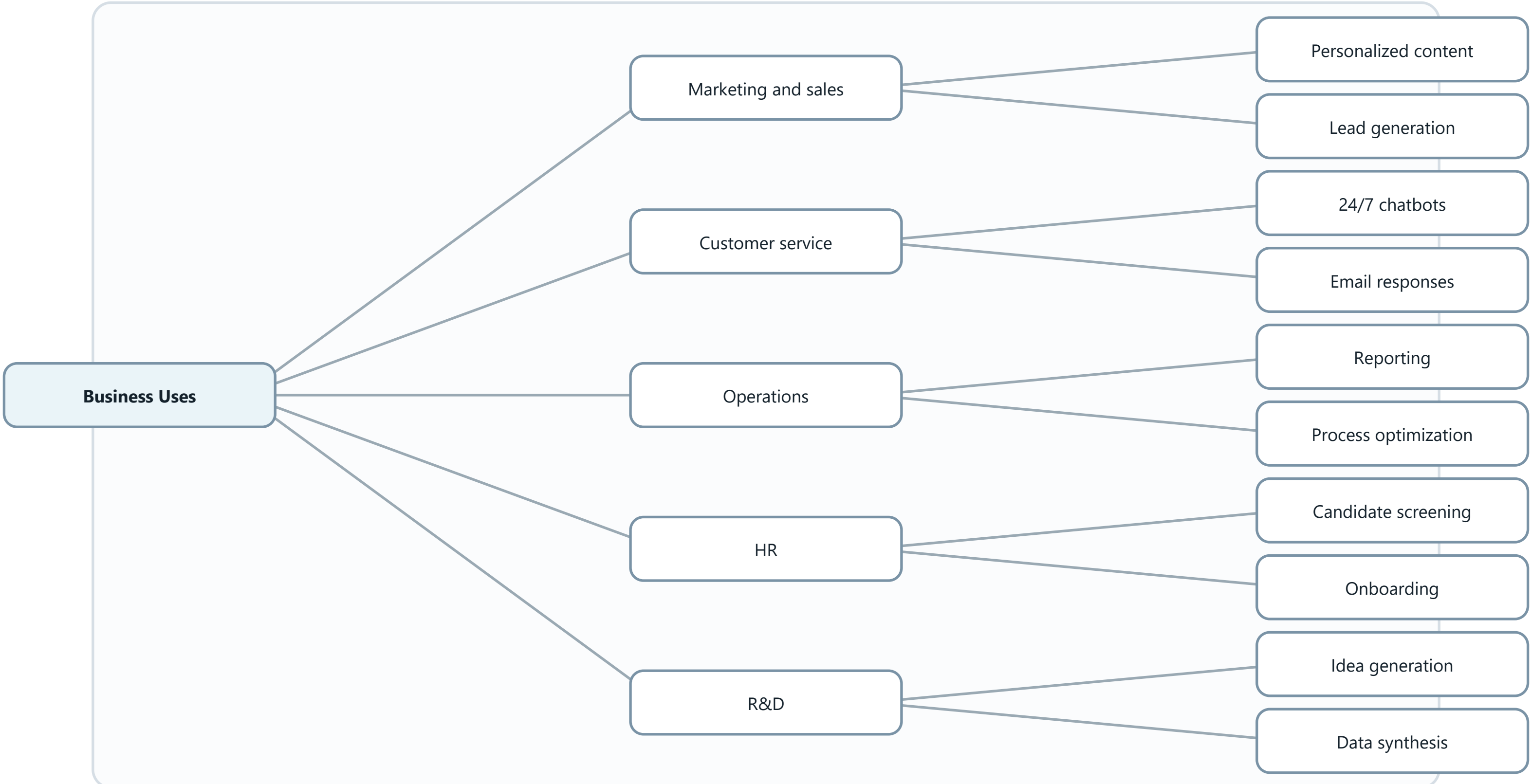
Natural Language Processing (NLP)

Diagram needed: Yes. NLP goes from words to meaning/context.



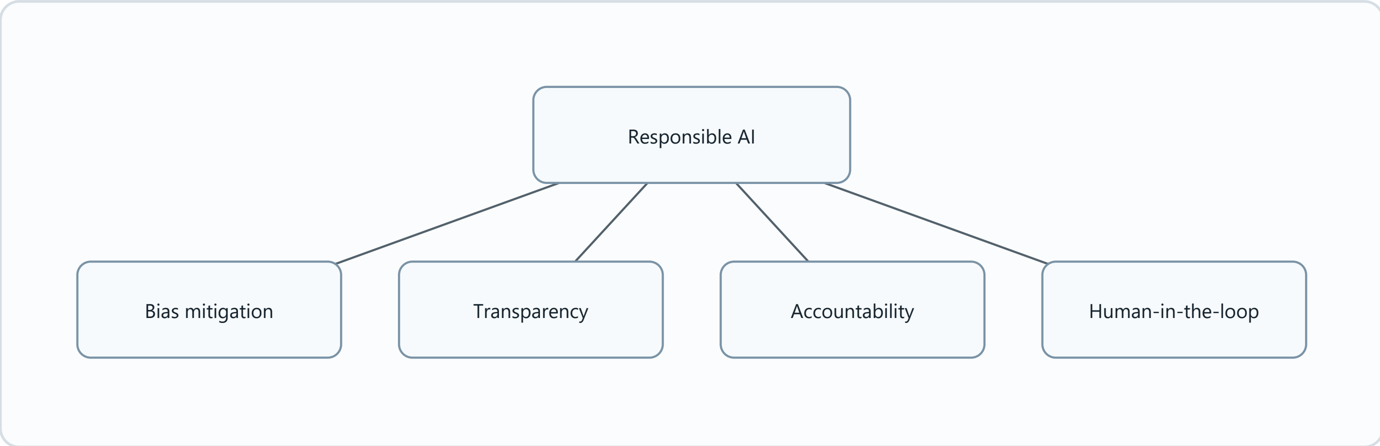
Business Applications of Generative AI and LLMs

Diagram needed: Yes. Applications are grouped by business function.



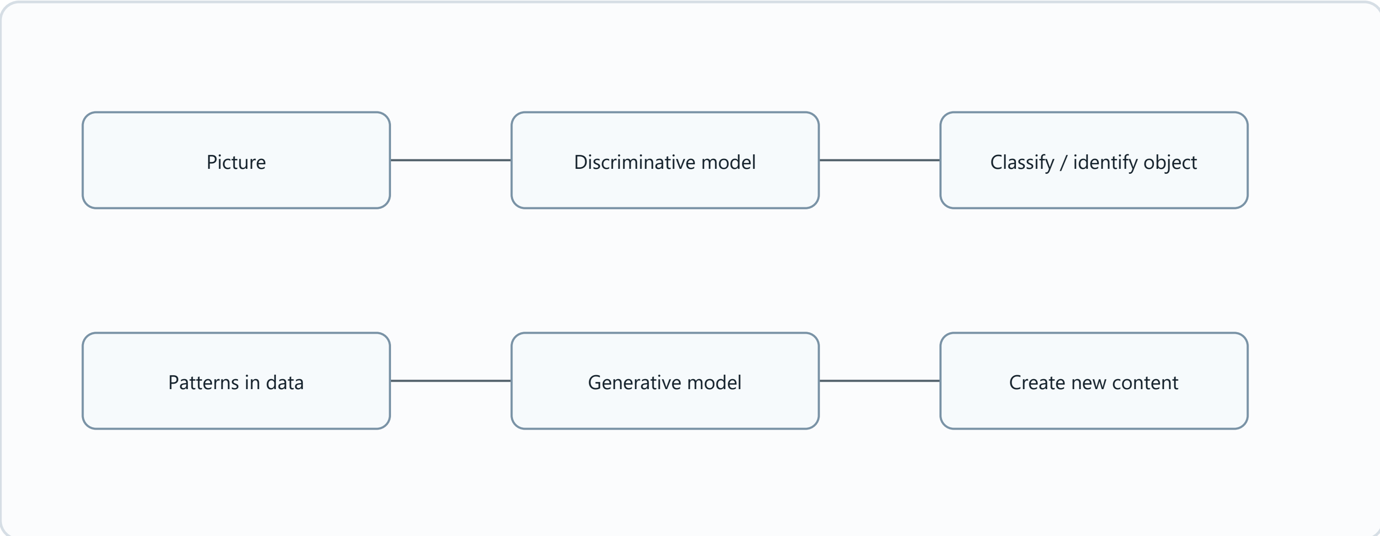
Ethical Considerations and Responsible AI

Diagram needed: Yes. Responsible AI has four key parts in the source.



Discriminative versus Generative

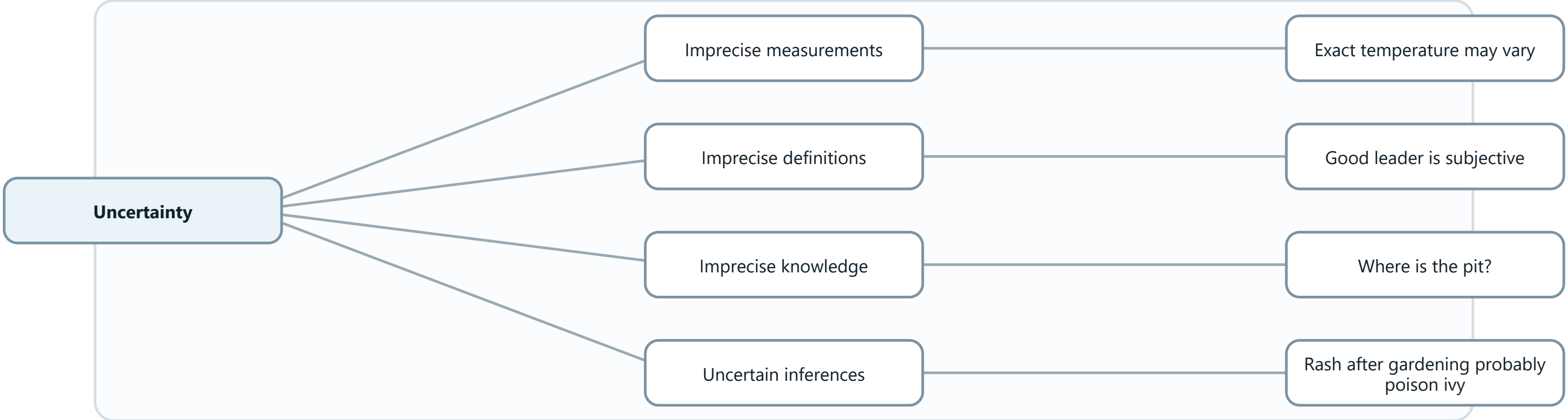
Diagram needed: Yes. The chapter repeats this core contrast.



Chap13-Other AI Approaches.pdf

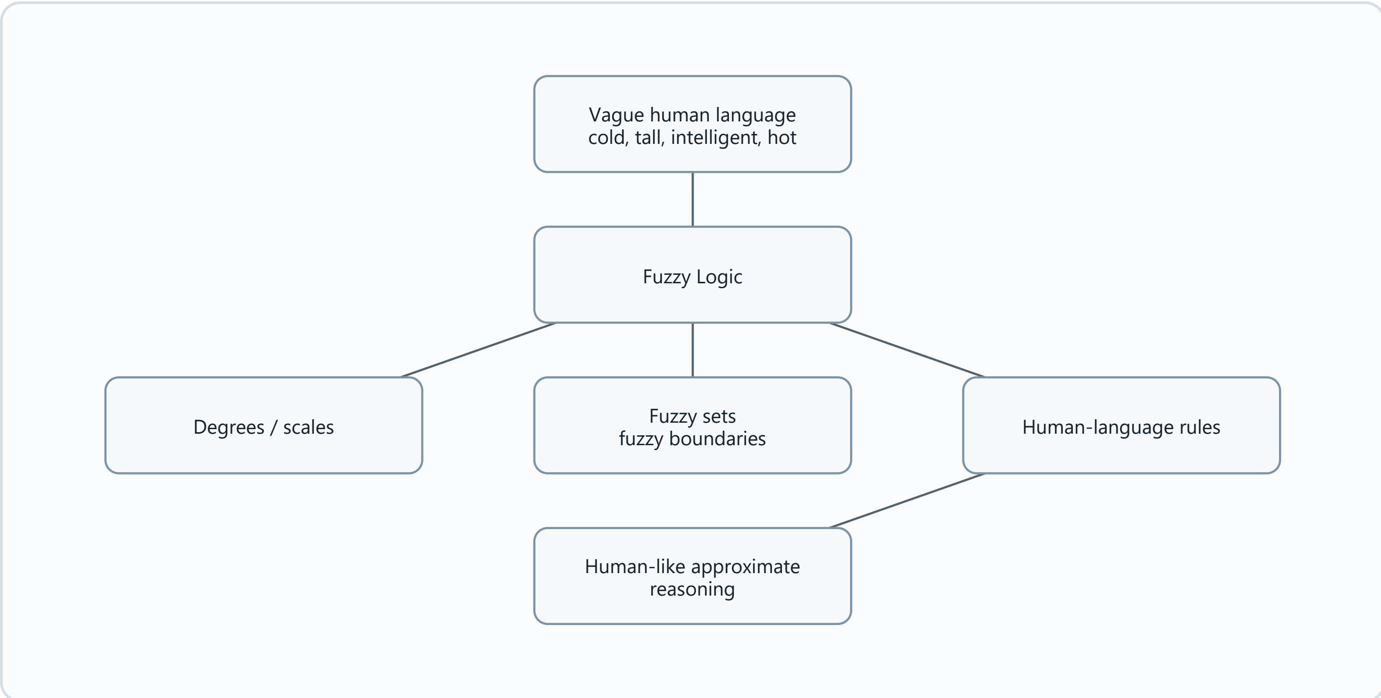
Uncertainty

Diagram needed: Yes. Fuzzy logic starts from uncertainty.



!!! EXAM TOPIC: Fuzzy Logic !!!

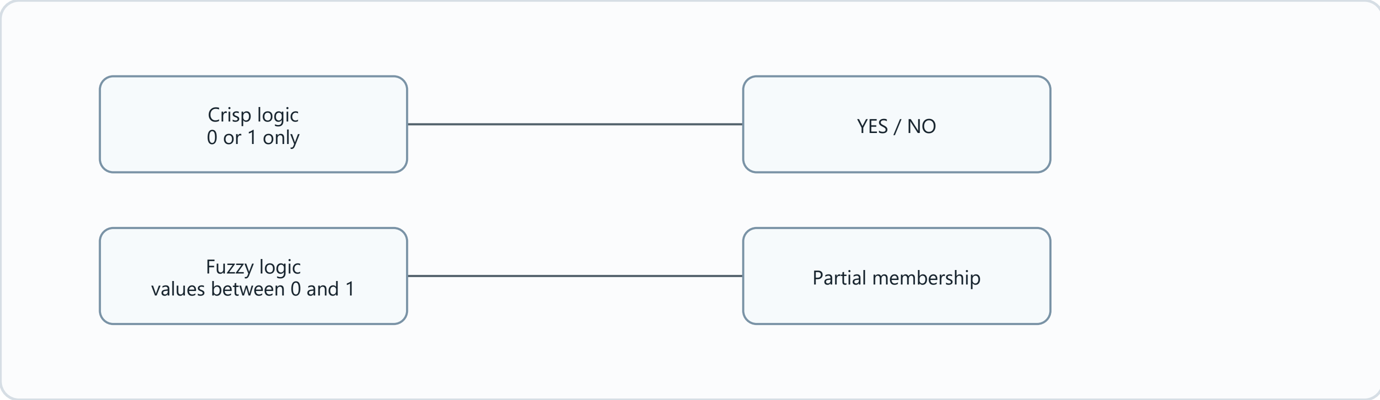
Diagram needed: Yes. Fuzzy logic is the most diagram-heavy topic.



Crisp Logic vs Fuzzy Logic

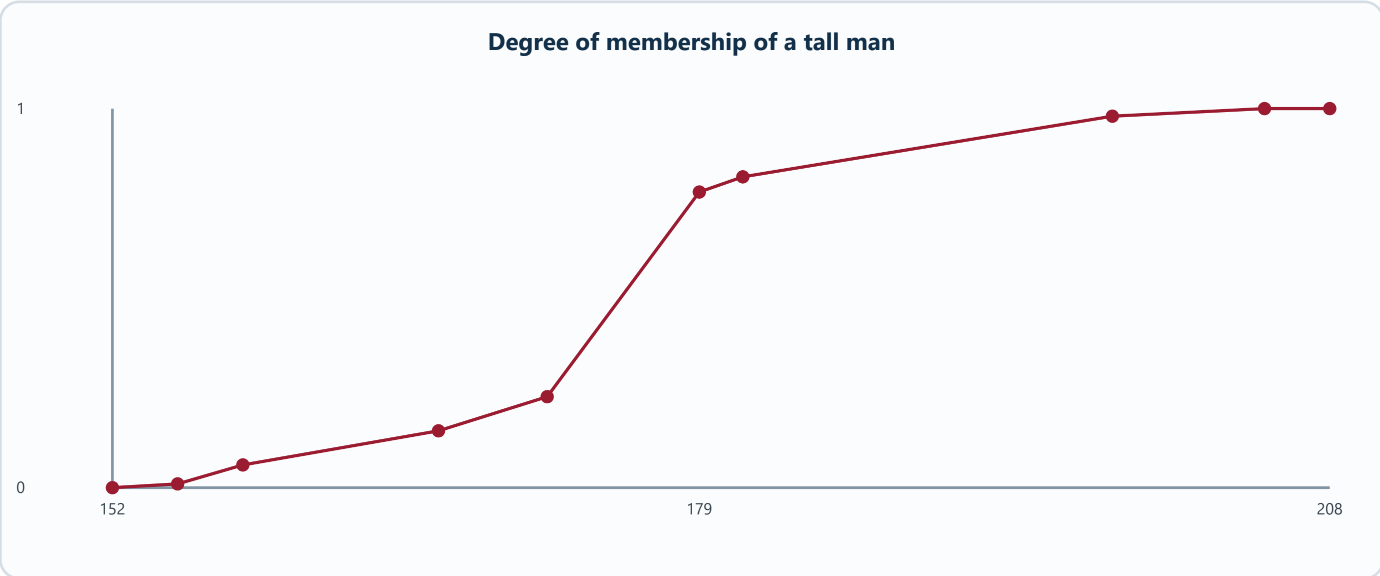
Diagram needed: Yes. This is the core comparison.

Crisp Logic	Fuzzy Logic
YES or NO.	YES gradually toward NO.
TRUE or FALSE.	TRUE gradually toward FALSE.
1 or 0.	1 gradually toward 0.
Full membership.	Partial membership.



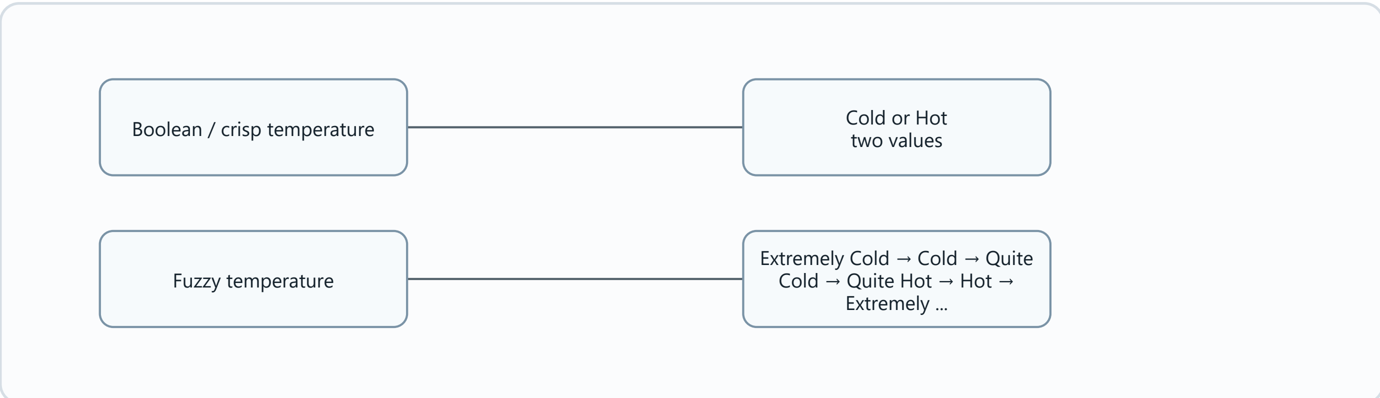
Degree of Membership: Tall Man

Diagram needed: Yes. This subtopic needs a membership chart.



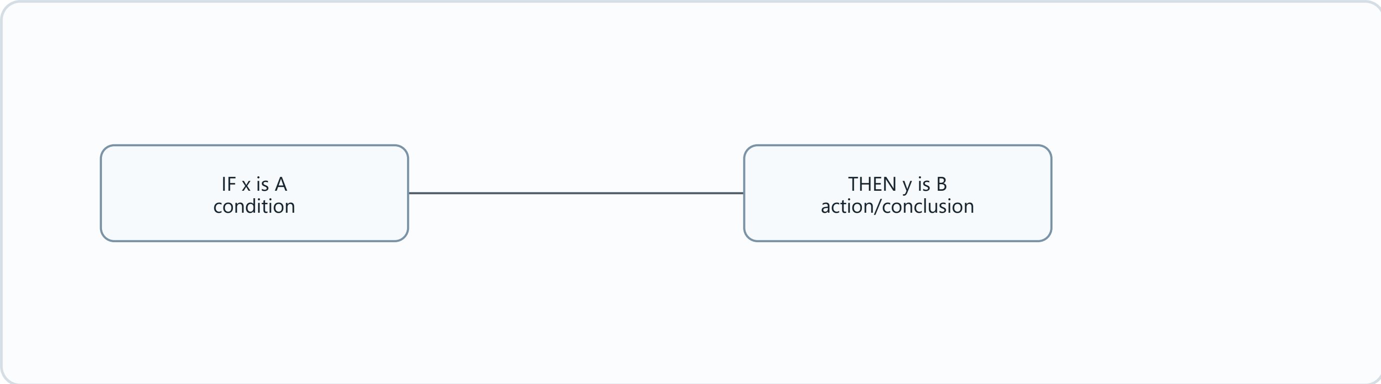
Boolean Temperature vs Fuzzy Temperature

Diagram needed: Yes. The temperature example explains smooth boundaries.



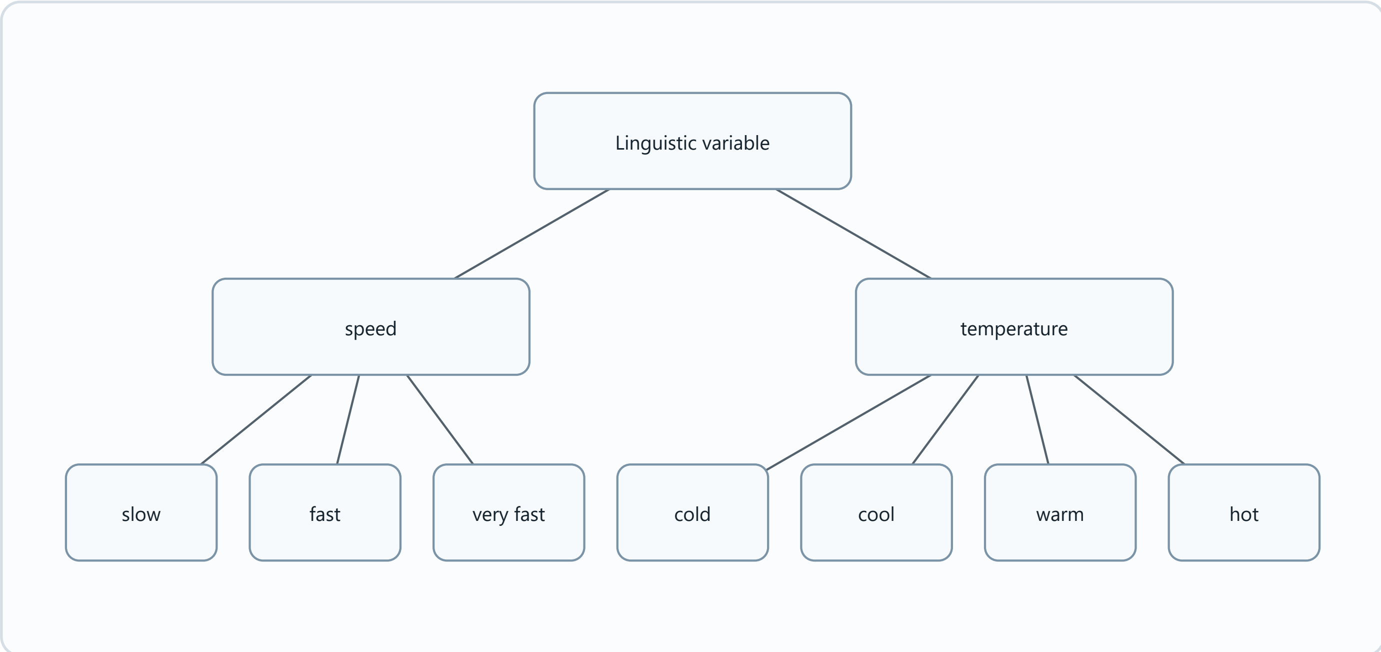
Fuzzy Rule

Diagram needed: Yes. Fuzzy rules are conditional statements.



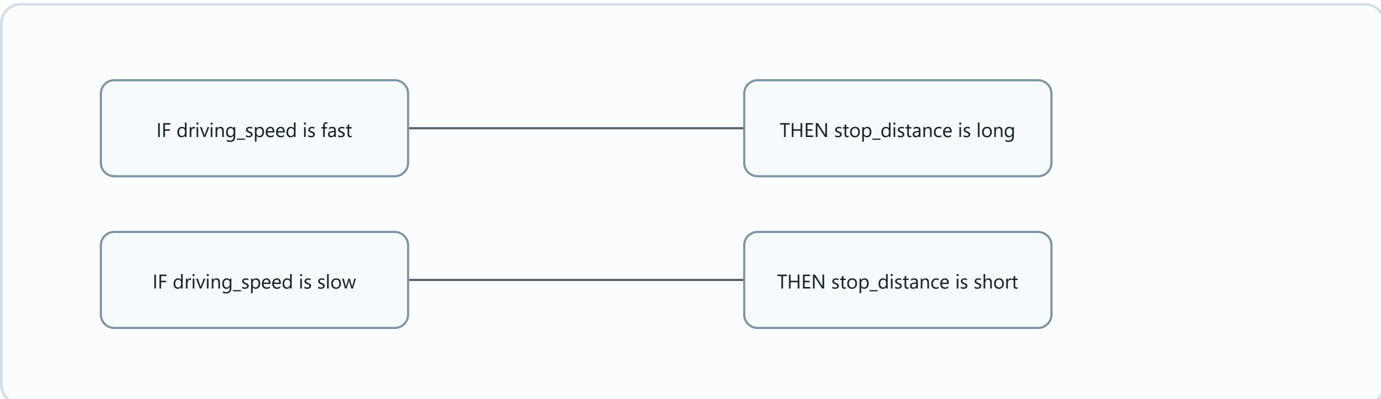
Linguistic Variable

Diagram needed: Yes. This term is abstract, so examples help.



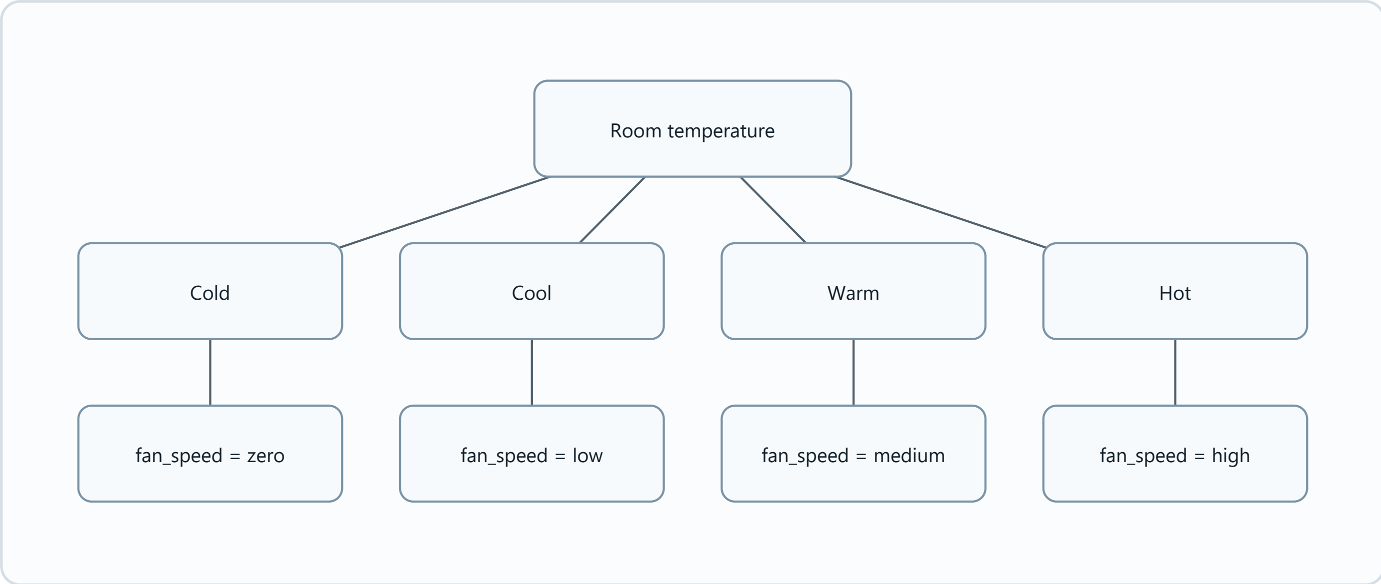
Fuzzy Rules: Driving Speed and Stop Distance

Diagram needed: Yes. This is a direct cause-effect relationship.



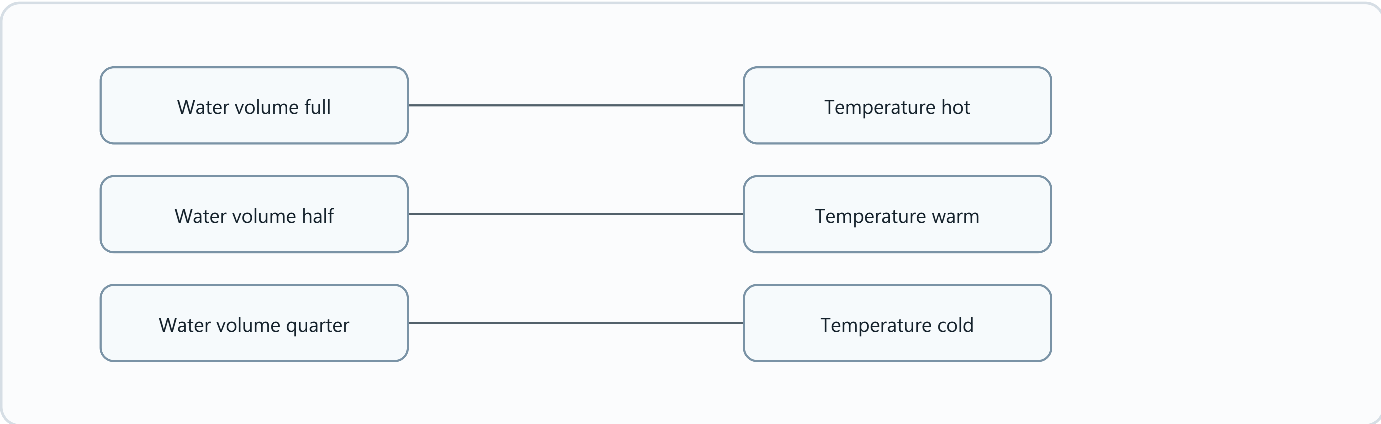
Air-Conditioner Fuzzy Rules

Diagram needed: Yes. It is a practical fuzzy-control example.



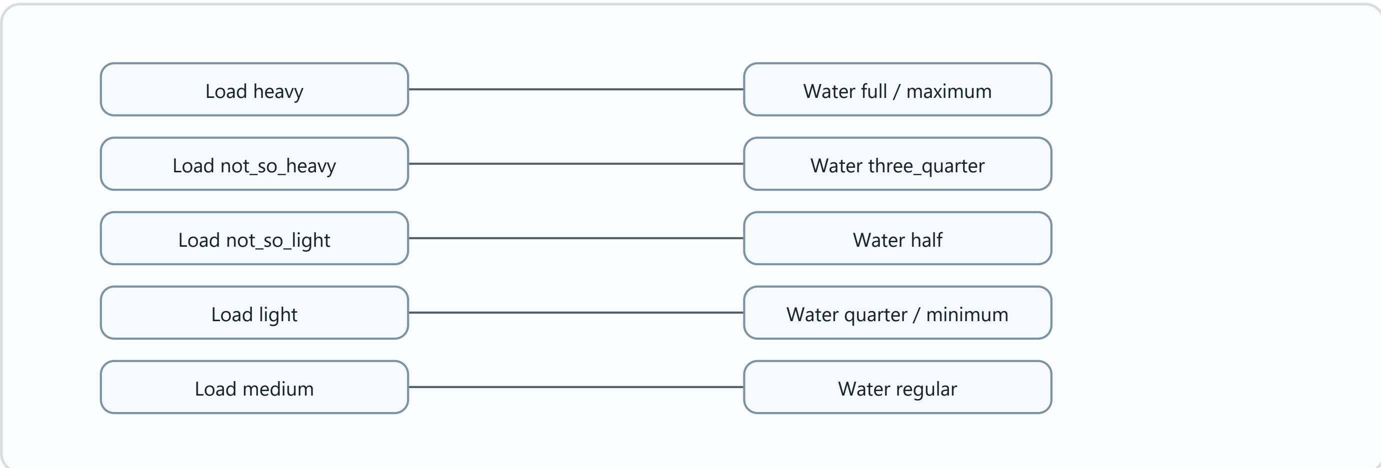
Bathroom Shower Fuzzy Rules

Diagram needed: Yes. It is a simple rule set.



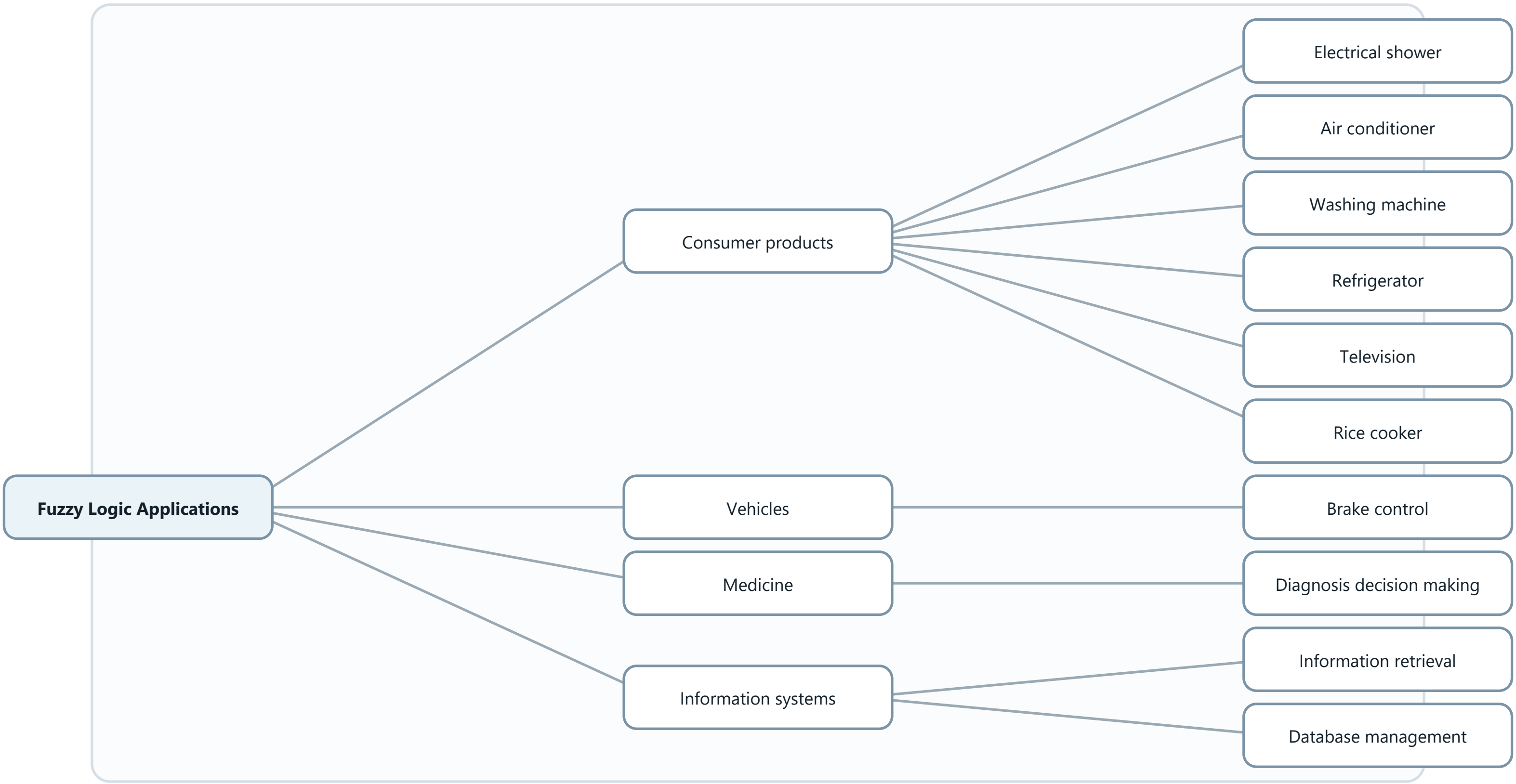
Washing Machine Fuzzy Rules

Diagram needed: Yes. It shows fuzzy control for load weight.



Fuzzy Logic Applications

Diagram needed: Yes. Applications are scattered across the slides.


Fuzzy Logic Applications
Consumer products
Electrical shower
Air conditioner
Washing machine
Refrigerator
Television
Rice cooker
Vehicles
Brake control
Medicine
Diagnosis decision making
Information systems
Information retrieval
Database management